Elliptic-Curve Index Calculus Against Pollard Rho:
A Survey of Results, a Measured Frontier with Every Cost Priced, and Open Problems

Anonymous

Abstract

Index calculus is the method that should exploit the most structure in an elliptic-curve discrete-logarithm instance, and in practice it has never beaten Pollard rho on a prime-order curve of cryptographic shape. This paper surveys where that question stands and measures it. We review the literature by field type—extension fields (Gaudry, Diem, FGHR), prime-degree binary fields and the contested first-fall-degree heuristics behind the subexponential claims (Petit–Quisquater, Semaev, and the evidence of Huang, Kosters and Yeo, Shantz–Teske and Galbraith–Gebregiyorgis), subfield curves (Galbraith–Granger–Merz–Petit) and prime fields (Petit–Kosters–Messeng, Amadori–Pintore–Sala, McGuire–Mueller)—and the generic lower bounds that frame them.

We then measure the frontier with one native harness that charges rho, baby-step giant-step, the kangaroo and several index-calculus pipelines in a single counted unit on one-target workloads with sealed, replayable records. Almost no cost in such records is complete, the rho baselines included, so we price every arm's unpriced native work alike across a band of weights and rank two methods only where the order holds across the band. On fixed ladders of eight Koblitz curves ($\log_2 r$ from 11 to 39), six NIST-shaped prime curves (20–40 bits) and five Kummer-tower curves—4,848 executions on public targets, parameters chosen on separate training targets, every session replay-audited—no index-calculus arm beats generic rho on any curve at any weight. The closest, a compact Frobenius-orbit factor base, is 3.8–15× signed-Frobenius rho with native work free; two-summand prime-field index calculus goes from 10.5× to 102× rho between 20 and 40 bits; no fitted cost exponent of a campaign index-calculus arm lies below rho's.

The measurements come from a sixteen-week engineering programme that we also document: root-finding decomposition oracles (a known mechanism; its root identity, Proposition 1, is proved in Lean, and the oracle is engineered to 383–1,535× over matrix F4 per decomposition), an $\mathbb{F}_p$ F4 brought to msolve parity on square systems, FGHR's D_4 symmetry and its weighted and Frobenius-reduced forms cutting one $k = 4$ decomposition by 48× and then 370–481×, factor bases closed under known-logarithm maps ($\mathbb{Z}[\tau]$, CM, integer and cofactor multipliers) that cut cost 1.3–20× at an unchanged exponent, and three weak families at or near rho: genus-three index calculus after the cubic GHS transfer (about $q^{1.4}$ against $q^{3/2}$; below the strongest rho at $q = 2^{16}$ under a preregistered rule and pricing, unresolved at other sizes to 2^{20}, above it under a heavier sensitivity pricing) and $k = 4$ index calculus over $\mathbb{F}_{(2^m)^4}$, whose gap to rho closes from 5,080× to 6.1× between $n \approx 2^{24}$ and 2^{56} at Gaudry's $n^{-1/4}$, with a crossover extrapolated near 2^{68}, and a Joux–Vitse genus-three cover of $E(\mathbb{F}_{p^6})$ (plain Nagao arm), at 1.6–2.0× a pooled rho constant at $p = 1009$ and closing as about p^{-2} (a lower-bound comparison). We also record the negative results, including a conservation lemma for isogeny transfers and an equal-advice audit that retires the programme's online wins over rho: Bernstein–Lange preprocessing rho with the same table is 53–317× faster, and at $n = 83$ it wins every pair run with a table matched on set-up time or memory. Separate sections cover Koblitz, random and prime-degree binary, and prime-field curves, and an appendix catalogues all 1,444 studies, benchmarks, problem records and notes of the programme by family, with each one's headline number and status. No result here is yet compared with a published index calculus at the same parameters; we state that reference arm, with the other open problems, as the experiment or proof that would settle it. Nothing here concerns a deployed curve or key recovery.

Contents

1 Introduction **3**

2 Background **5**

3 The theory, by field type **5**
3.1 What a decomposition costs . . . 5
3.2 Extension fields $\mathbb{F}_{q^k}$ with $k > 1$ fixed or small . . . 5
3.3 Binary fields $\mathbb{F}_{2^n}$ with n prime . . . 6
3.4 Subfield (Koblitz) curves . . . 6
3.5 Prime fields . . . 6
3.6 Lower bounds . . . 6

4 Measuring index calculus against rho **7**
4.1 The boundary ledger (10 September) . . . 7
4.2 ecbench (from 2 October) . . . 7
4.3 The pricing band . . . 8
4.4 Claim hygiene and change classes . . . 8
4.5 Corrections the accounting forced . . . 9

5 Chronology **10**

6 Root-finding decomposition **12**
6.1 Origin: a suggestion in an audited manuscript . . . 12
6.2 The algorithms . . . 12
6.3 Generalising the observation . . . 12
6.4 Results . . . 13

7 Gröbner-basis engines **15**

8 Koblitz curves **16**
8.1 Frobenius-invariant and compact-orbit bases . . . 17
8.2 The oracle sequence . . . 17
8.3 Against a stronger rho: the September reversals . . . 18
8.4 The one-target online ladder . . . 19
8.5 The $n = 83$ campaign . . . 20
8.6 Leaderboard Table A: the Koblitz rules . . . 20
8.7 ECC2K-130 . . . 21
8.8 Factor bases closed under $\mathbb{Z}[\tau]$ . . . 22

9 Random and prime-degree binary curves **22**
9.1 The degree question, measured . . . 22
9.2 Decomposition oracles . . . 23
9.3 Leaderboard rows . . . 24
9.4 ECC2-131 . . . 24

10 Prime fields **24**
10.1 Known-answer ladders and leaderboard rows . . . 25
10.2 The Kummer tower . . . 25
10.3 Structured bases on the P-256 field . . . 26
10.4 Small-abscissa bases and isogeny classes . . . 26
10.5 Lattices . . . 26
10.6 ECCp-131 . . . 26
10.7 Audits of standard curves . . . 27
10.8 Literature claims that did not survive . . . 27
10.9 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$ . . . 27

11 Covers, isogenies and other transfers **28**

12 What improved, and where the frontier stands **30**
12.1 The measured frontier: one campaign, every cost priced . . . 30
12.2 The engineering record . . . 32
12.3 The improvement ladder, 14–15 October . . . 32
12.4 The ladder continued, 16–20 October . . . 36
12.5 The ladder continued, 21–24 October . . . 39
12.6 Three frontiers . . . 41
12.7 What would move the cold frontier . . . 42

13 Open problems and proposed experiments **42**

14 Conclusion **44**

A Artefact index **45**

B Work in open pull requests **46**

C Catalogue of the programme's studies **48**
C.1 Koblitz curves . . . 48
C.2 Random and prime-degree binary curves . . . 57
C.3 Prime-field curves . . . 61
C.4 Extension fields, covers and transfers . . . 64
C.5 Rho, the harness and validation . . . 68
C.6 Engines, factor-base tooling and literature . . . 71
C.7 Out of scope . . . 77

1 Introduction

For a prime-order subgroup $\langle G \rangle$ of size r on an elliptic curve $E/\mathbb{F}_q$, the best generic algorithm for the discrete-logarithm problem is Pollard rho [22, 27], with expected cost $\sqrt{\pi r/(2A)}$ group operations when A automorphisms (negation, and on a Koblitz curve the Frobenius) can be folded into the walk [13, 28]. Index calculus, the method that breaks finite-field discrete logarithms in subexponential time, was brought to elliptic curves by Semaev's summation polynomials [23] and by Gaudry's and Diem's analyses over extension fields [5, 14]. Its open question has always been *where the crossover is*: on which curves, at which sizes and under which accounting does it cost less than rho?

This paper is a survey of where that question stands, in three parts. First, the literature (§3): what is proved, what rests on heuristics such as the first-fall-degree assumption, and what the experimental record says about those heuristics. Second, a measured frontier (§12): every method family—rho variants, baby-step giant-step, the kangaroo and several index-calculus pipelines—run on the same one-target workloads in one counted unit, on fixed curve ladders, with the native work that each method leaves unpriced priced alike on every arm across a band of weights, and a ranking reported only where it holds across that band. Third, open problems, each stated as the measurement or proof that would settle it (§13).

The measurements come from one open repository and a programme of work there between July and October 2026 (§5): about 6,900 commits and, on `main` at the paper's date, 784 study directories under `research/`, 205 benchmark rows, 39 open-problem records and about 400 research notes and programme documents, a boundary ledger (`docs/ic/boundary_targets.json`), a native harness (`src/cryptanalysis/ecbench/`), and the engineering results of §§6–11. It is easy, in a body of work this size, to improve a headline number by two orders of magnitude and establish nothing; this paper is organised to prevent that. Every result is stated with the boundary on which it was measured (oracle stage, complete solve, cold one-target cost, online cost after retained set-up), the unit, the sample, the interval, and the committed artefact that holds the records.

The paper is meant to be exhaustive for that repository. The family sections (§8 Koblitz, §9 random and prime-degree binary, §10 prime fields, §11 covers and transfers) discuss the lines of work that bear on the comparison with rho, and Appendix C lists all 1,444 entries—every study directory, benchmark row, problem record and note—by family, with each one's headline number, class and status, and a check that none is missing; every line with a result is either discussed or catalogued. Work in open pull requests is listed separately and not used (Appendix B). Other repositories of the same laboratory are out of scope.

The routes measured. The engineering lines, each a section below:

1. *Root-finding decomposition* (§6): replacing the Gröbner solve of the decomposition step by closed-form roots of S_3, then by meet-in-the-middle tables for more summands, then by GPU, tower and dense-index variants.
2. *Gröbner-basis engines* (§7): matrix F4, F5, an inherited F4 that re-uses its root echelon form, an index-calculus-specific F6-IC, an $\mathbb{F}_p$ F4 brought to msolve parity, and an audit that found and fixed one soundness bug.
3. *Koblitz and binary curves* (§8): Frobenius-invariant factor bases [12], their compact signed-orbit form, SAT oracles [26], and a ladder of one-target online comparisons with rho from $n = 53$ to $n = 83$.
4. *Prime and extension fields* (§10): $j = 0$ and $a = -3$ known-answer ladders, the Kummer-tower factor base of Petit–Kosters–Messeng [21], and Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$ with the D_4 symmetry of Faugère–Gaudry–Huot–Renault [9].
5. *Isogeny, transfer and structure routes* (§11): conductor gaps in the ECC2K-130 isogeny class, isogeny fibres, transfer-aware index calculus, GHS and Weil descent; mostly negative, with one lemma that explains why.
6. *Literature replication* (§10, §11): the *lit2026* programme, which turned each finding of a 2025–26 literature search into a measured change, a diagnostic, or a recorded negative.

Underneath all six sits the measurement work of §4: a single harness (*ecbench*) that charges rho, BSGS, the kangaroo and index calculus in one unit on one-target workloads, and the accounting corrections that the harness forced on earlier claims.

Contributions.

- A survey of elliptic-curve index calculus organised by field type, separating proved results from heuristic ones and recording where the first-fall-degree debate stands (§3).
- A measurement discipline: one unit (S = group-addition equivalents/$\sqrt{r}$), one-target cold and online boundaries, sealed and replayable records, and a pricing band that treats the unpriced work of rho and of index calculus alike (§4).
- A measured frontier on fixed Koblitz and prime ladders, with parameters chosen on separate training targets and fitted cost exponents per method (§12).
- The engineering record: root-finding decomposition (with its prior art), Gröbner engines, compact-orbit Koblitz bases, D_4 symmetry at $k = 4$, and the negative results (§§6–11), and the improvement ladder of 14–15 October measured against the index-calculus baseline, with the distance to rho left after each step (§12.3), continued through 20 October with the corrections of a repository-wide validation sweep (§12.4) and through 24 October with known-logarithm factor bases, a re-priced Koblitz reference and a third weak family near rho (§12.5).
- Open problems, each with the experiment or proof that would settle it (§13).

What this paper does not claim. No asymptotic sub-rho algorithm, no change to the Gaudry–Diem picture for prime fields, no statement about ECC2K-130, P-256 or any deployed curve beyond labelled arithmetic on public parameters, no key recovery. Every instance is a public synthetic or standard-parameter known-answer instance. Every change is classed, in the repository's vocabulary, as an ADVANCE (an exponent or a crossover moves under complete accounting), ENGINEERING (a constant moves), RELABELLING, or ACCOUNTING; with one modelled exception (§10.9) every improvement in this paper is ENGINEERING or ACCOUNTING. One measurement places index calculus below the strongest rho: genus three after cubic GHS at $q = 2^{16}$, under its preregistered pricing, in a weak class where theory predicts it, and not under the protocol's sensitivity pricing (§11); we report it as a measured crossing at one size, not as an advance. Nor does it claim to improve on published index calculus: every ratio is against rho or against this programme's own previous arm, and stage ratios against matrix F4 or Buchberger are against the repository's own engines. External engines were run as references only: msolve against the $\mathbb{F}_p$ F4 on the same systems, msolve and PolyBoRi computing full bases where our Boolean engines decide (a different computation), and CADO-NFS for the linear algebra (§7). A reference arm built from the published methods is open problem 10 (§13).

How to read the numbers. "$X\times$" always means baseline cost divided by candidate cost on the stated boundary. Stage ratios (one decomposition, one Gröbner call) are never presented as end-to-end speed-ups; wall-time ratios from hosts that did not earn an isolation level are labelled *descriptive*; a cost with unpriced work is labelled a *lower bound*. Appendix A lists the artefact behind each table.

2 Background

Rho and its floor. With A efficiently computable automorphisms acting on $\langle G\rangle$, a rho walk on orbit representatives has expected cost $\sqrt{\pi r/(2A)}$ group additions. $A = 2$ (negation) on every curve; $A = 2n$ on a Koblitz curve $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, where the Frobenius $\pi(x, y) = (x^2, y^2)$ acts on $\langle G\rangle$ as a scalar λ [13, 28]. ECC2K-130 is the case $a = 0$, $n = 131$ of this family; its record GPU rho attempt is a separate paper in this repository (`paper/ecc2k130/`).

Index calculus on elliptic curves. Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$. A *relation* is a decomposition $aG + bQ = R = P_1 + \cdots + P_m$ with $P_i \in \mathcal{F}$; once about $|\mathcal{F}|$ relations are found, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. The pipeline has five stages (factor base, decomposition oracle, relation yield, rank/linear algebra, verification) and a sixth, the comparison with rho on the same target. The *decomposition oracle* decides whether R splits over $\mathcal{F}$. Semaev [23] reduced it to the roots of the summation polynomial $S_{m+1}(x_1, \ldots, x_m, x_R)$, which vanishes exactly when $\pm P_1 \pm \cdots \pm P_m = R$ for points with those abscissae. Over $\mathbb{F}_{2^n}$ or $\mathbb{F}_{p^k}$ the factor base is chosen as the points whose abscissa lies in a subspace or subfield, and Weil restriction turns $S_{m+1} = 0$ into a polynomial system over the base field; the textbook oracle solves that system with a Gröbner-basis engine (F4/F5, [7, 8]) or a SAT solver [26]. Gaudry [14] and Diem [5] give the asymptotics over extension fields; for prime fields no factor base is known that makes the method competitive, and summation-polynomial systems over $\mathbb{F}_p$ are the subject of the Kummer-tower and rational-map bases of Petit–Kosters–Messeng [21].

Frobenius-invariant bases. Galbraith, Granger, Merz and Petit [12] observe that a factor base with $\pi(\mathcal{F}) = \mathcal{F}$ splits into Frobenius orbits of size dividing n, with $\log \pi^k(P) = \lambda^k \log P$. The unknowns collapse by $\approx n$, the matrix by $\approx n^2$, and ordered decompositions save $m!$—more than rho's $\sqrt{2n}$, but from a base too far behind to overtake it at deployed sizes. Invariant bases come from linearised polynomials attached to the factors of $x^n - 1$ over $\mathbb{F}_2$; the subspace dimension is $\ell = \mathrm{ord}_n(2)$, so the base size is fixed by n (Table 1 of `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`).

Symmetries of the system. The decomposition system is invariant under the symmetric group S_m; rational 2-torsion adds sign changes, and Faugère, Gaudry, Huot and Renault [9] show that in the coordinate Y with $Y(P + T) = -Y(P)$ the invariance group becomes the Weyl group D_m, with a much smaller quotient. We measure this at $m = 4$ in §10.9.

3 The theory, by field type

3.1 What a decomposition costs

Every index-calculus algorithm for elliptic curves in the literature spends its time in three places: finding about $|\mathcal{F}|$ relations (each a decomposition of a random combination $aG + bQ$ over $\mathcal{F}$), solving the sparse linear system over $\mathbb{Z}/r$, and, in some variants, a descent of the target. The second and third are well understood. The first is the crux: a uniformly random point decomposes over a base of size N into m summands with probability about $N^m/(m!\, r)$, so relations need about $m!\, r/N^{m-1}$ trials, and each trial costs whatever the decomposition oracle costs. Every result below is a statement about one of three things: how large N can be while membership stays algebraically testable, how cheap the oracle is, and how the two trade against the linear algebra.

3.2 Extension fields $\mathbb{F}_{q^k}$ with $k > 1$ fixed or small

Gaudry [14] takes $\mathcal{F}$ to be the points with abscissa in the base field $\mathbb{F}_q$ and decomposes into $m = k$ summands by Weil restriction of S_{k+1}, a system of k equations in k unknowns over $\mathbb{F}_q$. For fixed k this gives $\tilde{O}(q^{2-2/k})$, asymptotically below rho's $q^{k/2}$ for $k \geq 3$; the constant hidden in the Gröbner solve grows very quickly with k, which is what makes the crossover large in practice (to be quantified from the paper). Diem [5] extends the analysis to families where k grows, proving subexponential complexity over sequences of fields with suitably balanced q and k. Joux and Vitse's work on decomposition and cover attacks over small extension fields is to be added here once read. FGHR [9] exploit the action of S_m and of the 2-torsion to replace S_{m+1} by its invariant form, cutting the Gröbner work by large factors; our measurement at $k = 4$ (§10.9) is a direct instance.

3.3 Binary fields $\mathbb{F}_{2^n}$ with n prime

Prime n leaves no subfield, so the factor base is taken to be the points whose abscissa lies in an $\mathbb{F}_2$-vector subspace V of dimension ℓ. Weil restriction then gives a quadratic-to-low-degree Boolean system whose difficulty is governed by its degree of regularity. The whole debate about binary curves is a debate about that degree.

The subexponential claims. Petit and Quisquater [20] conjectured, from experiments and heuristic evidence, that these systems have degrees of regularity only slightly above the degrees of their equations (the *first fall degree assumption*), and deduced a heuristic $2^{c\,n^{2/3}\log n}$, $c < 2$, for a variant of Diem's algorithm, beating generic methods only for $n \gtrsim 2000$. Semaev [24] split summation polynomials into cubic Boolean systems, argued for regularity degree at most 4 under the same kind of assumption, and obtained a heuristic $2^{c\sqrt{n\ln n}}$ with $c \approx 1.69$; its first version concluded that several FIPS binary curves fall faster than Pollard rho, a conclusion its revision says does not hold under the weaker assumption that the regularity degree is $o(\sqrt{n/\ln n})$.

The evidence against. Shantz and Teske [25] tested the Petit–Quisquater heuristics experimentally. Huang, Kosters and Yeo [17] introduced the *last fall degree*, independent of the monomial order, gave computational evidence against the first fall degree assumption in characteristic 2, and constructed summation-polynomial Weil descent systems where it is unlikely to hold. Galbraith and Gebregiyorgis [11] improved the variable choice (binary Edwards curves), found symmetry-breaking factor bases and SAT solvers inconclusive, and concluded that rho remains much faster for prime-degree binary fields of reasonable size. The Galbraith–Gaudry survey [10] reaches the same verdict for characteristic-two curves over prime-degree fields.

Where it stands. No proof or refutation of a subexponential algorithm for prime-degree binary curves exists. The open quantity is the solving degree of Weil-descended S_{m+1} as n and m grow; recent work supplies diagnostics (saturation-exponent bounds, multi-graded regularity, tests for where systems stop looking generic) but no bound for these systems. Our own measurements on toy fields (§9.1) find first and last fall degrees of 3–4 at every reachable size while the solving degree grows with ℓ and stays within one of the semi-regular prediction; closure algorithms refute a degree or two earlier but rise too. That supports no subexponential claim and refutes none.

3.4 Subfield (Koblitz) curves

When the curve is defined over $\mathbb{F}_2$ the Frobenius acts on the group, and Galbraith, Granger, Merz and Petit [12] show that a Frobenius-invariant factor base saves a factor $\approx n$ in relations and $\approx n^2$ in linear algebra—more than rho's $\sqrt{2n}$—while concluding that Koblitz-curve security is not threatened at deployed sizes. Our Koblitz work (§8) builds on this.

3.5 Prime fields

Over $\mathbb{F}_p$ there is no subfield and no subspace, and no factor base is known for which membership is both dense enough and algebraically testable. Semaev's original small-abscissa base leads to systems of very high degree. Petit, Kosters and Messeng [21] construct bases from rational maps (including the Kummer tower μ_{2^t}) that give the summation system extra structure; Amadori, Pintore and Sala [1] build the base from the instance and need only one relation search, improving on both; McGuire and Mueller [19] avoid Gröbner bases altogether by fast evaluation of S_m through iterated resultants, and in a second variant by summing $m-1$ factor-base points and testing membership of the result—the same mechanism as the two-summand root-finding oracle of §6. All of these remain exponential and slower than rho; a 2024 analysis again concludes that no subexponential prime-field method is known.

3.6 Lower bounds

Generic-group lower bounds bound everything that does not use curve structure. With preprocessing, Corrigan-Gibbs and Kogan [3] show $ST^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T; this is the bound against which the retained-set-up (online) results of §8 must be read. A 2026 structured-generic-group bound shows that an algorithm exploiting structure on only a δ fraction of elements needs $\Omega(\min(\sqrt{q}, 1/\delta))$ time—for index calculus, the decomposable set must be dense *and* cheaply testable.

4 Measuring index calculus against rho

The programme's first lesson was that index-calculus numbers from different studies did not compose: one study timed a decomposition, another a relation, a third a whole multi-target batch with shared tables, each on a different host and in a different unit. Four mechanisms, built in this order, fixed that.

4.1 The boundary ledger (10 September)

`docs/ic/BOUNDARY_TARGETS.md` and its machine twin `docs/ic/boundary_targets.json` give each regime (Koblitz, binary, prime) and each pipeline stage (`factor_base`, `decomposition`, `relation_yield`, `rank`, `end_to_end_dlp`, `vs_rho`) a current record, a next target and acceptance gates. A row promotes only when its evidence JSON carries every field the measurement schema requires for that stage—for example, a `decomposition` claim that beats an algebraic frontier must report the first-fall degree, and a `vs_rho` claim needs one target, the identical public point for both methods, verified scalars, five exclusive IC online phases, the automorphism discount, matched resources and a host-isolation receipt. Missing fields fail closed. The primary `vs_rho` question is the online cost of *one* previously unseen target after reusable IC preparation; multi-target and amortised results are secondary and must be declared as such.

4.2 ecbench (from 2 October)

ecbench (`src/cryptanalysis/ecbench/`, `src/bin/ecbench.rs`, documented in `docs/ecbench/README.md`) runs every ECDLP method behind one interface, on the same single-target workloads, in one unit, into sealed records.

Unit.

$$S \;=\; \frac{\text{total group-addition equivalents (GAE)}}{\sqrt{r}}, \qquad S_{\text{floor}} = \sqrt{\pi/(2A)}.$$

An addition or doubling is one GAE; a scalar multiplication is charged as the additions it performs; native work (lookups, row operations, square roots, Artin–Schreier solves, word XORs) is priced at a pinned conversion where the repository has one and is otherwise recorded as *uncharged*, which marks the record's S as a *lower bound*. Plain rho has $S \approx 1.25$.

What a measurement is. One workload is one curve, subgroup, generator and target; nothing survives between executions, so every table a method builds is inside its cost (*cold*). Beside the cold S every record carries the one-target *online window*, split for index calculus into the five exclusive phases `target_query`, `target_PDP`, `target_relation_check`, `target_descent`, `target_recovery_check`. Targets are either planted or *public*—hashed to the subgroup so that nobody knows the logarithm—and the runner verifies $[k]G = Q$ in its own process; `ecbench verify` checks it again from the files. Failed, exhausted and timed-out runs are rows, never dropped.

Identity, sealing and replay. Every object (curve, factor base, method parameters, spec, session, host environment class) is named by a hash of what it is (`ECBS1h...` for sessions, `ECBND1h...` for bounds). Each execution appends one sealed record; `ecbench verify` replays measured records and emits an audit receipt whose SHA-256 is the replay certificate. Operation counts are deterministic, so a replay must match to the last digit; a mismatch is preserved as a failed audit (§4.5).

Isolation. A run earns L0 (recorded under a host capsule), L1 (pinned, affinity read back, no foreign SMT sibling, CPU/wall ≤ 1.05), L2 (quiet: zero steal, foreign busy time $\leq 1\%$, run-queue delay $\leq 0.5\%$, ≤ 50 preemptions/s) or L3 (isolated cores, fixed governor, bare metal). Wall time is admitted only at the spec's required level (default L2) with at least five admitted pairs and an A/A interval; otherwise it is printed as *descriptive*. Operation counts need no isolation. macOS hosts earn L0; a GitHub-hosted runner earned L1 on 72/72 runs and matched a macOS session's operation counts to the last digit. *isolab* prepares lab hosts for L3 and cross-host replay.

Statistics. The comparable statistic is the mean S (the floor is an expectation). Comparisons are paired by (workload, round) with shared seeds, reported as a ratio of totals $\sum S_B / \sum S_A$, never a mean of ratios, with a two-stage cluster bootstrap that resamples workloads, then pairs. An operations A/A is exactly 1 by construction and checks determinism. A claim about scaling fits exponents over at least four sizes.

Bounds and the frontier. `ecbench` also records sealed *bounds*—constant and exponent with intervals, per phase—and a challenge protocol whose verdict says whether a candidate advances, trades, matches or regresses an incumbent and at which level (exponent, constant, primitive). The incumbent for cold one-target Koblitz operations is `rho.signed_frobenius_strong` (bound `ECBND1h05a2ea918f3c`): $1.089\times$ the floor $[0.927, 1.261]$, fitted exponent $\alpha = 0.451$ $[0.399, 0.512]$, 144 runs at six sizes (`research/ecbench_frontier_20261013/`).

Definition 1 (comparability)**.** A performance point is keyed by the tuple

$$D = \big(\text{curve and subgroup},\ \text{target law},\ \text{cold} \mid \text{online} \mid \text{stage},\ \text{unit and unpriced work},\ \text{set-up amortisation } q,\ \text{memory envelope},\ \text{correctness/replay/isolation class}\big).$$

Two points are ratio-comparable only when every component of D except the method matches; a method is on the frontier of a domain when no admitted point is no worse on every axis and strictly better on one.

This rules out the four collapses we saw most often: online wall as cold cost; word XORs, probes, field operations and GAE as interchangeable; a lower bound as a complete cost; an L0 wall ratio as an admitted speed-up.

4.3 The pricing band

Almost no cost in a record is complete, and the rho baselines are no exception. Rho, BSGS and the kangaroo leave canonicalisations, partition hashes and table inserts and lookups unpriced. An index-calculus arm is complete only on curves that have a pinned timing calibration (`docs/ic/calibration.json`), which converts lookups, row operations, Artin–Schreier solves, Frobenius maps and solver word XORs to GAE at measured ratios of roughly 10^{-3} to 10^{-1}. A "complete" index calculus set against a lower-bound rho is therefore lopsided, and several earlier comparisons in this repository were of that kind.

Since 15 October every cross-method comparison uses a *pricing band* (repository rule, `AGENTS.md` §5a; native implementation `ecbench band`). For a record with charged cost G, calibration-priced native cost C over n_C operations, and U operations behind its unpriced counters,

$$S(w) = \frac{G - C + w\,(U + n_C)}{\sqrt{r}}, \qquad w \in \{0,\ 0.1,\ 1\} \text{ GAE per native operation},$$

applied alike to every arm, references included. Two arms are ranked only if their order is the same at every weight; otherwise the pair is reported as unresolved. An arm that names an unpriced counter without counting it is a lower bound and cannot be ranked. The band brackets the pinned calibrations from below and above, so a ranking that survives it does not depend on the host the calibration came from.

4.4 Claim hygiene and change classes

Every study directory states a protocol before it runs (frozen inputs are committed first), keeps failures, and closes with a decision in one of the four classes of §1. A controlled-experiment rule requires a negative control from the right null population, a positive control that proves the detector can see the effect, and the same instrument on every arm before any comparative statement. Blockers are recorded as named problems (`problems/`) with exit criteria rather than silently dropped. Measurement code is native (Rust); Python only lays out recorded values.

Generic baselines. The rho, BSGS and Pohlig–Hellman references were engineered alongside index calculus, which is why each round's distance is quoted against the strongest reference at the time. On Koblitz curves the strong signed-Frobenius rho fell from 1,196 to 679.7 instructions per step over four rounds with no established native-time gain (§12.4). On hyperelliptic Jacobians a negation fold and hash mixing each fell short of their registered gains, and full-width branch coefficients showed that the earlier 8-bit reference walk had been slow at large N (`benchmarks/20261018-hec-rho-negation/`, `benchmarks/20261019-hec-rho-branch-width/`). Generic DLP fixes cut Pohlig–Hellman steps 26.6× and BSGS time 1.3–2.0×, and the candidates run at 0.020–0.167 of PARI's `elllog` time, in shared-host wall time the validation sweep left unvalidated (`research/generic_dlp_memory_parallelism_20261017/`).

4.5 Corrections the accounting forced

Building the harness and the ledger exposed errors in earlier claims, including our own. We list them because they change how earlier numbers must be read.

1. *Results below rho that reversed against a stronger rho* (September; §8.3). A tournament candidate at 0.5347 of rho's instructions lost all five cells once rho used the same canonical form (`research/ic_triple_counted_20260923/`); an orbit-DLP ratio of 0.2403 became 1.923 against a strengthened rho and was withdrawn; a compact-orbit batch lead of 0.309 reversed against a batched-inversion rho.
2. *Online Koblitz ratios relabelled* (7–8 October, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`). Same-point online ratios of 124.85× ($n = 61$), 2,864.1× ($n = 71$), 1,189.7× ($n = 73$) and 8.2× ($n = 83$) had verified answers but no host-isolation receipt; they are *exploratory*. Only the later 64-target pinned panel at $n = 61$ is admitted (§8).
3. *ECC2K-130 extrapolation constants corrected* (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`): the structural model predicts $C = 0.25$ in probes/relation $\approx C\,r/(n^2K^2)$; the fitted C is 0.27 at $n = 61$ but 1.2–1.5 at $n = 71, 73$, an unexplained gap now stated rather than absorbed.
4. *"F4" that was Buchberger* (`research/ecbench_rootfinding_assessment_20261012/REPORT.md`): the NIST benchmark of the root-finding manuscript printed its baseline as F4 but called `groebner_f4::buchberger`; its 3,021–5,802× are ratios to Buchberger. The manuscript's claim that no implementation can use fewer than linear field operations per ideal is too broad once a reusable pair table is allowed.
5. *Unpriced word XORs* (`research/ecbench_rootfind_faster_20261014/`). Inherited F4's Boolean word XORs were counted but not charged, so its $S = 16.7$ at Koblitz $n = 13$ was a lower bound that looked competitive with rho (14.4). Pricing them gives $S = 421.4$.
6. *Cross-host counter drift* (`research/ecbench_frontier_20261013/audits/`): a replay-all audit of the root-finder session on x86_64 matched 120/168 records; all 48 inherited-F4 rows differed in solver word-XOR counts (e.g. 301,961 versus 300,531), with every scalar, trial count and charged GAE unchanged. The failed receipt is kept; the F4 stage comparison is historical until the drift is explained. A replay of the same session on current `main` (20 October) matches only 72/168, with mismatches in all four IC arms, root finding included. The cause was then isolated: an x86-only BMI2 rewrite in inherited F4 counts word XORs differently from the portable path; with it disabled the replay is 168/168 on x86 and on emulated aarch64 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). F4 counts are comparable across instruction sets only when the path is stated, which the method identity does not yet record.
7. *A soundness bug* (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` §2.1): a degree-bounded `f4_fp::solve` could certify solutions of an inconsistent system (over $\mathbb{F}_5$, $\{x^3 + 1, xy - 1, x + y\}$ returned two "solutions"). Fixed by evaluating every candidate on the input.
8. *F6-IC charged cost* (§7): a stage win in word XORs became a 64.6–1,353× loss in charged S once its batched geometry was counted—recorded as an accounting result, not an advance. With inherited F4's own word XORs priced the $m = 13$ loss is about 2.5× (§7).
9. *Lopsided pricing* (§4.3): comparisons of a calibration-priced index calculus with a lower-bound rho. On the frontier campaign's train panel at Koblitz $n = 13$ the compact-orbit arm, earlier quoted at 1.08× rho as a lower bound, is 0.93×, 1.53× and 5.7× the strong rho at $w = 0$, 0.1 and 1 (columns 2, four train targets, `research/ic_frontier_campaign_20261015/select/SELECTION.md`): below rho with native work free, above it otherwise, so the pair is unresolved inside the band rather than a win for either side.
10. *Wall-time figures recomputed* (validation sweep). Recomputing from the committed per-target records gives 2,321× for the $\ell = 12$ binary MITM oracle (quoted 2,187×), 98.1× for its $\ell = 8$ run 3 (quoted 100.7×), and 6.08× and 6.14 s for the quartic solve at $d = 6$ (quoted 6.4× and 5.9 s); all are shared-

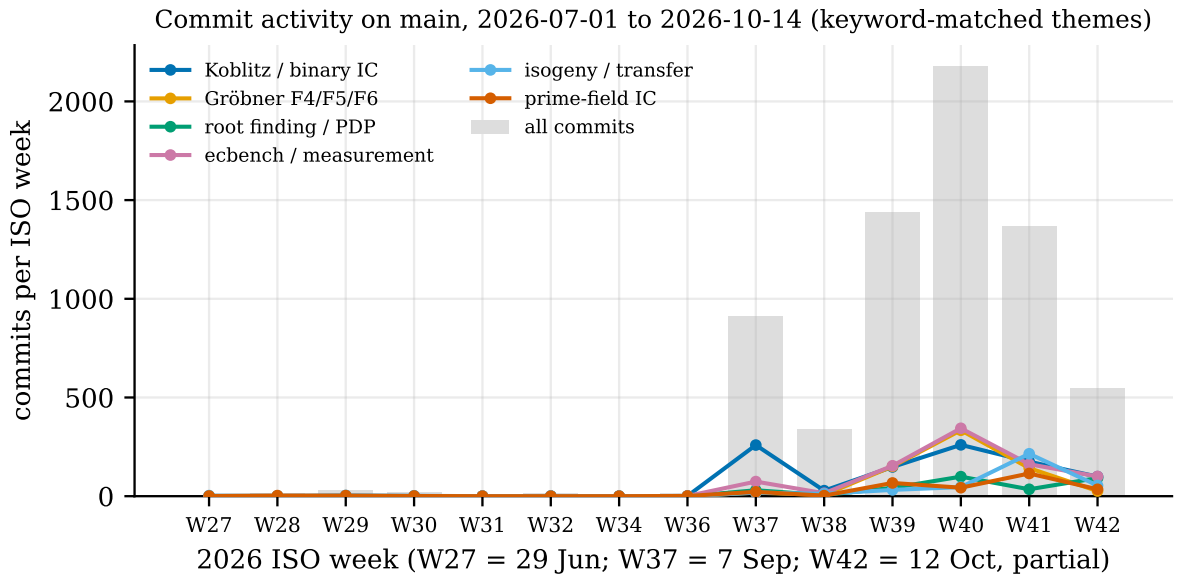


Figure 1: Commits per ISO week on `main`, 1 July–14 October 2026 (6,882 commits; week 42 partial; the figure has not been extended past 14 October). Lines: commits whose subject matches each theme's keywords. Source: `git log` at the paper's base commit; values in `figures/data.json`.

host wall times left unvalidated, and the counted evidence behind them stands (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

11. *An unmatched rho reference* (21–24 October). The Koblitz leaderboard's rule v3 priced its rho as built rather than at the canonical batched step. Re-pricing rho in the same process raised the cold distance 1.76–1.94×, to 9.9–43× (`research/ic_tool_program/rule/v4/README.md`); the ledger's rule v8, matching the reference its own way, raised it 1.30–1.58× and found that an apparent flattening of the gap's slope was mostly this artefact (`papers/ecdlp-ic-improvement-ledger/`). The same week withdrew a volcano-crater claim (IC 22–25% cheaper) whose reference should have been the cubic-automorphism rho, against which IC is 84.9× rho, and marked the plane-quartic S values unverified because their degree-3 reduction merges distinct classes.

5 Chronology

Figure 1 shows commit activity on `main` by ISO week. Theme lines are case-insensitive keyword matches on commit subjects (a commit may match several themes or none; the regular expressions are in the paper's `README.md`), so they indicate where effort went, not how much of it was index calculus. The grey bars include unrelated work in the same repository (the ECC2K-130 GPU rho campaign, an isogeny KEM, post-quantum benchmarks), which dominates week 37.

Prelude: July and August. The prime-field *ecdlp autolab* ran numbered threads on curve structure—universal order-2 Frobenius forms, the CM-73 family $\{19, 37, 79, 109\}$, sextic-twist pairs that are Howe-glueable (5 of 15)—and its draft "An Empirical Resistance Map for Prime-Field ECDLP" (`research/ecdlp_autolab/PAPER_DRAFT.md`). Those threads found structure but no solver advantage, which is what turned the programme towards index calculus, where structure is supposed to pay.

Phase 1, 6–14 September: Koblitz index calculus from scratch. The ECC2K-130 rho client and an audit of a public index-calculus benchmark's SAT/UNSAT labels (6 Sept) were followed by Frobenius-invariant factor bases with a matrix-F4 Semaev oracle, an XOR-native SAT oracle after Trimoska et al. (7 Sept), and a $\sim 50\times$ faster Semaev solve the same day. On 8 September a scaling experiment showed the SAT route does not scale and that the refutation frontier belongs to F4; pairs-and-solve replaced triple enumeration for S_4. The boundary ledger was codified on 10 September, together with a meet-in-the-middle oracle, incremental modular linear algebra and a CADO-style split into factor-base logs then descent. A burst of more than 100 numbered pipeline stages (12–13 Sept: PCLMUL field products, fused

pair collection, scalar-blind targets, rank-aware scans) took the multi-target $n = 53$ pipeline to a 1,024-target process-wall crossover against batched rho. On 13–14 September six pre-registered ECC2K-130 decomposition experiments ran ("exists, admissible, not findable"; no falsifier fired), and the repository began to require frozen benchmarks and full-pipeline evidence for any ECDLP speed claim.

Phase 2, 15 September–1 October: engines and the compact orbit. Inherited F4 (22 Sept) became the default algebraic oracle; a fast binary-field hot path, Itoh–Tsujii inversion and a parallel radix-sorted pair-table build landed (23–24 Sept); a long Boolean F4/F5 kernel series (AVX2/AVX-512 unpackers, compact columns, pivot orders; 26 Sept–3 Oct) ran in parallel. The compact signed-Frobenius orbit producer beat batched rho on multi-target $n = 61$ batches (25–26 Sept), and from 2 October carried the one-target online ladder to $n = 71$ and (6 Oct) $n = 73$ on 128-bit words, after $n = 63$ was closed as non-viable. A fixed-policy cold comparison on orbit-disjoint public points (1 Oct) was a measured no-go.

Phase 3, 2–10 October: one harness, and the audit it forced. *ecbench* landed on 2 October. F6-IC and the $n = 83$ campaign (145 entries, 4–21 Oct; §8.5), the registered $m = 3$ refutation-degree law (5–11 Oct), the ECC2K-130 feasibility extrapolation (5 Oct), the accounting correction plan, conductor-gap and novel-direction notes (7 Oct), and the transfer-aware, isogeny-fibre and Gröbner-engine audits (10 Oct) followed.

Phase 4, 11–14 October: literature, root finding, consolidation. The lit2026 programme (11 Oct) turned a literature search into twelve workstreams. A supplied manuscript on root-finding decomposition was frozen, audited and re-measured in ecbench (12 Oct), and the identity it rests on was generalised across PDP oracles, a CUDA kernel, Kummer towers, Koblitz engines and dense root indices (12–14 Oct). The admitted $n = 61$ online panel ran on two hosts (12 Oct). Gaudry's $k = 4$ solve was made generic and given the D_4 symmetry; the frontier synthesis and its replay audit closed the period (13 Oct); the last root-finding loops priced F4's word XORs and ran out of levers against rho (14 Oct).

Phase 5, 14–15 October: measured against the baseline. Two improvement rounds and a next-steps campaign paired each change against the index-calculus incumbent under frozen protocols (§12.3): genus-3 index calculus after the cubic GHS transfer (three rounds), the $k = 4$ ladder and its four-unknown solve, production free relations and oracle row harvesting on Koblitz curves, and a CADO-NFS-derived filter for the linear algebra. An equal-advice audit retired the online win (14 Oct). The owner then fixed the success criterion as iterative improvement over that baseline, with the gap to rho as distance remaining (15 Oct), and the priced frontier campaign of §12.1 ran the same day.

Phase 6, 16–20 October: audits and two closing gaps. A fourth genus-3 round brought the local exponent to 1.356; the $k = 4$ oracle became 370× cheaper over $\mathbb{F}_{p^4}$ and, in a Frobenius-reduced form, 481× in characteristic two, where the measured gap to rho closed to 6.1× at $n \approx 2^{56}$ (§12.4). The binary MITM oracle reached $\ell = 12$, the strong rho reference shed 40% of its instructions per step, and a precomputed-rho control at $n = 83$ (16 Oct) and a repository-wide validation sweep (19–20 Oct) retired the remaining online wins and reopened the root-finder replay, whose cause (an x86-only BMI2 path in inherited F4) it then isolated.

Phase 7, 21–24 October: folded bases, a third family, and the reference question. Four factor bases closed under known-logarithm maps ($\mathbb{Z}[\tau]$ on Koblitz curves, $\mathbb{Z}[\phi]$ on CM curves, integer multiples on random binary curves, the cofactor projection of LEAD-19) cut cost 1.3–20× against the programme's own arms without changing an exponent (§12.5). A two-round literature search drafted 19 point-decomposition leads and closed most at the desk. Re-pricing rho at the canonical step nearly doubled the Koblitz leaderboard's distance; a hybrid guess closed the $(3, 7)$ Weil-descent cell without moving the $m = 4$ row below rho; the Joux–Vitse cover family came within 1.6–2.0× of a pooled rho constant at $p = 1009$. No arm has yet been measured against a published index calculus at the same parameters; that reference arm is now open problem 10.

6 Root-finding decomposition

6.1 Origin: a suggestion in an audited manuscript

The textbook decomposition oracle builds Semaev's polynomial system and solves it with a Gröbner-basis engine. On 12 October a 13-page manuscript, *Point Decomposition by Root Finding*, was supplied for assessment. We froze it (`research/ecbench_rootfinding_assessment_20261012/source-paper.pdf`, SHA-256 `743d3d3e...7c01`) and audited it against its source branch before measuring anything. Its central suggestion is simple: for two summands, fixing one factor-base point turns S_3 into a quadratic in the remaining variable whose roots are already known.

Proposition 1 (roots of S_3)**.** *Let $E : y^2 = x^3 + ax + b$ over a field K of characteristic $\neq 2, 3$, and let $R, F \in E(K)$ be affine with $x_R \neq x_F$. Then*

$$S_3(x_R, x_F, X) = (x_R - x_F)^2 \, (X - x_+)(X - x_-), \qquad x_\pm = x(R \pm F),$$

so $S_3(x_R, x_F, X) = 0$ iff $X \in \{x(R+F), x(R-F)\}$.

Proof sketch. With $\lambda_\pm = (\pm y_F - y_R)/(x_F - x_R)$ the chord slopes, $x_\pm = \lambda_\pm^2 - x_R - x_F$. Expanding $(X - x_+)(X - x_-)$ and eliminating y_R^2, y_F^2 with the curve equation gives exactly Semaev's $S_3/(x_R - x_F)^2$. The identity is machine-checked over an arbitrary field in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`, theorems `SemaevS3.S3_factor` and `SemaevS3.S3_root_iff`; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`). □

Hence the two-summand system has a parametrised variety: $R = \pm F_i \pm F_j$ iff $x(R \mp F_i) \in x(\mathcal{F})$. No polynomial system is solved; the oracle is one subtraction and one table probe per base point.

Prior art. Neither the identity nor the oracle is new. The identity is a direct consequence of how S_3 is defined: it is the polynomial whose roots are the abscissae of $P_1 \pm P_2$. The oracle mechanism—sum $m-1$ factor-base points in the group and test whether the result lies in the base—is the second algorithm of McGuire and Mueller [19], who also evaluate S_m by iterated resultants instead of solving it, building on the instance-derived bases of Amadori, Pintore and Sala [1]. What this section adds is engineering and measurement: a machine-checked proof, a fail-closed verification protocol, binary pair-sum tables for up to four summands, GPU, Kummer-tower and dense-index variants, and complete-cost measurements in one harness against rho.

What the audit kept and corrected. The identity is exact under the stated generic affine condition, and a returned witness can be re-added in the group for soundness. Completeness of the exhaustive enumeration holds only with an explicit sign and factor-base convention (the binary table stores full point keys over a base containing both signs; the manuscript's schematic x-only table needed that convention stated). The manuscript's NIST baseline was Buchberger, not F4 (§4.5), and its statement that no implementation can use fewer than linear field operations per ideal fails for repeated queries against a reusable pair table. Larger-summand enumeration has a growing exponent in $|\mathcal{F}|$, and the manuscript did not benchmark it end to end. All of this is recorded in `research/ecbench_rootfinding_assessment_20261012/REPORT.md`.

6.2 The algorithms

Prime fields (Algorithm 1, `src/cryptanalysis/prime_fast.rs`): $\Theta(N)$ field operations and $2N$ lookups per target. *Binary fields* (`src/cryptanalysis/koblitz_index_calculus.rs`, `DecompositionStrategy::PairTable`, made the default): a precomputed table of pair sums $F_i + F_j$ turns an m-decomposition into 1, N or N^2 lookups for $m = 2, 3, 4$—the meet-in-the-middle half of a birthday search, exact and complete like exhaustive search, at the price of $\Theta(N^2)$ set-up and memory. Both are verified fail closed: every decomposition is re-added in the group, S_3 is checked to vanish, Gröbner-variety membership and agreement on refutation are checked against F4, and planted positive controls must be found.

6.3 Generalising the observation

Over three days the identity was carried through every layer of the pipeline. Table 1 lists the steps; each is its own study with a protocol registered before it ran.

Algorithm 1 Prime two-summand oracle by direct subtraction

Require: target R, base $\mathcal{F} = \{F_1, \ldots, F_N\}$, hash set H of abscissae (or full points) of $\mathcal{F}$

1: $d_i \leftarrow x_R - x_{F_i}$ for all i; invert all d_i with one field inversion and $3(N-1)$ multiplications (Montgomery's trick)
2: **for** $i = 1..N$ **do**
3: **for** $\varepsilon \in \{+1, -1\}$ **do**
4: $\lambda \leftarrow (y_R + \varepsilon y_{F_i})\, d_i^{-1}$; $x_3 \leftarrow \lambda^2 - x_R - x_{F_i}$ ▷ $x(R - \varepsilon F_i)$
5: **if** $x_3 \in H$ **then** recover the sign of F_j from y_3; emit $R = \varepsilon F_i + \tau F_j$
6: **end if**
7: **end for**
8: **end for**

Table 1: Root-finding generalisations, 12–14 October. Every row is ENGINEERING: no slope fell below its generic floor.

Step	What it adds; artefact
Prime subtraction, binary pair table	Prop. 1; batched inversion; $m \leq 4$ tables. `research/semaev_root_vs_groebner_20261012/`
NIST P-192…P-521	standard parameters, 6- and 64-point bases; dedicated P-256 field 11–12× over generic big integers. `research/semaev_rootfind_nist_20261012/`
End to end vs msolve	complete DLPs with only the oracle swapped. `research/semaev_rootfind_e2e_20261012/`
PDP oracle family	`rf-mitm`, `mitm3`, `mitm4` (2-sum table); `rf-pairs`, `pairs3` (pairs then solve by the group law); `subtract2`; binary rungs $\ell = 8..14$, prime Kummer towers $m = 2, 3, 4$. `research/pdp_root_finder_20261012/`
CUDA kernel	one thread per (target, slice); per-thread Montgomery inversion; SoA base; shared-memory hash for $\lvert\mathcal{F}\rvert \leq 2048$; 64-bit and P-256 fields. Validated in host emulation only (no GPU available). `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`
ecbench engines	`pdp3-koblitz:m=2|3,engine=rootfind`, `rootfind-subtract`, later `rootfind-window` and `rootfind-negfold`. `research/ecbench_rootfind_engine_20261012/` and successors
Kummer-tower base	PKM's μ_{2^t} base with the root-finding oracles, 20–28-bit primes. `research/pkm_tower_ecbench_20261013/`
Dense root index, $n = 83$	packed five-byte S_3 root table, prefix widths K1271–K1798, 8 GiB of slots. `research/n83_dense_root_20261013/`
Prime tuning	pair-table size, arity, batched table inversions, synchronised lanes. `research/ecbench_rootfinding_\{tuning,arity,batchtable,lanes\}_2026101*/`

6.4 Results

Oracle stage. Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $\lvert\mathcal{F}\rvert = 55$, 254 targets) and 1,535× ($\mathbb{F}_{2^{13}}$, $\lvert\mathcal{F}\rvert = 4{,}005$, 256 targets) faster than matrix F4; 3.5–$4.7 \cdot 10^4$× faster than `groebner_f4` on 16–24-bit prime curves ($\lvert\mathcal{F}\rvert = 6$, 40 targets each); and 3,021–5,802× faster than Buchberger on the decomposition ideal of the five NIST curves (Figure 2). Every binary verdict agreed with F4; every prime solution was in F4's variety and every prime refutation was F4's unit ideal. In the PDP study, `rf-mitm` refutes a binary target in 3.1 μs at $\ell = 8$ and 482 μs at $\ell = 14$, against the PS-0 baseline's 9.3 ms and 55.5 s (3,007–115,160×), costs $5\lvert\mathcal{F}\rvert - 3$ multiplications and $2\lvert\mathcal{F}\rvert$ lookups a target, and sits on the pair-table floor at 1.000 ± 0.027 bits per rung; at prime $m = 3$, $N = 18$ it uses 197 multiplications where F4 uses $8.99 \cdot 10^{13}$ multiply-adds; 707/707 brute-force agreements (`research/pdp_root_finder_20261012/`).

Complete solves. End to end, the gain is two orders of magnitude smaller and not one-sided: 50–105× against msolve on 16–24-bit prime curves; 122× on the one binary instance ($n = 23$) large enough to amortise the pair table; and F4 *faster* at $n = 5, 9, 13$, where a solve needs too few decompositions to pay for the table. Inside ecbench, with every group operation charged, the root-finding IC beats inherited F4 in solve wall by about 350× at $n = 13$ (admitted) and about 10^4× at $n = 23$ (descriptive), yet stays 2.9–25,116× the strong rho in S on binary curves and 29–141× rho on prime curves, with the gap widening (`research/pdp_root_finder_ecbench_20261013/`).

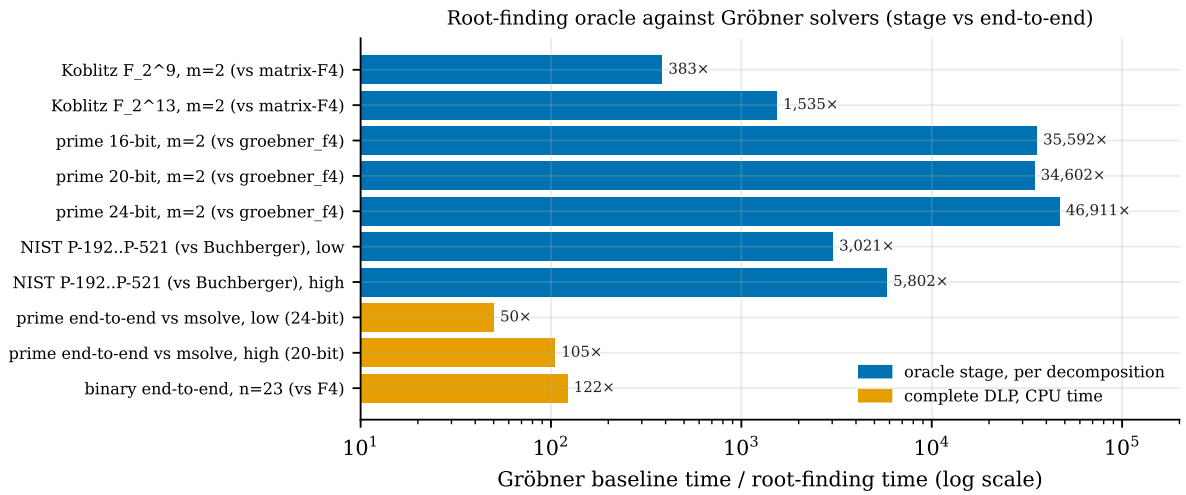


Figure 2: Root finding against Gröbner solvers on identical targets. Blue: one decomposition (oracle stage only). Orange: complete discrete-log computations with the oracle swapped and everything else fixed (prime: CPU time against msolve, 20/24-bit totals composed from sampled per-query cost; binary: one cold solve including table build). Sources: `RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`, `papers/rootfinding-decomposition/README.md`.

The Koblitz loops. Figure 3 follows the best complete-count root-finding arm at Koblitz E_0, $n = 13$ through four registered improvement loops (8 public targets × 3 rounds each, L0 operation counts). The first engine had $S = 38.82$ ($m = 3$) and 28.38 ($m = 2$). A windowed third-summand scan cut $m = 3$ by a paired 0.828 $[0.783, 0.863]$; a fold over negation (`rootfind-negfold`) took $m = 2$ to 23.22 and then 22.97. Two further cycles of denser and sparser bases, windowed negfold and plain root finding were dead ends (`research/ecbench_rootfind_vs_rho_20261014/`). With F4's word XORs priced, the paired ratio negfold/F4 is 0.0545 $[0.047, 0.064]$, a complete-against-complete comparison; against strong signed-Frobenius rho ($S = 14.82$) the best complete arm is still 1.55×. A compact-orbit base with a counted Frobenius pair table reaches $S = 15.55$, below the then-unpriced F4 (16.75), but it is itself a lower bound (factor-base construction natives unpriced), and it is 1.08× rho. Amortised over 64 targets that arm reaches $S \approx 11.2$, below rho; by the ledger's rules that is a secondary, multi-target result, not a cold claim.

Prime tuning in ecbench. On a 20-bit prime curve, the pair table's counted S is 0.018861 $[0.018547, 0.019242]$ of direct subtraction (24 pairs; prime native work unpriced, so a bounded ratio), and 0.057047 $[0.041159, 0.092069]$ on a 17-bit Koblitz curve, fully priced, where the table set-up is about 68% of the arm's cold cost. Doubling the table from 32 to 64 abscissae gives 0.830 $[0.697, 0.962]$; three summands instead of two costs 1.431× $[1.225, 1.716]$; batching the table inversions leaves counted GAE unchanged (1.000; 4,096 versus 64 inversions, set-up wall 243 versus 128 μs, descriptive); and a train-set lane selection did not resolve a hold-out winner (0.979 $[0.919, 1.038]$; 1,088/1,088 verified executions) (`benchmarks/README.md`, rows of 12–14 Oct).

Dense root index at $n = 83$. Widening the prefix base from K1500 to K1798 cut mean probes per relation from 300,021.9 to 169,069.8 (paired difference 130,952, 95% [52,058, 209,991]) and gave a one-point online ratio of 18.528 over same-point rho (IC 63.3 ms, rho 1,173.3 ms), with 339.0 s of reusable preparation and an 8 GiB table outside the window. The host was shared, so the result is exploratory; two K1500 runs hit the 900 s timeout and are retained. The probe reduction stands; the online ratio does not: it was taken against rho with an empty table, and rho with a precomputed table matched to the index calculus's set-up is faster in every matched $n = 83$ pair (validation sweep, 20 October; §12.4).

Reading. Root finding is the largest constant-factor change of the programme at the oracle stage, and the one with a proof behind it. It does not change the asymptotic count of index calculus: its cost is $\Theta(|\mathcal{F}|^{m-1})$, every measurement is at $m \leq 4$, and it leaves relation yield, factor-base size and linear algebra untouched. Its effect on the cold frontier is to move the decomposition stage from dominant to minor, which exposes the stages that now are.

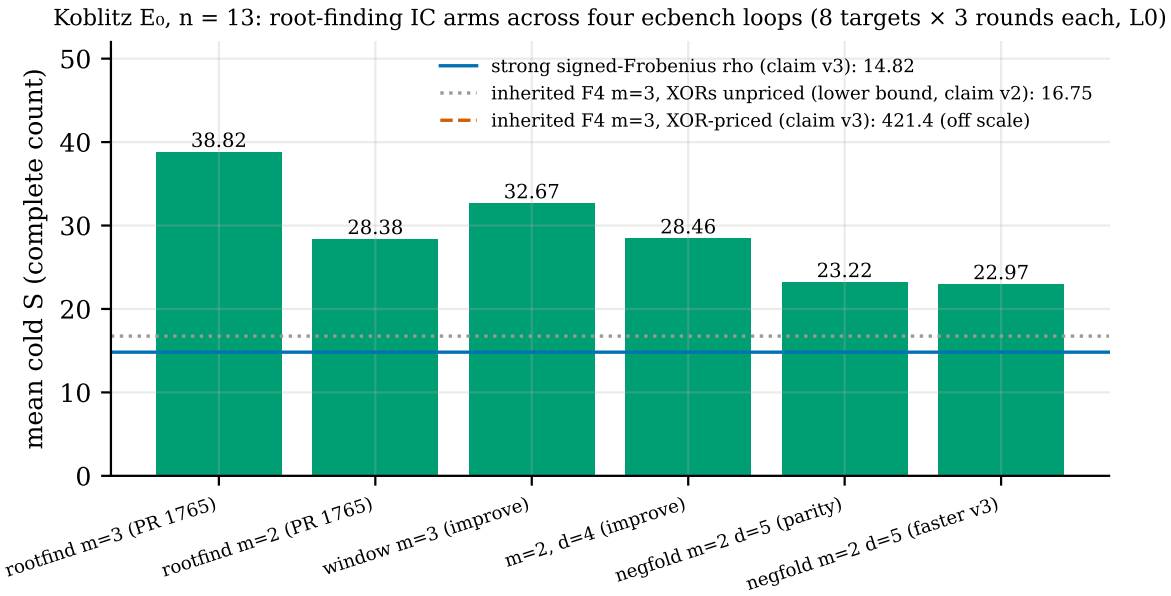


Figure 3: Mean cold S of root-finding IC arms at Koblitz E_0, $n = 13$, $d = 5$ unless stated, across the engine, improve, parity and faster loops. All bars are complete counts. Sources: `research/ecbench_rootfind_engine_20261012/`, `research/ecbench_rootfind_improve_20261013/`, `research/ecbench_rootfind_parity_20261013/`, `research/ecbench_rootfind_faster_20261014/`. Each loop had its own seeds and rho arm (rho 14.12–14.82).

7 Gröbner-basis engines

Before root finding, the decomposition oracle was a Gröbner solve, and for $m \geq 3$ over binary fields it still is. The repository grew a family of engines (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`, §1; file names below are under `src/cryptanalysis/`):

Matrix F4 and F5 over $\mathbb{F}_2$
(`koblitz_groebner.rs`, `matrix_f5_f2.rs`): Macaulay matrices of the Weil-restricted Boolean system, reduced by a shared Four-Russians kernel (`gf2_elim.rs`, four 8-bit tables, AVX-512 or AVX2).

Inherited F4
(`inherited_f4.rs`, from 22 September, the default algebraic oracle): a DPLL-style tree over partial assignments in which each child *inherits* its parent's echelon form and re-reduces only what the assignment changed, instead of building a fresh Macaulay matrix per node.

F6-IC
(opt-in): inherited F4 plus a factor-base domain gate (a fully assigned summand whose abscissa is not a base point refutes; partial support pruning refutes earlier), and meet-in-the-middle leaves (with $m - 1$ summands fixed the last is looked up; at $m = 3$ and $|\mathcal{F}| \leq 256$ every compatible second point is added and the third looked up).

$\mathbb{F}_p$ F4
(`f4_fp.rs`, `f4_fp_tower.rs`, `sig_fp_tower.rs`): degree-bounded F4 with sparse-by-reducer then dense reduction, a Faugère–Lachartre block for the Kummer-tower oracle, and a signature variant.

Correctness. The audit pinned every engine against every other and against brute force (88+15+9+14+13+12+21+6 passing library tests in the touched modules), and found one soundness bug (§4.5, item 6), two contract defects and several latent holes, all fixed with regression tests.

Engineering results. Figure 4 collects the headline ratios; all are ENGINEERING with identical outputs.

- *Inherited versus matrix F4*: matrix work 198.6× (K13) and 222.1× (random-13) the inherited engine's on the same targets (the two can return different decompositions, so their splitting trees may differ; only matrix F4 and F5 share a tree by construction); at $m = 19$ every matrix F4/F5 run timed out at 1,800 s while inherited F4 finished in about 6 s (`research/f4_f5_f6_cross_family_20261011/`).
- *F5 versus F4*: word-XOR ratios 0.6465 / 0.6414 (K13 / random-13), wall 0.79–0.86.

- *F6-IC versus inherited F4*: word XORs about 0.82 at $m = 13$ and 0.77–0.78 at $m = 19$, relation wall 1.7–1.9× faster. But once its batched group additions are charged, F6-IC's S was 64.6–68.8× inherited F4's at $m = 13$ and 1,303–1,353× at $m = 19$: an accounting loss, not a frontier move. Those ratios used inherited F4 before its word XORs were priced (#1774); priced, at K13, F6-IC's S is 1,572.1/634.12 ≈ 2.5× inherited F4's, still a loss, while the R13 and $m = 19$ cells remain lower bounds (`research/f4_pricing_jump_20261018/`). At $m = 31$ both engines hit the 3,600 s cap on the first target.
- *Kernel routing and rewrite*: routing every echelon call to the Four-Russians kernel gave 1.75–3.22× on the oracle's own Macaulay matrices; a word-level `pext`/`pdep` rewrite plan made F4 and F6-IC 10–14% faster at $n = 23$; eager rewrite of kept rows beat lazy rows by 1.40–1.62× (`research/inherited_f4_eager_rewrite_20261011/`).
- *Instruction and allocation work*, measured by paired A/B/A′ instruction counts on a frozen eleven-system suite: fused construction and readback, −5.28% (F4) and −11.02% (F5) instructions; native hashing, −3.715% F5 [3.714, 3.726] and −45.1% for the standalone criterion; bounded direct indexing in the F5 packer, −3.947%; row allocation, 42–50% fewer requested bytes. Elapsed time was not claimed for these.
- *Quotient-curve model*: solving the $m = 2$ system directly on the quotient curve instead of a symmetrised control cut F4 word operations by 1.340–1.642× with identical root digests in five interleaved cells (`research/lit2026_w8_summation_models_20261011/`).
- *The $\mathbb{F}_2$ engine campaigns* (about 90 directories, 22 September–17 October; Appendix C, clusters "boolean", "gf2-elim", "f5-kernel", "f5-support-cut", "f4-f5-shared"). Each micro-change was gated against the strongest reference, and most failed. Kept as defaults: a branchless AVX2 strip (1.210× one thread; `research/gf2_branchless_strip_20260929/`), a bitmap seen-set in Boolean F4 (1.199× the full F4; `research/boolean_f4_bitmap_20260926/`), batched pair inserts, five-column tables, and direct packing in F5 (default since 17 October). The combined F5 options reached 1.530× (`research/f5_combined_2x_20260928/`) and a fixed support cut 3.02–3.07× on one complete call (`research/f5_support_cut19_confirm_20261003/`); about thirty unpack, pivot, layout and packing variants were refuted, and graded reuse in F5 reached 1.78–1.84×, short of the campaign's 2× goal (`research/boolean_f5_graded_batch_20261002/`). These are stage speed-ups of the decision engine; none changes an index-calculus distance to rho.
- *$\mathbb{F}_p$ F4 to msolve parity* (`research/f4_fp_msolve_parity_20261011/`): Gebauer–Möller criteria, packed monomials, interned symbolic preprocessing and a parallel symbolic pass made `f4_fp` 29.8–54.1× faster than its predecessor on square generic systems, with msolve/ours at 0.96–1.22 single-threaded; a later blocked pair reduction made ours 0.75–0.78× msolve's time on the two largest square systems but 1.11–1.49× on four others (`research/f4_fp_blocked_pairs_20261016/`). A day earlier msolve had been 12–24× faster (`research/gb_cas_comparison_20261011/`); every reduced basis agrees with Singular or msolve. On Boolean decomposition systems ($n = 17$, 8 targets) our oracle decides in 7–24 ms per target, while msolve and PolyBoRi spend 40–52 s computing a full basis and Singular exceeds 120 s—a difference of task (decide versus compute the basis), not of engine quality.

What did not work. Degree-2 early-abort probes after a 2025–26 proposal: word-XOR ratios 1.00083–1.00378 (no change) and 0.99714–0.99928 (worse). Gate-first geometry: 1.57–1.78× fewer word XORs but lane-batched F4 roots cost 2.6–4.0× a sequential control. Convolution splitting of the summation polynomial: 1.21–2.07× *more* word operations despite lower degree; Frobenius auxiliary variables: 57–701× more (`research/lit2026_w\{4,5,8\}_*_20261011/`). Symmetry-breaking clauses in SAT did not help (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). A registered law for the $m = 3$ refutation degree resolved all 68 units but ended with the verdict "neither": one variant met its generic degree window on 32/32 draws, the other fell below it on 8, and the heaviest degree-10 Macaulay solves were $2.4 \cdot 10^5 \times 2.0 \cdot 10^5$ matrices at 42–51 minutes each (`research/ic_semiregular_law_20261005/`).

8 Koblitz curves

Koblitz curves $E_a : y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, $a \in \{0, 1\}$, carry the Frobenius endomorphism τ, which acts on the prime-order subgroup as a known scalar λ; ECC2K-130 is the $a = 0$, $n = 131$ member. They were the programme's main family: the compact-orbit producer, the $n = 83$ campaign, leaderboard Table A and the ECC2K-130 cost model all live here. Random and prime-degree binary curves without τ are in §9, and the Weil-descent degree studies that price ECC2K-130's algebraic route

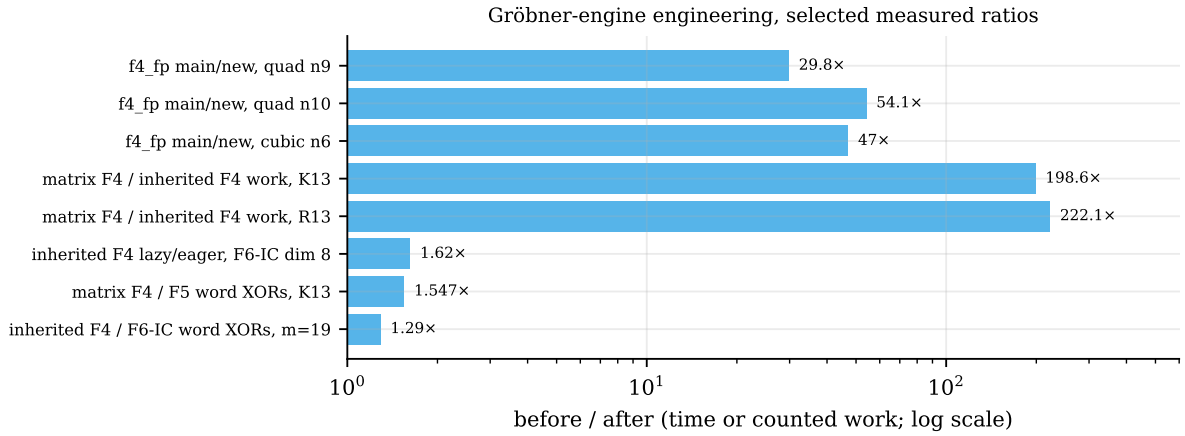


Figure 4: Selected engine ratios (before/after or baseline/candidate): `f4_fp` successive versions (single-threaded wall, L0); matrix versus inherited F4 (counted matrix work); lazy versus eager inherited rewrite (median wall); F4 versus F5 and inherited F4 versus F6-IC (word XORs, stage only). Sources in `figures/data.json`.

are there too, because their systems use random subspaces rather than Frobenius structure. Every study in this section is listed in Appendix C.

8.1 Frobenius-invariant and compact-orbit bases

The Koblitz implementation (`src/cryptanalysis/koblitz_index_calculus.rs`) builds GGMP's invariant base from the linearised polynomial of a factor of $x^n - 1$, checks $\pi(\mathcal{F}) = \mathcal{F}$ point by point, rewrites each relation into orbit columns with known λ^k weights, and solves the cofactor-corrected system $ha + hbd \equiv \sum \lambda^{k_i} x_{o_i}$. On small fields the orbit collapse is as predicted: $|\mathcal{F}|/\#\text{orbits}$ of 7.86, 10.89, 12.96 at $n = 9, 11, 13$ for relations and 61.7–168.0 for linear algebra, against rho's 4.24–5.10 (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). Because $\ell = \text{ord}_n(2)$ fixes the invariant base's size, the programme moved to bases sized to the instance: divisor bases, SAT-optimised bases, a rank-guided base at $n = 53$ (220 orbits, a complete implicit S_5 formula of 22,887 clauses), and finally the *compact signed-Frobenius orbit* base, which stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). In the ecbench factor-base suite it is the best measured Koblitz construction: $S = 9.114$ at $n = 17$ (272 points, 8 columns), 9.985× matched signed-Frobenius rho, against full orbit 18.7× and subspace bases 47× (`research/ecbench_frontier_20261013/`).

8.2 The oracle sequence

The binary oracle went through SAT (XOR-native CDCL after Trimoska et al. [26]; decides the $n = 19$, $\ell = 6$ corpus but does not scale), matrix F4 (whose refutation range was the real frontier), pairs-and-solve for S_4 (validated to $n = 21$, $\ell = 7$, still $O(2^{2\ell})$), meet-in-the-middle pair tables, inherited F4 and F6-IC (§7), and root-finding engines (§6). The boundary ledger's current Koblitz decomposition record is an $n = 31$, $m = 2$ F4 cell with an exact block-6 M4RI acceleration; at $n = 53$ an exact Frobenius-orbit S_5 compression removes the coordinate ROM but every unrestricted solver arm remains UNKNOWN.

The September toy studies that fed this sequence are catalogued as campaigns. At $n = 13$ and $n = 19$, rotated subspace bases covered 1,591 of 2,003 targets with five spaces against 61 with one, and joint rank recovered 58/64 targets against 29/64, without passing a 10% yield gate at $n = 19$ (`research/notes/ecc2k130/`, clusters "rotated-n13" and "rotated-n19"). Finite-fibre S_3 semantics were proved exact and an O-aware CNF made 252.7× smaller at $n = 13$, but no existing solver interface was admitted as an arm; three SAT engines decided 5 satisfiable and 27 unsatisfiable targets at $n = 13$, with the five negatives DRAT-certified. At $n = 131$ a symbolic full-point DAG replayed a single-edge DIMACS instance but could not be admitted past the 64-bit ceiling, and the exact Frobenius census of the balanced $(m, d) = (10, 13)$ rotated base has 1,048,134 columns, the only shape with as many columns as the field. A frozen test of the Frobenius quotient found that toy fields saturate and so test nothing; an independent SAT reproduction solved $n = 13$ and none of four planted targets at $n = 19$ in 1,200 s (`research/frobenius_quotient_sat_20260930/`). At $n = 13$ the presentation of the curve does not

matter: yield spread is explained by cofactor balance (correlation 0.850) and there is no curve effect at $n = 19$ (intraclass correlation ≤ 0.006; `research/koblitz-isogeny/`).

8.3 Against a stronger rho: the September reversals

Three September results that put Koblitz index calculus at or below rho were reversed when rho was strengthened, which is why every later row is quoted against the strongest available rho (§4.5).

- *Candidate tournament.* The 15 September tournament's best candidate used 0.5347 of a worker rho's retired instructions and a triple table 0.409 at $n = 43$, $a = 1$. Giving rho the index calculus's own canonical form made rho win all five cells, IC at 1.19–2.88× (`research/ic_triple_counted_20260923/`; the tournament README, `research/ic_candidate_tournament_20260915/`, carries no correction note).
- *Matched rho for the orbit DLP.* An IC/rho of 0.2403 in retired instructions ($n = 53$, 1,024 targets) used a rho with software carry-less multiplication on x86-64. Strengthening rho in four rungs moved the ratio $0.240 \to 0.317 \to 1.294 \to 1.923$, and the preregistered sweep then read 0.794, 1.520, 1.923, 2.297 at $n = 37, 41, 53, 61$, rising with n (`research/notes/index-calculus/RESEARCH_STRONG_RHO_LADDER_20260929.md`). Against a strong single-target rho the direct arm costs 44.0× $[39.4, 45.4]$, which withdrew a selected-panel crossover of 0.8392 (`research/notes/index-calculus/RESEARCH_SINGLE_TARGET_STRONG_RHO_N53_20260930.md`).
- *Compact-orbit batch lead.* A point panel had compact orbit at 0.309× a corrected rho at $n = 41$, $L = 1{,}024$ targets. A 32-walk batched-inversion normal-basis rho reversed it (2.081 at $n = 37$, 3.086 at $n = 53$ for $L = 1{,}024$; `research/notes/ecc2k130/compact_orbit_strong_rho_20260929/RESULT.md`), and the orbit-disjoint cold confirmation ended with six eligible no-gos, compact/rho CPU 1.314 ($n = 41$, $L = 1{,}024$) to 22.675 ($n = 53$, $L = 1$) (`research/notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md`). Batched compact IC does use fewer instructions than rho (0.5165× at $n = 37$, 0.6221× at $n = 41$, $L = 1{,}024$) while its cold CPU stays above rho, a gap traced to memory behaviour (`research/notes/ecc2k130/compact_ir_ledger_20260930/RESULT.md`). Some sixty notes on S3 batch inversion, swap quotients, Bloom prefilters, window screens and K grids each moved a constant and none crossed rho (cluster "compact-orbit-cold").

The boundary ledger of the same weeks reached the same place from the other side. Re-pricing every row against a matched negation rho raised the vs-rho ratios 2.1–6.5× (`research/ic_rho_reference_20260923/`); the best batch row was 4.86× $[4.21, 5.55]$ at $r = 2^{39}$, and a page $r^{1/6}$ law was falsified at the small end (`research/ic_exponent_20260926/`); single-target online IC beat table-less rho at all six sizes (8.84–17.5×) while cold rho won at all six (1.98–31.6×; `research/ic_single_target_20260930/`). At $n = 37$ in ecbench the cold IC/rho was 3.64× in instructions and 13.6× $[10.6, 18.7]$ in counted S, and the $n = 37$ parameters did not transfer to $n = 41$ or $n = 53$ (`research/ecbench_n37_shared_rank_20261004/`, `research/ecbench_n41_n53_shared_rank_20261005/`). An $m = 4$ exponent audit measured a cost slope of 0.985–1.060 bits per unit of n at $n \leq 19$, against the 0.25 the lead needed (`research/ic_m4_exponent_audit_20260928/`).

Other Koblitz lines. The remaining Koblitz campaigns moved constants inside index calculus and are catalogued in Appendix C. Instruction-level rounds on the Gröbner stage left the whole relation log at 964× rho on K_0 over $\mathbb{F}_{2^{13}}$ (`research/groebner_instructions_20260925/`); a September autolab campaign cut cold instructions by 47.7% and left the candidate at 4.98× matched rho's instructions (`research/ic_autolab_next30_20260915/`); at $n = 37$ a full-rank compact-orbit gate passed with IC at 22.6× rho (`research/notes/ecc2k130/full_rank_compact_orbit_ic_20261003/RESULT.md`). Pipeline kernels gave laned canonical keys 1.09–1.31× (`research/canon_laned_keys_20261008/`), u64 root batching 3.26× whole-process wall (rank cell corrected to 2.47×), and aimed gathering 1.037× set-up (`research/ic_aimed_gather_20261023/`). Transporting a base through ECC2K-130's degree-263 and degree-7 isogenies was built and certified, but on toy fields the transported and descendant bases cost 3.6–4.0× the original (`research/ecc2k130_factor_base_pilot_20260924/`); and τ-adic known-scalar arithmetic is 0.413 of the binary cost at $m = 131$ with no measured rho benefit (`research/notes/ecc2k130/tau_adic_arithmetic_20260929/README.md`). On the rho side, GPU iteration redesigns for the ECC2K-130 walk did not beat the shipping distinguished-point walk (`research/ecc2k130_iteration_redesign_20261012/`, `research/dp32_sigma_table_20261012/`).

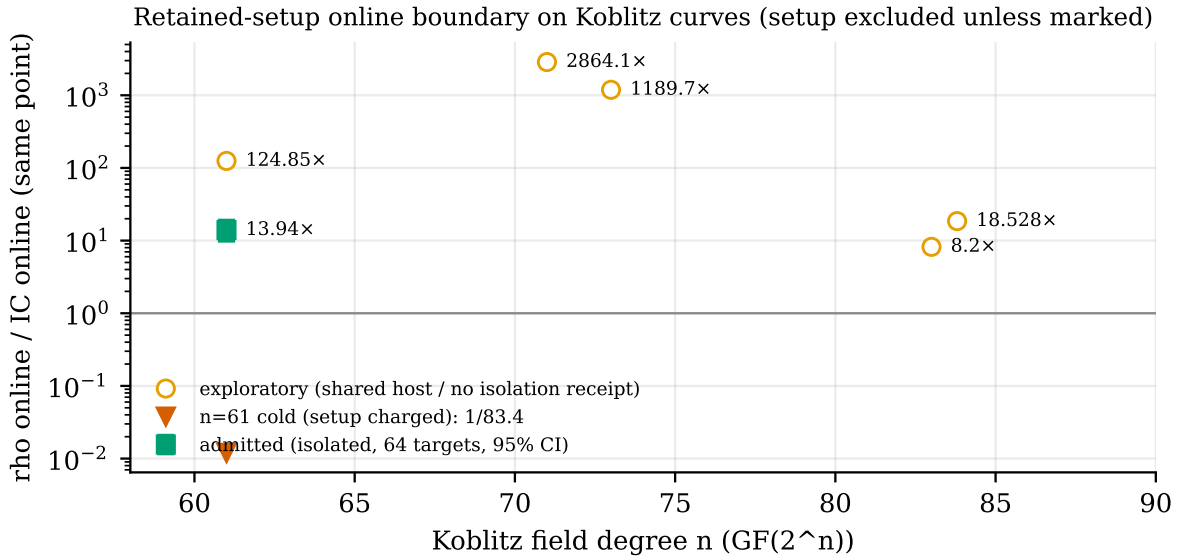


Figure 5: Rho online time over IC online time on the identical public point, reusable IC preparation excluded. Open circles: exploratory (verified answers, no qualifying isolation receipt; three repeats of one point or a single point). Filled square: the admitted 64-workload panel with its 95% bootstrap interval. Triangle: the same $n = 61$ candidate with set-up charged (cold). Sources: `docs/ic/boundary_targets.json`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `research/n83_dense_root_20261013/`.

8.4 The one-target online ladder

The ledger's primary `vs_rho` question is: with reusable IC preparation already done, how long does one unseen public target take, against rho with the automorphism discount on the same point? The ladder (Figure 5):

- $n = 53$: guided shared-log precompute with an explicit pair table (9.34 GB peak RSS); earlier, a 1,024-target shared-log batch beat batched rho in all 15 paired blocks (secondary, multi-target).
- $n = 61$: compact orbit, 600 columns, 73,200 points, 2.73 GB; fresh reproduction 124.85× (exploratory).
- $n = 71$: first rung past 64-bit packing (u128 words), 85,200 points, 2,864.1× median of three repeats of one point (exploratory).
- $n = 73$: 87,600 points, 1,189.7×; the guided-rank precompute took 5.1 h at 57–70 million probes per relation (exploratory).
- $n = 83$, $a = 1$: IC online 5.6–7.9 s against rho 43–71 s, 8.2× (exploratory). Re-measured on 12 October against a default strong rho on five repeats of the same target, IC was *slower* online, 1.382× [1.036, 1.882], and 218.0× rho in fresh-process wall time (`research/n83_evaluation_20261012/`). The later dense-root reading of 18.5× is in §8.5.
- $n = 61$, **admitted** (12 October): $a = 0$, $K = 500$ orbit columns, 61,000 points, 64 fresh public one-target workloads per host, identical candidate manifest on two hosts. Pinned x86_64: median online speed-up 13.94×, 95% [9.81, 19.13], claim-check 64/64; unpinned arm64: 12.87× [7.30, 19.08]. Set-up of about 28 s is outside the window; with it charged, IC costs a median 83.4× rho.

The exploratory ratios are large and irregular because a one-point online window measures the luck of one relation search against one rho walk; the admitted panel's range over 64 workloads is 1.06–648.6×. The admitted point is a constant-factor online result at $n = 61$ (47-bit subgroup) with preparation retained; it is not a cold result, not a scaling law, and not an asymptotic claim.

Retired as an index-calculus win (14 October). Every rung of this ladder compares index calculus with a retained table against rho with none. Given a distinguished-point table of equal or smaller size, Bernstein–Lange preprocessing rho [2] is 53–317× faster online than the compact-orbit arm on the same 64 targets at $n = 41$, 53 and 61 (§12.3). At $n = 83$ (§8.5) rho with a table matched to the index calculus's set-up time or memory won every pair that was run. The ladder therefore measures what a table buys, and generic rho buys more per byte. The validation sweep later found `docs/ic/BOUNDARY_TARGETS.md` still printing the 13.94× as a win; it is superseded (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

8.5 The $n = 83$ campaign

Degree 83 was the programme's largest Koblitz target with a working pipeline: 145 study directories between 4 and 21 October (Appendix C, clusters beginning "n83" and "f6-n83"). In order:

- *F6 micro-studies, 4–5 October* (24 directories, almost all on the $a = 0$ curve). They kept kernel gains (queries up to 2.998× faster, resident memory −52%) and shrank SAT encodings, but every ordinary solve timed out; four-summand coverage was at most $4.662 \cdot 10^{-12}$ of the subgroup and no natural relation was found. On the 81-bit $a = 0$ gate curve, SAT and XCNF decompositions succeed only with a known witness, and the prebuilt-base question is an open problem record.
- *The $a = 1$ point, 7–12 October.* The archived K600 point read 8.2× online and was re-measured slower (above).
- *Root-index capacity, 13–16 October* (25 directories). Root-index layouts grew the $a = 1$ base from K512 (84,992 points) to K3596 (596,936 points) within 14 GiB. The dense root index cut probes per relation from 300,021.9 to 169,069.8 (K1500 to K1798; paired difference 130,952 [52,058, 209,990]), a within-IC result that stands; its one-point online 18.528× is retired (`research/n83_dense_root_20261013/`).
- *Factor-base selection, 14–16 October* (56 directories). Root4, root11 and root21 layouts, Bloom filters, sparse and canonical variants and K screens from 100 to 20,000 points selected within index calculus; the best means are about 26–29 ms online (K3425, K3450, K3500), and many screens end BOUNDED_UNKNOWN. A three-hash 512 MiB Bloom filter cut a K3300 online mean from 34.77 to 31.32 ms (`research/n83_root11_bloom_results_20261015/`); a K3350 win of 7.18% did not replicate.
- *Tier-1 online readings, 14–16 October* (14 directories): median rho/IC of 54.1, 210.3, 113.0, 104.2, 35.1, 413.3 (one point) and 208.7× (K3072, five fresh points, [38.3, 2,510]; `research/n83_tier1_online_20261016/`), all against a rho with no table and with set-up excluded.
- *The control, 16 October.* Bernstein–Lange rho given a distinguished-point table matched to the index calculus's set-up time or memory was faster online in every pair run: IC/rho 3.74 [2.65, 11.05] (K3300, time-matched), 40.61 (K3300, memory-matched), and 81.5–395.5 (K600) (`research/20261016-n83-precomputed-rho-control/`). Only the K3300 and K600 bases were paired; the K3072 reading was not, so its retirement rests on the same mechanism rather than on its own measurement. The validation sweep and the ledger audit retired all the online readings on 19–20 October.
- *Cold, 16–21 October* (11 directories). The cold one-target process fell from about 150 s (K1024) to 119.8 s (K1120, holdout) and 113.8 s (K1107; `research/n83_cold_base_confirm_20261021/`), against a rho online time of about 2.2–2.6 s on the same hosts. No matched cold rho was run, so no cold ratio is admitted; the hosted single-target benchmark's 133.7× is the nearest cold figure (`benchmarks/20261014-n71-n83-hosted-single-target/`).

The open pull requests at the time of writing include further $n = 83$ work ($m = 4$ queries on a retained 16,627-column base, AWS replays, shared-binary panels); none is on `main` (Appendix B).

8.6 Leaderboard Table A: the Koblitz rules

Table A of `docs/ic/LEADERBOARD.md` prices one unseen public target on Koblitz curves at $n = 41$–61 ($\log_2 r = 36.6$–47.2), cold, against rho on the same point. Its history is a sequence of rules, each a pipeline baseline plus a rho reference (`research/ic_tool_program/`):

- *Rule v3*: IC/rho 5.23–21×, the figure still printed in Table A, which remains unvalidated (pre-gate).
- *Baselines v4–v8* (16–17 October): head-of-main, scan-key (v5, 1.20–1.24× cold), scan-fold (v6, 1.33–1.40×), build streams (v7, 1.07–1.13×) and admitted prefetch (v8, 1.06×), together 1.90–1.98× cheaper cold than v4 (`benchmarks/20261017-ic-r12-admitted-prefetch/`). Seven further candidates (prefetch gates, filter slots, huge pages, a wider base) did not clear their rounds.
- *Rules v6, v7*: cold rho still faster at every size, 3.78–12.06× and then 3.43–11.51×; but their rho references ran at 0.57–0.85 of the strong walk per step, so the distance was understated (`benchmarks/20261016-ic-rule-v7/`).
- *Rule v8* (19 October): with a matched GFNI rho the cold distance is 4.67 [4.34, 5.07] to 18.07 [15.98, 20.58]×, 1.30–1.58× the v7 control, with a fitted slope of 0.168 per $\ln r$ (`benchmarks/20261019-ic-matched-rho-v8/`). The run is archived as incomplete although its row is listed.
- *Rule v4 recipe* (24 October; §12.5): a paired retuning improved one size of six; with rho priced in the same process at the canonical batched step the distance is 9.9–43×. Its validation file is withheld because its rho reference may not be the strongest, so it is not an admitted row either.

Table C prices a 32-target batch on nine Koblitz curves ($\log_2 r = 18$–47.2) against batch rho on the same targets: 3.62–14.9×, a historical diagnostic under the one-target rule. All nine failed the 24 October validation gate, and Table D, the Joux–Vitse historical study, is withheld.

The validation sweep found that Table A understates its gap to rho by 1.7–2.0×, and the 24 October validation gate did not check Table A (`research/ic_validation_gate_20261024/`). Rules v8 and v4 measure the same sizes against differently built rho references and are kept apart (Definition 1).

8.7 ECC2K-130

ECC2K-130 is the $a = 0$, $n = 131$ member of the same family, with a 130-bit prime subgroup and rho at about $2^{60.81}$ group operations. The compact producer already solves $n = 41, 53, 61, 71, 73$ of the family; about 300 notes under `research/notes/ecc2k130/` price the routes to $n = 131$ itself. No route prices below rho.

- *Extrapolating the producer.* The fitted probes-per-relation law $C\,r/(n^2K^2)$ with $C = 1$ gives $1.1 \cdot 10^{18}$ core-years at $K = 600$ and $6.8 \cdot 10^8$ at $K = 10^{12}$ columns (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`, with the unexplained factor-of-5 gap in C of §4.5).
- *Decomposition oracles.* Decompositions exist, but no oracle finds one for less than 2^{131} per family; using unions of Frobenius orbits lowers the family's bottom from $2^{132.58}$ to $2^{124.99}$, still $2^{64.18}$ times rho, and needs a materialised base of $2^{44.53}$ points at $m = 3$ (`research/notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION_TARGETS.md`). Relation sweeps over m give a best single-relation row of +4.98 bits at $m = 4$ (with $2^{70.79}$ bytes) and a best honest $m = 8$ row of +13.59 bits with the Frobenius quotient (`research/notes/ecc2k130/RESEARCH_ECC2K130_RELATION_SWEEPS.md`). Crossbred, RR-solver and WDSat oracles are engineering only: Crossbred is 0.023 of F4's bit operations at $m = 3$ but slower end to end, and the RR solvers' advantage over Semaev at $d = 6$ is worthless against brute force.
- *Linearisation.* Plain linearisation leaves a gap of $70.19 - b_{\max}$ bits to rho; the best bilinear XL setting, with each Macaulay solve charged, is 71.82 bits at $\omega = 2$ (78.15 at $\omega = 2.81$), and a trilinear variant is worse (87.16 bits) (`research/linearization_reach_20260930/`, `research/xl_bilinear_core_20261006/`, `research/xl_trilinear_t_unknown_20261006/`).
- *Weil descent.* The descent-degree note's rows are $m = 3$ with a free oracle at $2^{68.58}$; $m = 4$ at degree 4, $2^{68.43}$ (+7.62 bits, now contradicted); and $m = 4$ at degree 5, $2^{71.13}$ (+10.32) at $w = 1$ and $2^{106.52}$ (+45.71) at $w = 2$, in unconverted column touches (`research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md`). The studies behind these rows, and the hybrid guess that puts the $m = 4$ row at +8.43 to +58.27 bits, are in §9.1.
- *Covers.* Over $\mathbb{F}_{2^{131}}$ every ordinary curve has GHS magic number 1 or 131, so GHS descends ECC2K-130 to genus 1 (a zero map on the subgroup) or to genus about 2^{130}, and the Hess–Menezes–Teske walk has nowhere to land. One question is open: $\langle G \rangle$ is the group of a simple 130-dimensional abelian variety, and if that variety were isogenous to a Jacobian with an evaluable isogeny, index calculus on the genus-130 curve would cost $2^{37.17}$, $2^{23.64}$ times *less* than rho; no such curve is known, the Torelli codimension at genus 130 is 8,128, and CM constructions with prescribed Frobenius stop at genus 3 (`research/notes/ecc2k130/RESEARCH_ECC2K130_HYPERELLIPTIC.md`, with the magic-number correction of `docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md`). The $N = 130$ counterfactual is genus 255 at a modelled $2^{55.60}$ against rho's $2^{60.31}$, and measuring one relation trial turns that margin negative (§12.5). The best explicit curve construction reaches genus $2^{28.92}$ (`research/notes/ecc2k130/RESEARCH_ECC2K130_CURVE_CONSTRUCTION.md`). The isogeny class has 262 curves of conductor gap 263 and no measured level effect (§11); rho on the degree-263 leaves keeps only the negation map, an 11.45× penalty (`research/notes/ecc2k130/isogeny_rho_speed_20260930/`).
- *Rho itself.* The separate GPU rho estimate is about 4.3 RTX PRO 6000 card-years (`paper/ecc2k130/`); on one CPU core the ECC2K-130 client walks 13.2 million steps per second (`research/certicom131_rho_ic_bench_20261022/`). Signed σ steps cut modelled GPU time to 0.961–0.970 (`benchmarks/20261014-ecc2k130-signed-sigma/`).

Index calculus is not a route to ECC2K-130 on this evidence: every route that has been constructed or priced is at least about 5 bits above rho (the $m = 4$ single-relation row, at $2^{70.79}$ bytes), most by tens of bits, and the one hypothetical route below rho needs a genus-130 curve that no known method can construct.

8.8 Factor bases closed under $\mathbb{Z}[\tau]$

The $\mathbb{Z}[\tau]$ small-norm multiplier base of 23 October (`research/ic_improve_20261023/README.md`) is the latest Koblitz change: 1.9–4.4× child CPU at $n = 41$–61 and 6.2–10.5× at $n = 71$, against the programme's own producer, exponent unchanged; it is discussed with the other known-logarithm bases in §12.5.

9 Random and prime-degree binary curves

This section covers binary curves $y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^n}$ with n prime, attacked with random subspace bases that use no Frobenius structure, whether the curve is random or Koblitz: the Weil-descent route and the degree question behind the subexponential claims (§3), the decomposition oracles, the random-binary leaderboard rows, and the 131-bit ECC2-131 target. Many of the descent studies were priced for ECC2K-130 because it was the programme's named target, but their systems are built on prime-degree fields with random subspaces and use no Frobenius structure (the systems at $n = 19, 21$ say so explicitly), so they belong here. The Gröbner and $\mathbb{F}_2$ engines they run on are in §7; every study is listed in Appendix C.

9.1 The degree question, measured

The subexponential claims for prime-degree binary curves rest on the first-fall-degree heuristic: that Weil-descended summation systems have a solving degree close to their first fall degree (FFD), which stays small. The programme measured both, on toy fields, in about thirty preregistered studies from late September to 24 October.

First fall degree stays low; the solving degree does not. Every Weil-descended S_3 system for $n = 3$–7 has first fall degree 3 (`research/notes/index-calculus/RESEARCH_FFD_MEASUREMENT.md`), and the HKY last fall degree d_F is 3 or 4 at every reachable size—$m = 2$ to $n = 32$, $m = 3$ at $\ell \le 6$, $m = 4$ at $\ell \le 3$, $m = 5, 6$ to $N = 54$ unknowns—while the plain refutation degree rises from 5 to at least 7 and random controls rise with it (`research/fall_degree_bounds_20261008/`). At $m = 2$ the Semaev d_F sits one below the semi-regular degree at $n \le 18$, and the gap does not grow over $n = 10$–18.

The refutation degree tracks ℓ, within one of semi-regular. For chained $m = 3$ systems at fixed surplus the refutation degree reads 5 at $\ell = 2$, 6 at $\ell = 3, 4$, and exactly 7 at $\ell = 5$ on all four draws of the $(10, 5)$ cell, with FFD 3 throughout (`research/dreg_ell_grid_20260925/`, `research/dreg_surplus_control_20260925/`, `research/dreg_degree7_ell5_20260930/`). At $m = 4$ it reads 5, 6 and at least 7 at $\ell = 1, 2, 3$, flat in n at fixed ℓ (`research/dreg_m4_chain_20261006/`). In 10 of 14 exact cells the semi-regular D_{reg} equals the measured degree and in all 14 it is within one. With a dense in-tree engine at $\ell = 6$ ($n = 19$, both Koblitz curves) the x_4 formulation refutes at 11 and the RR formulation at 8 on every draw, 4 above the sharp first-fall bound of 7, rising one degree per rung from $\ell = 3$ (`research/ic_dense_ladder_20261004/`). A registered semi-regular law then fitted none of seven discriminating cells, with x_4 inside $[d_{\mathrm{reg}}, d_{\mathrm{reg}} + 1]$ on all 32 draws (`research/ic_semiregular_law_20261005/`).

Closure algorithms refute earlier, but rise. A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4 where XL needs 5–7, and none of 345 satisfiable draws (`research/dreg_closure_mutantxl_20261016/`). It stays at degree 4 to $N = 42$ at $m = 4$, $\ell \le 4$, but rises past 4 at $m = 3$, $\ell = 7$ on 8 of 8 draws (`research/dreg_closure_scale_20261016/`); the number of unknowns is not the driver, and of two candidate explanations only "$\ell = 7$" survives a run at $(m, \ell) = (4, 6)$, $N = 54$, which refuted at degree 4 (`research/dreg_closure_driver_20261016/`, `research/dreg_closure_m4l6_20261018/`). Guessing one or two variables restores degree 4 at $(3, 7)$ on 10 of 10 draws at 1/116–1/260 of the projected degree-5 cost (`research/dreg_hybrid_guess_20261024/`).

What the measurements price. Applied to ECC2K-130's $m = 4$ row, the closure results contradict the degree-4 floor (+7.62 bits over rho) and leave degree 5 as the optimistic case, +10.32 bits at $w = 1$ and +45.71 at $w = 2$; the hybrid gives +8.43 bits if the number of guesses stays flat in ℓ and +58.27 if it grows linearly, which only a run at $\ell = 8$ can separate (§12.5). The $m = 4$ cost slope on this engine is 0.985–1.060 bits per unit of n against the 0.25 an exponent lead would need, and a symmetrised

refutation degree rises about one per unit of ℓ (`research/ic_m4_exponent_audit_20260928/`, `research/ic_m4_head_engine_20260929/`, `research/ic_symmetry_lever_slope_20260929/`).

Levers that did not move the degree. Decomposition trees cut on the curve's own structure have the same sampled cut-out degree as random sets in all 102 cells (`research/ic_tree_split_structured_v2_20261011/`); nonlinear factor-base labels refute nowhere through degree 6 (`research/nonlinear_factor_labels_20261002/`); Hamming-weight ideals answer correctly but resolve only after 7–18 of n coordinates are assigned (`research/hamming_ideal_pdp_20260930/`); the one measured subspace-shape effect on the degree was a trace artefact (`research/notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md`), and frozen shape cells were killed at a gap of zero (`research/shape_search_g1_g6_20261015/`). The offensive first-fall-degree thread found no lever within 9 bits of the 2^N boundary (`research/notes/index-calculus/RESEARCH_DEGREE_REDUCTION.md`). A census over $\mathbb{F}_{2^7}$ found non-subfield quasi-subfield polynomials below the HKPYY Lemma 4.1 bound, which answers a tightness question but moves no cost (`research/notes/index-calculus/RESEARCH_QUASI_SUBFIELD.md`).

Reading. On toy fields, the evidence says the solving degree of Weil-descended summation systems grows with ℓ and stays within one of the semi-regular prediction, while the first and last fall degrees stay at 3–4. That is the behaviour of a generic system, and it supports no subexponential claim. It does not refute one: the sizes are small, the closure algorithms bend the curve by a degree or two, and the $(4, 6)$ and $\ell = 8$ cells that would decide the $m = 4$ question have not run. The PKM analogue over prime fields (§10) is in the same state.

9.2 Decomposition oracles

The pair-table floor. On the non-Koblitz curve $y^2 + xy = x^3 + zx + 1$ the meet-in-the-middle pair table is the floor every oracle is measured against. Over Gaudry's pairs-and-solve it is 94–277× at $\ell = 8$, 2,474× and 4,244× at $\ell = 9, 10$, and 928× and 2,321× at $\ell = 11, 12$, with no disagreement on 128 targets per rung (`research/binary_mitm_oracle_l8_20261015/`, `research/binary_mitm_oracle_l9_l10_20261015/`, `research/binary_mitm_oracle_l11_l12_20261016/`; the $\ell = 12$ figure is the validation sweep's recomputation from the committed per-target medians, the study quotes 2,187×). These are wall times on a shared host and unvalidated; the counted evidence—8,068 and 16,092 point additions against 2.1 and 8.4 million quartic solves—carries the result. An earlier pairs-and-root-find oracle was 926× faster than triple enumeration at $\ell = 8$ and decided $\ell = 12$ (11.5 billion triples) in 15 s (`research/notes/index-calculus/RESEARCH_SEMAEV_DECOMPOSITION.md`).

Algebraic oracles against the floor. No algebraic arm beats the pair table: the cheapest is 303× the floor at $n = 13$ and $3.8 \cdot 10^4$× at $n = 31$ for $m = 3$, and the slopes in n are steeper (F4 0.534, F6-IC 0.583, floor 0.207; at $m = 4$ a wide chained solver 0.805 against 0.333; `research/ic_m3_algebraic_vs_mitm_20261014/`).

SAT, SMT and WDSat are enumerators. Every SAT, SMT and WDSat back end spends about one conflict per candidate tuple (CDCL 0.97–1.02 per triple), so the best enumerate-and-root-find oracle beats every solver by 10^2–10^4 (`research/notes/index-calculus/RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md`). The SAT pipeline decides all 11 satisfiable $n = 19$, $\ell = 6$ corpus instances in 17.8 s; SMT runs at 0.44–2.63× the native CDCL+XOR solver (`research/pdp_smt_oracle_20261008/`); WDSat solves $n = 5$ (47 against 51,425 conflicts with and without XOR-Gauss) and crashes at $n = 9$ on a static allocation limit, where CryptoMiniSat takes 0.4–2.6 s (`research/wdsat_rr_20260920/`). The SAT route is closed as an exponent lead. Isogenous and random-b neighbours give identical Macaulay ranks and FFD 3 (`research/pdp_isogeny_probe_20261008/`).

The PDP oracle programme. The point-decomposition speed-up programme (`research/pdp_oracle_speedup_20261011/`), run over $\mathbb{F}_{2^{3\ell}}$, measured the oracle in steps before root finding replaced it: a baseline (PS-0; 338–452 ns per pair across five rungs, A/A spread $\leq 8.8\%$), a gate-count model of the group-law form (PS-1; 7,410–14,608 fused ternary-logic issues against 70,695–190,275, 9.5–13.0×), a scalar group-law kernel (PS-2; 29.8–40.8 ns per pair, 0.067–0.123 of baseline, 400 corpus root sets agreeing), and a bit-sliced kernel (50–62× the baseline on two hosts, about equal to the scalar kernel). The root-finding oracle then ran 3,007–115,160× the PS-0 baseline at $\ell = 8$–14, at the generic floor's

slope (`benchmarks/20261012-pdp-root-finder/`). A split/hybrid solving-degree study (Koblitz $b = 1$) projected its closest relation phase at 22.8× strong rho, 391× signed-Frobenius rho (`research/pdp_split_dreg_20261011/`); with Frobenius closure the priced S fell to 0.35–0.54 of the standard base but the best arm stayed at 36× strong rho at $n = 11$ and 889× at $n = 19$ (`research/pdp_split_slots_20261012/`). A shape-replay switch cost more on all 9 cells and stays off (`research/pdp_shape_replay_20261015/`).

Norm and circuit searches at $n = 83$. The 17–19 October autolab cycles built a native norm predicate over GF(2^{83}): a cutoff-8 circuit cuts AND gates 60.5% and clauses 32.9%, an affine support limit of 2 cuts internal coordinates from 58,176 to 45,427, and 1,237 of 1,800 positive-dimensional cells were certified empty. Every hard search returned UNKNOWN with zero verified witnesses, and a primary-source check found the hybrid ANF/CNF approach already established in Bosphorus, SMFF and PolyBoRi (`research/autolab_cycle_20261019_v1/`, `research/autolab_cycle_20261019_v3/`). The companion draft `papers/norm-rootfinding/` reports the same: no measured speed-up.

Toy end-to-end. On toy fields the fastest index calculus, a cached Nagao arm, was 2.33× plain rho, and S_4/S_3 with CryptoMiniSat 14.5–14.7× (`research/toy_ic_e2e_candidates_20260922/`). Frobenius transport preserves zero and lift sets; a constant term c changes zero counts only; neighbouring curves show no consistent F5 advantage; symmetry quotients help three summands (0.26–0.31 of the ordered cost) but not two (1.06–1.32) (`research/toy_binary_elimination_20260924/` and the toy cluster).

9.3 Leaderboard rows

Table B of `docs/ic/LEADERBOARD.md` has five random-binary rows, $\mathbb{F}_{2^{15}}$ to $\mathbb{F}_{2^{27}}$ ($\log_2 r = 14$–24.4), one target, $m = 2$, folded pair table with walked targets: IC is 13.1, 10.6, 18, 23 and 38.2× matched rho. All five passed the validation gate (commit `a8d8e3cd38`; `research/ic_validation_gate_20261024/`), which the ledger paper, written a day earlier, records as no row passing. The cofactor-projected base of LEAD-19 cuts three of these rows by 1.29, 1.58 and 3.36× in cold S, leaving 6.1–14.4× to matched rho (`research/lead19_projected_base/`); it is not yet a leaderboard row. The integer-multiple closure of 23 October cuts S 3–4× against plain seeds at degree 26–28 and remains 6–12× the framework's rho reference (§12.5).

9.4 ECC2-131

The Certicom ECC2-131 target ($\log_2 r = 130.0$, rho $2^{64.83}$ GAE with the negation map) has a cost model, gates and two falsifiers (`docs/ic/CERTICOM131_IC_PROGRAM.md`), all on toy instances. With a free oracle the $m = 3$ floor is $2^{65.2}$–$2^{67.2}$, above rho from linear algebra alone; $m = 4$ is the first usable arity and needs a solving degree of at most 3 (generous pricing) or 1 (plain). On random-b curves at $m = 3$, $n = 5$–11, the measured degree is $D = 4, 5, 6$ at $\ell = 1, 2, 3$, equal to the semi-regular reference with zero spread over 60 draws, and equal to the Koblitz $b = 1$ value on 52 of 52 paired draws; extrapolated to $n = 131$ it is between about 6–7 (if it plateaus) and 39 (semi-regular). One CPU core walks ECC2-131 rho at 8.50 million steps per second with a batched pclmul walk, and 11.46 million bit-sliced (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`); a CUDA port compiles but has not run on a GPU. The bench's IC ladders on smaller random-binary curves read 11–24× matched rho; it is not an admitted row.

sect113r1. The 113-bit SECG curve is weak by size alone ($2^{56.33}$ generic work); its two degree-5 isogeny routes transfer the problem without an established speed-up, and the GHS and Semaev routes reached no conclusion (`research/sect113r1-weak-curve-20261006/`).

10 Prime fields

Over $\mathbb{F}_p$ there is no subfield to descend to and no Frobenius to fold, so index calculus must find structure in the factor base itself (§2). The programme tried known-answer ladders, root-finding and pair-table oracles, the Petit–Kosters–Messeng Kummer tower, structured bases on the P-256 field, isogeny-class walks on the NIST curves, and lattice reformulations. No prime-field arm comes within a single-digit factor of rho except on the smallest Table B rows, and the gap grows with size. The section ends with Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$, the one odd-characteristic extension field the programme measured end to end. Every study is listed in Appendix C.

10.1 Known-answer ladders and leaderboard rows

Over prime fields the ledger tracks staged known-answer ladders (`docs/ic/boundary_targets.json`, regime `prime`). On $j = 0$ curves the staged driver recovers a 20-bit logarithm ($p = 1{,}048{,}609$) in 56.8–64.9 s of IC wall against 150–153 ms of rho on the identical public point, with split timers and three seeds; running the relation search on 14 threads made that stage 3–4.4× faster in wall time with counted work unchanged (`research/j0_parallel_relations_20261016/`). On the generic $a = -3$ shape (P-256 and CryptoPro-B class), a Semaev-S_4 three-decomposition family with Cantor–Zassenhaus root finding yields one relation per trial at $p = 65{,}519$ with a 120-point base; at fixed base size, median trials per relation grow from 2 to 28 to 379–506 at 16, 20 and 24 bits, about 16× per four bits. The prime `vs_rho` row is *not achieved* at any measured size.

Table B of `docs/ic/LEADERBOARD.md` has eight prime rows, $\log_2 r = 9.7$–23.4, one target, $m = 2$, folded pair table with walked targets: IC is 6.34, 5.47, 6.43, 6.71, 5.82, 8.3, 10.5 and 13.5× matched rho, the ratio rising from about 18 bits. All eight passed the validation gate on 24 October (`research/ic_validation_gate_20261024/`). In ecbench the root-finding pair table is 16–20× matched rho in counted GAE, and 7.46× [5.82, 9.63] with a lane collector, all lower bounds (`research/ecbench_rootfinding_lanes_20261013/`); a planner that picks the base, stop rule and oracle beat every one-change alternative on six curves at $w = 0$, the root-find-and-subtract variant by 171–439× (`research/ic_planner_validation_20261020/`). The frontier campaign's two-summand prime arm goes from 10.5× rho at 20 bits to 102× at 40 bits (§12.1).

Root finding against Buchberger on NIST shapes. Per query, root finding of Semaev's S_3 was 3,021× (P-256 shape) to 5,802× (P-224) cheaper than the comparison arm at a six-point base (`research/semaev_rootfind_nist_20261012/`); the arm was `groebner_f4::buchberger`, not F4, so the ratio is against Buchberger (§4.5). It is an oracle ratio with no complete solve. A host emulation of a GPU decomposition kernel passed six configurations on 40–61-bit P-256-class curves (`research/gpu_rootfind_decomposition_20261012/`); no device ran it. Three frozen SMT formulations of a 20-bit decomposition all returned UNKNOWN at 120 s, where the native reference needed 146 probes (`research/prime_field_smt_20261005/`).

10.2 The Kummer tower

Petit, Kosters and Messeng's base $V = \mu_{2^t}$ (the roots of $x^{2^t} - 1$) gives prime-field summation systems extra structure for an algebraic solver. Its claim rests on the solving degree of those systems staying bounded as $N = m\ell$ grows; the programme measured that degree for four weeks.

The solving degree. The pilot found F4's solving degree at 4–5 for $m = 2$ through $N = 18$ and read the preregistered "grows like N" as failing at toy size (`research/pkm_tower_pilot_20260924/`). Round 2 showed it is not bounded: at $m = 2$ it is 5 to $N = 18$ and 6 at $N = 20, 22$, at $m = 3$ it reaches 7 at $N = 15$, and the null family behaves the same (`research/pkm_tower_round2_20260925/`). At $m = 4$, $N = 16$ finished at $D = 7$, as at $N = 12$ (`research/pkm_tower_round3_20260925/`); at $m = 3$ the line reads $6, 6, 7, 7$ over $N = 9$–18 (`research/pkm_tower_round7_20261006/`). Signature engines agreed with F4 but needed one or two degrees more and 1.7–28× the memory, so F4 stayed (rounds 4–5); a full-echelon exit cut multiply-adds 39.8% at $m = 4$, $N = 12$ without moving D (round 6). The torus and split Dickson presentations give the same D in every cell, and the Kummer D does not depend on the 2-adic valuation (`research/pkm_dickson_torus_20261015/`). At $m = 3$, $N = 21$ two engines finished every degree-7 step without reaching the unit ideal and stopped inside degree 8 on memory—the native `f4_fp_tower` at a 317 GB matrix cap after a 67 GB peak, msolve at 99.6 GiB on 16 threads (`research/pkm_lead1_m3n21_20261016/`, `research/pkm_m3_n21_msolve_20261018/`). That implies $D \geq 8$ at $N = 21$, which both studies read as inconclusive because no degree-8 step completed. The Certicom-131 programme's gate needs $D \leq 3$ at $m \geq 4$; the toy plateau at 7 is above it, at 2^{89}–2^{98} (generous) and 2^{129}–2^{139} (plain) at 131 bits (`docs/ic/CERTICOM131_IC_PROGRAM.md`). The exponent sum of decompositions on the tower is indistinguishable from random at $k = 1$–6 (`research/ic_exponent_falsifiers_20261013/`).

In ecbench. We ran the tower end to end with the root-finding oracles on five curves of 20–28 bits (896 runs, all verified, all replay audits OK; `research/pkm_tower_ecbench_20261013/`). Index calculus costs 34.0, 41.9, 29.4, 96.6 and 140.6× rho-negation in S (95% intervals in Figure 6), growing by 0.26 ± 0.28

bits of S_{IC}/S_ρ per bit of r, consistent with the $p^{1/6}$ the floors predict. The tower base costs the same as a plain base of the same size to within 1% (ratios 0.999–1.012): PKM's structure helps an algebraic solver, not a table. An isogeny-tower variant costs 1.00–1.33× the plain base once its auxiliary-curve search is charged, and every arm is 17–137× rho (`research/pkm_isogeny_tower_ecbench_20261015/`); for a P-192-sized tower that search would cost about 2^{74} GAE, or 11.2 core-years, while a supersingular auxiliary curve gives the needed point in 7 ms with the same solving degree (`research/pkm_isogeny_tower_p192_20261015/`, `research/pkm_supersingular_aux_20261015/`). A 30-bit row of 24 October is 158.9× rho (§12.5). Swapping in an A11 presentation could not be priced: ecbench has no method that builds a tower summation system (`research/a11_pipeline_pricing_20261019/`).

10.3 Structured bases on the P-256 field

Twenty-four rounds between 4 and 6 October (`research/icv1-fp256-t89188191154553853111372247798585809583-f188c491_affine_bitbox_degree_round1_20261004/` and its 23 successors; cluster "icv1-fp256") searched for factor bases on the P-256 field whose summation systems have lower degree. They found degree reductions on split systems: Dickson-torus branching lowered the solving degree of components from 4 to 3, image trees for S_9 and S_{17} reached local degree 2, and an indexed quadratic image cut one stage 498×. At the P-256 modulus the images hit their generic size ceilings (width ratio 1.000000), so the degree reduction transferred and the compression did not (round 14). No round found a relation for a target that was not planted: planted length-17 relations were recovered, hash-public targets were not, and a log-transport round found zero relations among 14.4 million images (round 21). Projected end to end, the best schedule is 16.2–19.4 bits above rho in the same unit, after a symmetric projection of $2^{144.625}$ was withdrawn for exceeding its domain (round 20); the final geometric bases are 392–404× rho, and a known-log constant of 0.964× rho in round 24 is a lower bound that the study itself says is not an attack.

10.4 Small-abscissa bases and isogeny classes

Two-point small-x bases have an exact pair yield of $2B^2/(n-1)$ with no pair-sum collisions: on P-192 a base of 8,192 points needs 2^{178} trials for enough relations against rho's 2^{96}, and on all 33 SECG curves the route loses to rho by at least $2^{40.9}$ in trials (`research/prime_small_x_p192_20261014/`, `research/secg_small_x_20261014/`). At $p = 2^{61} - 1$, 1,228,800,000 target trials over 1,024 curves of a horizontal isogeny class found zero relations, while a 20-bit positive control found 33 and recovered the scalar (`research/prime61_factor_base_20261016/`, `research/prime61_disk_extend_20261018/`). Native walkers enumerated 1,064,000 distinct curves isogenous to P-256, 20,000-curve classes of P-224 and P-192, and verified isogenies of degree up to 10,453 (P-192) and 1,471 (P-224); no preregistered structural screen fired on any walked curve, so no factor-base test was ever selected (`research/p256_isogeny_million_20261006/`, `research/p192_p224_large_degree_isogenies_20261009/`). By Lemma 1 a homomorphic transfer could change only bounded constants in any case. The class of a 61-bit curve has a certified order of 358,094,304 with adjacency still open (`research/p61_terminal_closure_20261023/`).

10.5 Lattices

Every lattice intrinsic to $(E, P, [k]P)$ has rank 2 (4 over $\mathbb{Z}[\tau]$), and relation lattices have a derived ceiling of $p^{5/6}$ (`research/notes/ecdlp-general/RESEARCH_LATTICE_ECDLP_ROUTES.md`). A Coppersmith reformulation of the $m = 4$ oracle via S_5 reaches $\delta^* = 1/80$ against about $1/5$ needed and is closed (`research/lit2026_w10_s5_coppersmith_20261011/`); on S_3, a frozen falsifier recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3 (`research/coppersmith_s3_theta2_20261023/`).

10.6 ECCp-131

ECCp-131 ($\log_2 r = 130.19$, rho $2^{64.92}$ GAE) has exact parameters wired into the BSGS and IC command paths; a three-point small-x probe found zero natural relations and rank zero at a 128-point base, which blocks any full-parameter IC solution and is an open problem record (`research/eccp131_bsgs_ic_20261020/`, `problems/eccp131-full-parameter-index-calculus.md`). Structured prime bases for the Certicom programme were a null result (N/E 0.969–1.000 against a random control of 0.974). One CPU core walks ECCp-131 rho at 1.23 million steps per second with a 32-bit Montgomery batched walk, and

15.06 million with an AVX-512 eight-lane walk (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`).

10.7 Audits of standard curves

Several studies examined the NIST, SECG and CryptoPro curves themselves. None found an ECDLP weakness. The NIST prime-curve seeds were regenerated as 1997–98 `drand48` output, with the skip rule and base points still unexplained (`research/nist_seed_provenance_20261016/`, `research/nist_basepoint_provenance_20261017/`); a claimed P-256 $n-1$ outlier was not established after controls, and the published b sits at the median of every statistic over a million seed samples (`research/nist_cyclotomic_controls_20261011/`, `research/p256_trapdoor_round2_20261012/`). The $j=0$ SECG curves have sextic twists of weak order (rho $2^{65.5}$ for secp256k1's), a fact about twists, not about the curves (`research/safecurves_trapdoor_slots_20261017/`). Special prime shapes change neither the distribution of $\#E$ nor the embedding degree, so only small-embedding-degree pairing curves open an SNFS-type route (`research/snfs_prime_families_20261023/`, `research/solinas_distribution_20261023/`). A P-192 "BREAKS" verdict applies to a singular nodal cubic, an implementation weakness (`research/p192-weak-curve-20261006/`), and the P-224 directory is an invalid-curve attack on an unvalidated ECDH oracle (`research/p224_attack/`). The CryptoPro-B and CP6-782 work is constructive: GLV-2 scalar multiplication 1.25–1.50× faster via isogeny-chain endomorphisms (`research/cryptopro_b_chain_sweep_20261006/`, `research/endo_native_bench_20261016/`). Cheon's attack with auxiliary inputs beats rho from 24 bits up, but only for the DLP with auxiliary inputs (`research/torsion_auxiliary_inputs_20260918/`).

10.8 Literature claims that did not survive

The lit2026 programme (`research/lit2026_speedup_program/`) ran each finding of a 2025–26 search as a measured change against an interleaved baseline binary. Workstream W12 implemented the zero-minor guess-and-determine attack (Mahalanobis, arXiv 2607.09814v2): it solved 48/48 planted targets on 16–26-bit curves, but its search phases scale with slope 0.905 $[0.769, 1.021]$ in $\log_2 r$ and its cost over rho grows from 260× to 6,715×—exhaustive in r, as the source's own table says. A second flagged ePrint (Hu, 2025/015) did not specify an implementable algorithm and is recorded as blocked. On the generic side, W11 cut BSGS from 16.75 to 3.71 multiplication equivalents per visited point with paired steps (wall 1.15–1.67×), which moves rho's and BSGS's constants, not index calculus.

10.9 Gaudry's $k=4$ index calculus over $\mathbb{F}_{p^4}$

Over $\mathbb{F}_{p^k}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, a relation needs a k-decomposition solved by a Macaulay system. We made the `gaudry_quartic` solve generic in the number of unknowns and faster (median whole solve 1.96/4.19/6.08× at $d=4,5,6$, 37.3 → 6.1 s at $d=6$, as recomputed by the validation sweep from the committed records on a shared host; the S_5 solve from about 30 minutes to 164–203 s; every count and solution identical). Then we built FGHR's D_4 variant (`research/gaudry_quartic_d4_20261013/`): on a curve with rational 2-torsion T, the coordinate $Y=(x-x_0-c)/(x-x_0+c)$ satisfies $Y(P+T)=-Y(P)$, so S_5 in Y is invariant under the Weyl group D_4 (order 192). In the invariants $\sigma_k=e_k(Y^2)$, $\pi=\prod Y_i$, the system has Bézout number 512 instead of 4,096, Macaulay degree 14 instead of 29, and a 13,412 × 11,628 matrix instead of 41,780 × 40,920.

At $p=269$, the cost C_4 of one decomposition falls from 1.21–1.23 $\cdot 10^{12}$ to 2.54–2.55 $\cdot 10^{10}$ $\mathbb{F}_p$ multiplications, 47.5–48.4×; a solve takes 4.2–5.6 s instead of 164–203 s. Decompositions agree with an independent meet-in-the-middle enumeration on 19 of 19 solved targets; one target in 20 is degenerate (quotient dimension 517) and is discarded, adding about 5% to the relation phase. The Y-line base decomposes about 1/8 as often (pooled ratio 0.133 over 8,000 targets; prediction $1/8 \pm 30\%$ held on one of two curves), but each column covers $\{\pm P, \pm(P+T)\}$, halving the unknowns.

Inserted into the repository's crossover model (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md` §11.16), the measured C_4 and rate move the $k=4$ crossover with rho from $2^{149.4}$–$2^{152.7}$ to $2^{135.1}$–$2^{135.8}$ (cofactor $h=2$, forced by the 2-torsion), and the asymptotic ratio past it from 0.34–0.63 to 0.12–0.22 of rho. This is the one place in the programme where a crossover moves; it moves in an *extrapolated model* ($n^{1/4}$ relations, $n^{1/2}$ linear algebra and rho) fed by a measured constant, so it is classed as ENGINEERING, not an advance. The validation sweep notes that this crossover was not re-derived against a rho with the negation map; against the negation walk, a later note puts the plain prime-order $k=4$ handover at $2^{156.1}$ and a ψ-folded one at $2^{146.0}$ (`research/notes/index-calculu`

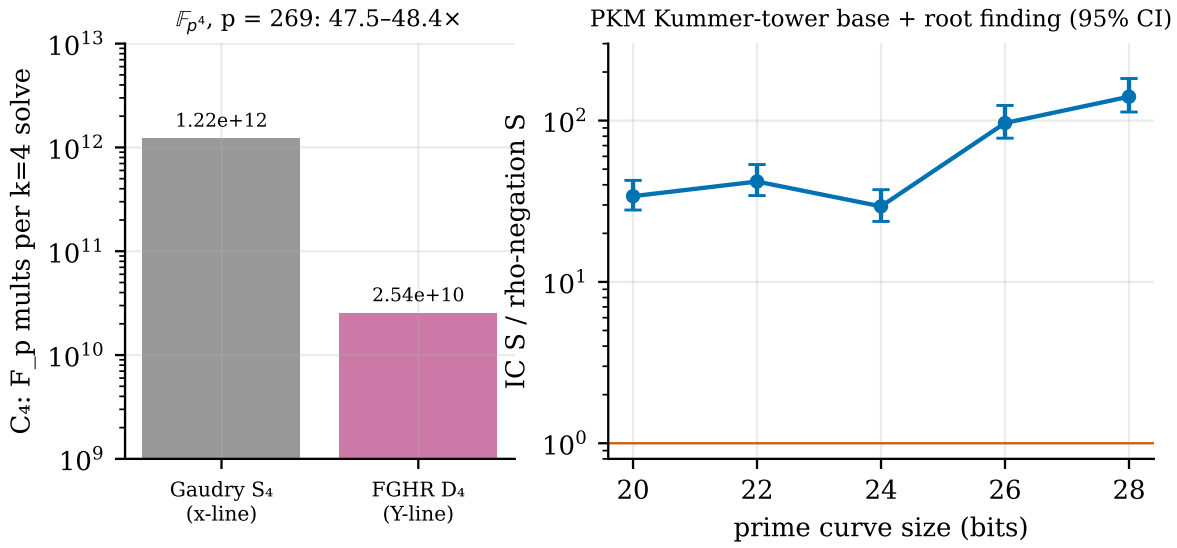


Figure 6: Left: cost C_4 of one $k = 4$ decomposition over $\mathbb{F}_{p^4}$, $p = 269$, Gaudry's S_4 system on the x-line against FGHR's D_4 system on the Y-line (`research/gaudry_quartic_d4_20261013/`). Right: index calculus with the PKM tower base and root finding over rho with negation, eight one-target workloads × three rounds per size, paired 95% cluster-bootstrap intervals (`research/pkm_tower_ecbench_20261013/`).

`s/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`). The measured characteristic-two $k = 4$ ladder is in §12.3. Extension fields of degree four are not the fields deployed curves use.

11 Covers, isogenies and other transfers

A recurring hope was that some other representation of the same instance—an isogenous curve, a curve at another level of the isogeny volcano, a Jacobian, a Weil descent—would make index calculus cheaper. These routes produced the programme's clearest negatives, and one lemma that explains most of them.

Lemma 1 (transport conservation, `docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §1)**.** *Let $\varphi : E \to E'$ be an $\mathbb{F}_q$-rational isogeny (or any homomorphism of rational points), $\kappa = \ker\varphi \cap E(\mathbb{F}_q)$, $R \in E(\mathbb{F}_q)$, $\mathcal{F}' \subset \varphi(E(\mathbb{F}_q))$ a factor base on the codomain and $\mathcal{F} = \varphi^{-1}(\mathcal{F}') \cap E(\mathbb{F}_q)$. Then the number of ordered m-decompositions of R over $\mathcal{F}$ is $|\kappa|^{m-1}$ times the number of ordered m-decompositions of $\varphi(R)$ over $\mathcal{F}'$.*

Proof. Images of decompositions of R are decompositions of $\varphi(R)$. Conversely, for $\sum Q_i = \varphi(R)$ choose rational preimages P_i of the Q_i; then $\sum P_i - R \in \kappa$, and of the $|\kappa|^m$ choices exactly $|\kappa|^{m-1}$ sum to R (translate P_1 by the kernel element). □

So a homomorphic transfer changes nothing but three curve-specific inputs: the rational kernel symmetry $|\kappa|^{m-1}$ (bounded by the rational torsion; trivial on prime-order curves), the factor-base geometry available on the destination, and the cost of arithmetic there. Sequences of isogenies compose to one homomorphism; a loop is an endomorphism and acts on the subgroup as a known scalar, useful only as a known-log multiplier. Multi-curve factor bases glued by isogenies are single-curve bases in disguise.

Conductor gaps in the ECC2K-130 class. Jao, Miller and Venkatesan [18] prove random self-reducibility among curves with the same endomorphism ring, leaving open whether curves whose conductors differ by a large prime could differ in hardness. The ECC2K-130 class has conductor gap 263; we computed its structure (`docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`) and ran several rounds of experiments. Native counted-S index calculus on two fixed CM pairs gave floor/crater ratios 0.975 (99% $[0.603, 1.451]$) at $\ell = 8209$ and 1.127 $[0.922, 1.342]$ at $\ell = 131101$: no level effect (192 verified records, 96 exact replays; `research/ecbench_conductor_gap_20261012/`). Two-torsion symmetrisation is a quotient isogeny: at a matched base of abscissae it cut F4 row-reduction work 4.1–4.3× ($|H| = 2$) and 17.6–19.6× ($|H| = 4$) over 1,128 systems—but a correction the next

day showed that at a matched number of *columns* the saving disappears; it was the cost of a redundant orbit base. A combined Weil descent used 234–529× the F4 operations of the blind system (`research/isogeny_conductor_gap_round3_20261011/`).

Fibre-aware index calculus. Isogeny fibres are torsors whose arithmetic is controlled by one cohomology class (`RESEARCH_ISOGENY_FIBERS_ECDLP.md`); relation generation is isogeny-equivariant at zero lifting cost, and a fibre-closed pair table was the best lever of the conductor-gap rounds (12.8× rho in its best cell). Measured against matched rho across sizes at fixed cofactor, the envelope IC/rho ratio grows from 8.4–8.8 to 35.1–37.0 ($h = 4$) and from 13.4–20.1 to 60.4–63.1 ($h = 72$), as $\#E^{1/6}$: the gap widens, it does not close (816 audited records; `research/fiber_ic_scaling_20261011/`).

Other structure. Symmetrised u-frame bases saved 0.84× at $n = 17$ and 0.76–0.77× at $n = 23$, but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`). GHS and hyperelliptic covers: ECC2K-130 does map to hyperelliptic curves, but the genus that would pay is missing. In odd characteristic the cubic GHS transfer from $E_0(\mathbb{F}_{q^3})$ to a genus-3 Jacobian now runs end to end with double-large-prime index calculus, at about $q^{1.4}$ against rho's $q^{3/2}$, within 1.15× of a measured negation-map rho at $q = 2^{20}$ (§12.3). The isogeny-loop GLV/GLS work produced a measured 1.09–1.15× faster four-dimensional scalar multiplication (`research/isogeny_loop_glv_20261012/`) and a $\sqrt{\text{élu}}$ kernel cost exponent 0.822 $[0.805, 0.838]$; these speed up curve arithmetic, not the discrete logarithm.

Covers, 21–24 October. The Joux–Vitse genus-3 cover of $E(\mathbb{F}_{p^6})$ is now within 1.6–2.0× of a pooled rho constant at $p = 1009$; measuring the per-trial cost on a genus-255 binary Jacobian turned the $N = 130$ GHS counterfactual's modelled margin negative (-1.83 bits); and two censuses bound what covers exist at toy sizes (genus floor 13 for odd-order $E/\mathbb{F}_{q^4}$; a 2^{-l} genus-3 fraction over $\mathbb{F}_{2^{3l}}$). All are in §12.5. None touches a prime field or a prime-degree binary field, where the GHS genus is out of reach.

Genus three against the strongest rho. Rounds three and four of the genus-3 work ran without a rho arm; a 23 October study supplied one (`research/ghs3_vs_rho_20261023/`; it has a frozen protocol and tables but no README, and no review is recorded in it). On round 4's instances, $q = 2^{14}$ to 2^{20}, three curves per size, in round 4's unit and binary, the reference was an ideal negation rho priced with Karatsuba products and 32 walks per inversion (R_{ideal}, the primary reference, the most favourable to rho) and a measured negation walk (R_{meas}). The frozen rule declares "IC below the strong rho" at a size only if every curve is below both references at every weight of the band. At $q = 2^{16}$ it is: IC/R_{ideal} is 0.70, 0.71, 0.70 at $w = 0$ and 0.64 and 0.38 (geometric means) at $w = 0.1$ and 1. At the other six sizes the decision is *unresolved*, with geometric means 0.97–1.52 (1.27 at 2^{20}). Three things qualify the result. The registered unit prices one subgroup multiplication as one $\mathbb{F}_q$ multiplication ($c_\ell = 1$); the protocol's own sensitivity, $c_\ell = 9$, puts every instance at 2.7–12.4× rho. The ratio carries $\sqrt{c}$ of the cofactor, because IC depends on q alone and rho on $\ell = \#E_0/c$: the three curves at 2^{16} all have cofactor 4, and a post-hoc fit gives exponent 0.474 in c (theory 0.5) and -0.081 per bit of q at fixed c (theory -0.167). And the pooled trend over all 21 instances is flat (0.034 $[-0.047, 0.105]$ per bit), so no crossover size is determined. This is the programme's only preregistered decision of index calculus below the strongest rho. It is a measurement of a known pipeline in a structured, weak class ($E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, never prime order), where theory predicts $q^{4/3}$ against $q^{3/2}$; it depends on the pricing of one operation; and it is not yet reviewed. It updates the ledger paper's "1.15× at 2^{20}" and bears on that paper's prediction of a crossover at $q \le 2^{22}$ once rho is measured: below rho at one size, unresolved at the others, with no measurable pooled trend.

Joux–Vitse: two arms of one cover. The Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (full rational 2-torsion, a weak class) has been measured with two solvers, and their comparisons with rho do not agree. The *plain Nagao arm* with a sparse-FGLM linear stage is 1.59–1.98× a pooled rho constant at $p = 1009$ and closes as about p^{-2} (`research/jv_cover_p_scaling_20261024/`; §12.5); it is a lower-bound comparison and current. The *sieve arm* (`jv_cover -exp sieve`) measured 0.024 $[0.019, 0.031]$ of a strong rho at $p = 1009$ and 0.0042 at $p = 1823$ on 19 October, but its 20 October review withheld every comparison (no numeric acceptance for the controls, uncharged rho hashes, incomplete accounting; `research/jv_sieve_strong_rho_20261019/`); a replication with fixed thresholds failed its acceptance rule and is in an open pull request, and a prospective re-run is registered and has not run. A 9 October whole-route replay from a non-weak challenge curve, through an isogeny walk to a weak curve and its cover, read 0.058 of a

pooled, unmatched rho at $p = 1009$ on one seed per prime, with an earlier revision of the solver whose source includes the sieve module (`research/jv_cover_e2e_30bit_20261009/RESULT.md`). The roughly thirtyfold gap between the arms at $p = 1009$ is unexplained on `main`. The cover-decomposition note still states a measured sieve crossover near $p \approx 430$ (`research/notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md` §11) with no erratum; it should be read through the review. Leaderboard Table D, the historical study, is withheld. On the method side, the sparse-FGLM stage made the plain arm 1.28× cheaper end to end (`research/jv_cover_sparse_fglm_20261018/`); relation collection is 99.99% of S (`research/ic_cover_leads_20261018/`); depth-1 isogeny classes hold no weak curve over 4,871,062 curves, while 86–95% of deeper classes hold one (`research/jv_weak_class_law_20261007/`); and the $(n-1)$-decomposition variant is $4.5 \cdot 10^7$–$1.8 \cdot 10^8\times$ rho (`research/joux_vitse_n1_20261015/`).

Cover censuses. The minimal cubic GHS genus over an isogeny class follows $\#E(K)$ mod 4: 13 for odd order, 5 for 2 mod 4, 3 for 0 mod 4, and genus 3 is the only value below rho's $q^{3/2}$ (`research/ghs_cubic_cover_census_20261012/`); at $n = 5$ the floors are 33, 33, 25 and 5 by 2-torsion type (`research/ghs_cover_census_n5_20261013/`). No deployed prime-field or prime-degree binary curve has a usable subfield; only the RFC 2409 Oakley groups 3 and 4 are reachable, at genus 15–16, where exponent models disagree in sign and the measured genus-16 relation phase alone is 5.2 bits slower than rho (`research/cover_route_prime_binary_20261008/`, `research/ghs_oakley_extended_20261023/`). All 15,791 cover maps in the registry were assessed: 165 binary maps have genus-3 arithmetic and the 15,626 odd-characteristic genus-2 covers lack a Jacobian implementation (`research/cover_transfer_assessment_20261013/`). An inventory of about 90 cover and decomposition studies found every route on prime, Koblitz and prime-degree binary curves closed, negative or constant-factor (`research/cover_decomposition_gap_search_20261018/`). Smith's $(2, 2, 2)$-isogeny route was reproduced natively (full success 0.18638 against a theoretical 0.18574 over 10^6 curves) and its sign bias shrinks like $1/\sqrt{p}$ (`research/smith_prop24_native_20261014/`, `research/smith_sign_scaling_20261016/`).

Volcano levels and generators. A 403,168-run census found the level effect equivalent in 33 of 35 comparisons and different in none (`research/volcano_ecdlp_census_20261017/`), and a large-sample rho study found the floor/crater ratio 0.998 $[0.986, 1.010]$ (`research/conductor_gap_rho_large_sample_20261018/`). The one real level effect is the automorphism group: $\sqrt{3}$ on the $j = 0$ crater, confirmed at 1.86–1.92× (`research/drastic_prime_conductor_gap_round3_20261012/`). At $\ell = 2$, floors give 5.8% fewer relations than craters, and 99% of the contrast is explained by the base's cofactor classes (`research/isogeny-volcano-cost-v2_20261017/`, `research/generator_effect_ic_20261019/`).

Other transfer lines. Index calculus on genus-2 to -4 Jacobians over $\mathbb{F}_p$ (the ADH–Gaudry attack, measured against rho on the same Jacobian) read below a distinguished-point rho in early rounds, but the reference was later found slow and the ratios are lower bounds, unranked under the band (`research/notes/index-calculus/RESEARCH_HYPERELLIPTIC_IC_RHO.md`); later rounds are in review. A weak-curve atlas of 922 exact models and 36 published cases ranks covering-model and pairing signals but leaves the selected-subgroup advantage unmeasured (`research/weak_curve_leaderboard_20261012/`); three complete isogeny volcano graphs (24,823 vertices) were built and their graph-wide IC surveys halted at checkpoints (`research/full_volcano_graphwide_ic_20261017/`). Weil restriction of ECC2K-130 to extension fields costs at least 2^{131}, $2^{70.19}$ times rho (cluster "transfer-boundaries").

Named obstructions. Blocked requirements are kept open as problem records (`problems/README.md`), among them the complete-cost calibration of root-finding IC (`problems/rootfinding-complete-cost-calibration.md`), with exit criteria rather than abandoned.

12 What improved, and where the frontier stands

12.1 The measured frontier: one campaign, every cost priced

The frontier synthesis of 13 October assembled ratios from studies that used different curves, sizes and accounting. To replace it we ran one campaign (`research/ic_frontier_campaign_20261015/`, protocol frozen before any measured run). Every method family ran on the same fixed curve ladders: eight Koblitz curves chosen by subgroup size ($\log_2 r \approx$ 11–39, $n \in \{13, 17, 19, 23, 37, 41, 43, 47\}$), six NIST-shaped $a = -3$ prime-order curves of 20–40 bits, and the five 2-adic curves of the Kummer-tower

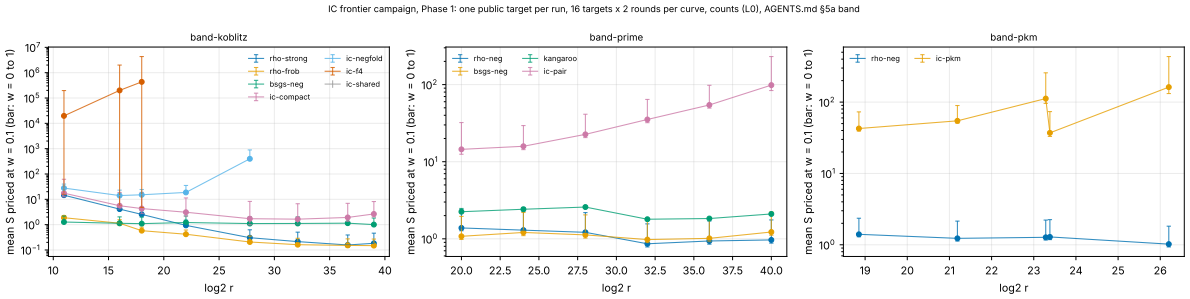


Figure 7: The campaign frontier: mean S per arm and curve at $w = 0.1$ GAE per native operation, the bar spanning $w = 0$ to $w = 1$; one public target per run, 16 targets × 2 rounds per curve, operation counts. Sources: `research/ic_frontier_campaign_20261015/analysis/band-*.json` (`ecbench band`).

Table 2: Index calculus over the cheapest constant-memory generic arm on the same curve (signed-Frobenius rho on Koblitz curves, rho with negation on prime curves), mean S ratio at three band weights. Every run verified (32/32 per cell). Source: `research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv`.

Curve	$\log_2 r$	Arm	$w = 0$	$w = 0.1$	$w = 1$
Koblitz $n = 13$	11.0	compact orbit	6.9	9.3	27.1
		negfold	14.4	14.8	17.5
Koblitz $n = 17$	16.0	compact orbit	3.8	4.9	11.8
Koblitz $n = 19$	18.0	compact orbit	4.9	7.5	23.6
Koblitz $n = 23$	22.0	compact orbit	5.5	7.4	18.4
Koblitz $n = 37$	27.8	compact orbit	5.2	8.3	23.3
		negfold	1806	1927	2512
Koblitz $n = 43$	32.1	compact orbit	7.4	10.2	23.1
Koblitz $n = 47$	36.6	compact orbit	9.9	12.7	25.3
Koblitz $n = 41$	39.0	compact orbit	15.0	17.8	30.3
prime 20 bit	20	pair table	9.8	10.5	13.5
prime 24 bit	24	pair table	12.1	12.2	12.9
prime 28 bit	28	pair table	18.6	18.6	18.9
prime 32 bit	32	pair table	40.9	40.9	41.1
prime 36 bit	36	pair table	58.1	58.0	57.5
prime 40 bit	40	pair table	95.2	101.8	131.4
PKM 20 bit	18.9	Kummer tower	30.3	30.4	31.0
PKM 28 bit	26.2	Kummer tower	141.3	158.7	238.7

ladder. Each curve had 16 public hash-to-subgroup targets nobody knows the logarithm of, 2 measured rounds and a warm-up, and every answer was checked outside the solver. That is 4,848 executions, 3,232 of them measured: 3,178 verified and 54 exhausted, all of the latter in the shared-rank arm at $n = 41$ and $n = 47$. Every session is audited by re-executing every measured run and requiring identical counts. Index-calculus parameters (compact-orbit columns, negfold subspace dimension, prime base size) were chosen per size on a disjoint set of training targets, with grids widened until the optimum was interior. Every cost is reported across the pricing band of §4.3.

What the campaign shows.

- *No index-calculus arm beats generic rho on any curve, at any weight.* Signed-Frobenius rho is below every index-calculus arm on all eight Koblitz curves and across the whole band; rho with negation is below every prime-field arm on all eleven prime curves (Table 2). The one unresolved pair in the campaign is compact orbit against the *strong* rho reference at $n = 13$, whose scalar-multiplication set-up dominates on an 11-bit subgroup; the cheaper signed-Frobenius walk is below both.
- *Compact orbit is the closest index calculus.* It sits 3.8–15× the best rho with native work free and 4.9–17.8× at $w = 0.1$; the ratio does not shrink with size.
- *Prime-field index calculus falls further behind with size.* The two-summand pair-table arm goes from 10.5× to 101.8× rho (at $w = 0.1$) between 20 and 40 bits, the $r^{1/6}$ widening its $r^{2/3}$ cost against rho's $r^{1/2}$ predicts; the Kummer-tower arm is 28–159× rho.

- *Negfold root finding stops scaling.* It is 12–45× rho up to $n = 23$, about 1,900× at $n = 37$, and does not complete within $5 \cdot 10^7$ trials from $n = 43$ on (PROTOCOL A4b).
- *Inherited F4 is not competitive once its work is priced*: S rises from 16 (word XORs free) to about $2 \cdot 10^4$ at $w = 0.1$ at $n = 13$. The band weighs one 64-bit word XOR like a table lookup, which overstates F4 (its pinned calibration is about 0.002 GAE per XOR); even with XORs free it is above rho.
- *BSGS has the lowest time on several curves* because it spends $\sqrt{r}$ memory; it is reported on the time–memory Pareto set, not ranked against constant-memory methods.

Cost exponents. Fitting total cost $\propto r^{\alpha}$ per arm over each size tier with `ecbench bound fit -weight` (two-stage bootstrap, 95% intervals; a tier needs at least four sizes for an exponent):
- prime pair table $\alpha = 0.620$ $[0.557, 0.663]$ at $w = 0$ against rho with negation 0.435 $[0.331, 0.501]$; separated at $w = 0$ and $w = 0.1$, overlapping at $w = 1$ (0.594 $[0.482, 0.665]$ against 0.444 $[0.334, 0.517]$). Index calculus grows faster than rho (the predicted 2/3 against 1/2), but under the band rule the difference is not established at every weight;
- Koblitz compact orbit, $n = 37$–47: 0.589 $[0.518, 0.743]$ against signed-Frobenius rho 0.436 $[0.358, 0.508]$ at $w = 0$, overlapping at $w = 0.1$ and $w = 1$; on $n = 13$–23 the two overlap at every weight (0.25 against 0.29 at $w = 0$, both with wide intervals because the automorphism count $2n$ changes with the size);
- Kummer tower 0.717 $[0.358, 0.940]$ against 0.438 $[0.332, 0.599]$, overlapping.

No index-calculus arm has a fitted exponent below rho's at any weight, so no crossover with rho is supported by these data, extrapolated or otherwise. The fitted rho exponents (0.435–0.444) lie below the theoretical 1/2, which reads as a toy-size effect (fixed per-run overheads and the size-dependent automorphism count) rather than a property of rho; so the measured widening of the gap is a fit on small sizes, and the conclusion that it widens as $r^{1/6}$ rests on the generic-oracle bound, not on these fits alone. The sixty-three fits are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

Limits. Operation counts only (isolation L0); wall time and a second-host replay certificate need the planned RunPod phase. Parameters were selected on four training targets per size. The shared-rank arm leaves three categories of work uncounted and is reported as a lower bound only. Fitted cost exponents per arm and tier are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

12.2 The engineering record

Table 3 collects the measured improvements of the programme on the boundary each was measured on. Ratios are baseline/candidate; "stage" means one oracle call or one solver run; "complete" means a verified discrete logarithm with the stated phases.
Derived entries: $53.0 = 1/0.018861$; $17.5 = 1/0.057047$; $1.21 = 1/0.828$ with interval from $[0.783, 0.863]$; $18.3 = 1/0.0545$ with interval from $[0.047, 0.064]$; $1.77 = 300{,}021.9/169{,}069.8$; $1.20 = 1/0.830$ with interval from $[0.697, 0.962]$; $1.87 = 18.714/9.985$ (one session; the compact arm's S is a lower bound, so the ratio is an upper bound); 1.69 and 25.2 compare means from different sessions with different seeds and are descriptive; F5 and F6-IC ratios are reciprocals of the recorded candidate/baseline word-XOR ratios.

12.3 The improvement ladder, 14–15 October

The repository's owner fixed the criterion for success on 15 October: a verified, measured improvement over the current index-calculus baseline is a success; each one becomes the new baseline; the gap to the matched rho is reported as the distance remaining, not as a pass/fail gate; and relabelling or accounting changes do not count. In the two days before that, two improvement rounds and a next-steps campaign landed about twenty measured changes, each paired against its own incumbent under a protocol frozen before data. Figure 8 and Table 4 list them, with the distance to rho that remains after each one. None is in the frontier campaign of §12.1, which predates them, and none was priced across the band; the boundaries are the ones the studies used, which differ from row to row.

Genus three: an exponent below rho's. The cubic GHS transfer [15] maps the DLP in $E_0(\mathbb{F}_{q^3})$ to the Jacobian of a genus-3 hyperelliptic curve over $\mathbb{F}_q$, where double-large-prime index calculus [16] costs $\tilde{O}(q^{2-2/g}) = \tilde{O}(q^{4/3})$ against rho's $q^{3/2}$. That is known; what is new is a priced, end-to-end, verified

Table 3: Measured improvements, July–October 2026. Class: E = ENGINEERING, A = ACCOUNTING. Boundary: St = stage, Cx = complete solve outside ecbench, Cold = ecbench cold one-target S, On = online window after retained set-up. Sources: Appendix A.

Change	Bnd.	Cl.	Ratio	Scope
Decomposition oracle: root finding				
S_3 roots, binary pair table vs matrix F4	St	E	383–1,535	$\mathbb{F}_{2^9}$, $\mathbb{F}_{2^{13}}$, $m = 2$
S_3 roots, prime vs `groebner_f4`	St	E	3.5–4.7·10^4	16–24-bit, $\lvert\mathcal{F}\rvert = 6$
S_3 roots vs Buchberger, NIST curves	St	E	3,021–5,802	P-192…P-521
`rf-mitm` vs PS-0 oracle	St	E	3,007–115,160	binary $\ell = 8..14$
Root-finding DLP vs msolve DLP	Cx	E	50–105	16–24-bit prime
Pair-table DLP vs F4 DLP	Cx	E	122	binary $n = 23$ (F4 wins $n \leq 13$)
Pair table vs subtraction	Cold	E	53.0	20-bit prime (bounded)
Frobenius pair table vs subtraction	Cold	E	17.5	Koblitz $n = 17$, priced
Root-finding K13 arm, first to best	Cold	E	1.69	38.82 → 22.97, across sessions
Windowed $m = 3$ scan	Cold	E	1.21	$[1.16, 1.28]$
Negfold root finding vs XOR-priced F4	Cold	E	18.3	$[15.6, 21.3]$, K13
Dense root index K1798 vs K1500 (probes)	On	E	1.77	$n = 83$, one point
Pair-table size 64 vs 32 abscissae	Cold	E	1.20	$[1.04, 1.43]$
Gröbner engines				
Inherited vs matrix F4 (matrix work)	St	E	198.6–222.1	K13, R13
F5 vs matrix F4 (word XORs)	St	E	1.55–1.56	K13, R13
F6-IC vs inherited F4 (word XORs)	St	E	1.22–1.30	$m = 13, 19$
`f4_fp` new vs previous	St	E	29.8–54.1	square systems, = msolve
Four-Russians routing	St	E	1.75–3.22	oracle matrices
Eager vs lazy inherited rewrite	St	E	1.40–1.62	dim 8–9
Quotient-curve $m = 2$ model	St	E	1.34–1.64	five cells
Factor bases and symmetry				
Compact orbit vs full orbit	Cold	E	≤ 1.87	Koblitz $n = 17$; compact is LB
FGHR D_4 vs Gaudry S_4, C_4	St	E	47.5–48.4	$\mathbb{F}_{269^4}$
`gaudry_quartic` solve	St	E	1.96–6.08	$d = 4..6$ (sweep recompute)
One-target online				
Compact-orbit IC vs strong rho	On	E	13.94	$n = 61$; rho without a table; retired (§12.3)
Accounting (costs that went up when priced)				
F4 word XORs priced	Cold	A	1/25.2	16.75 → 421.4, K13
F6-IC batched geometry priced	Cold	A	1/64.6–1/1,353	$m = 13, 19$; F4 unpriced (about 1/2.5 priced)

Table 4: Changes measured 14–15 October 2026 against their own incumbents. Ratio: candidate/incumbent, < 1 better. Class, against the index-calculus incumbent (not the paper's ADVANCE, which is against rho): Ie = improves the exponent, Ic = improves the constant, E = ENGINEERING on one stage, R = regresses, T = trades, A = ACCOUNTING. "To rho": the distance left to the matched rho on the study's own boundary; "d" marks a derived or extrapolated value. Sources under `research/`: `ic_next10_20261014/`, `ic_improve_20261014/`, `ic_improve_20261015/`, `ic_improve3_20261015/` and the named study directories.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer, $E_0(\mathbb{F}_{q^3})$, cost in $\mathbb{F}_q$ multiplications			
Double large primes vs full base (04)	Ie	0.036 $[0.035, 0.036]$ at 2^{18}; $q^{1.910} \to q^{1.385}$	2.58× negation rho model at $q = 2^{20}$
Large primes as columns, SGE, Wiedemann (01)	Ie	0.653 $[0.647, 0.660]$ at 2^{20}; $q^{1.398} \to q^{1.380}$	—
$\kappa(q)$ schedule + block Wiedemann (06)	T	0.929 at 2^{21}	1.15× measured negation rho at 2^{20} (ref.)
Cheaper relation primitives + re-tuned κ (r3-01)	Ic	0.617 $[0.616, 0.619]$ at 2^{22}	no matched rho in this unit
$k = 4$ over $\mathbb{F}_{p^4}$, operation counts			
D_4 Y-line vs plain S_4, $p = 101$–1033	Ic	0.0548 $[0.048, 0.061]$	handover 2^{126}–2^{137} (d)
D_4-weighted S_5, four unknowns, vs five	E	1/18.5–1/17.8	handover $2^{135.6} \to 2^{118.9}$ (d)
First-dependency stop (`d4-ed`)	Ic	0.979 $[0.973, 0.986]$	no $k = 4$ run below rho
Double large primes, tuned	R	1.025 $[1.016, 1.037]$ (LA 0.827)	—
Koblitz, compact orbit and Gröbner oracle			
Free relations in production, $n = 41, 53, 61$	Ic	0.960 $[0.946, 0.976]$	3.8–15× rho-frob ($w = 0$, §12.1)
Linear-row harvest, $(23, 12)$, $m = 2$	Ic	0.858 $[0.803, 0.916]$	oracle stage; default on
Root-rank harvest gate, $(29, 15)$, $m = 2$	Ic	1/1.41 $[1/1.52, 1/1.32]$; end to end 1/1.20	oracle stage; default on
Gated fall-row augmentation	Ic	1/11.9 on barrel $n = 13$; 1 in production	production no-op
Equivariant Macaulay oracle	T	4.0 ($n = 5$) … 1/15.5 ($n = 13$)	not applicable to production
`stride-renew` targets, $n = 13$	E	0.531 $[0.506, 0.557]$	4.7× rho-strong at $n = 19$
Cyclotomic γ-weight base, K1	R	1.022 $[1.021, 1.023]$	yield-neutral
Trace-zero pre-filter	R	1.00003	vacuous on odd order
Prime fields and shared stages			
PKM isogeny tower, search charged	A	1.002–1.331	17–137× rho
Binary MITM pair-table oracle, $\ell = 8$	E	1/277–1/94 per target	no exponent claim
Filter + Wiedemann LA, $D = 2^{18}$	E	1/10.40; 1/1.93 vs CADO-NFS	LA is 0.1–1.9% of runs

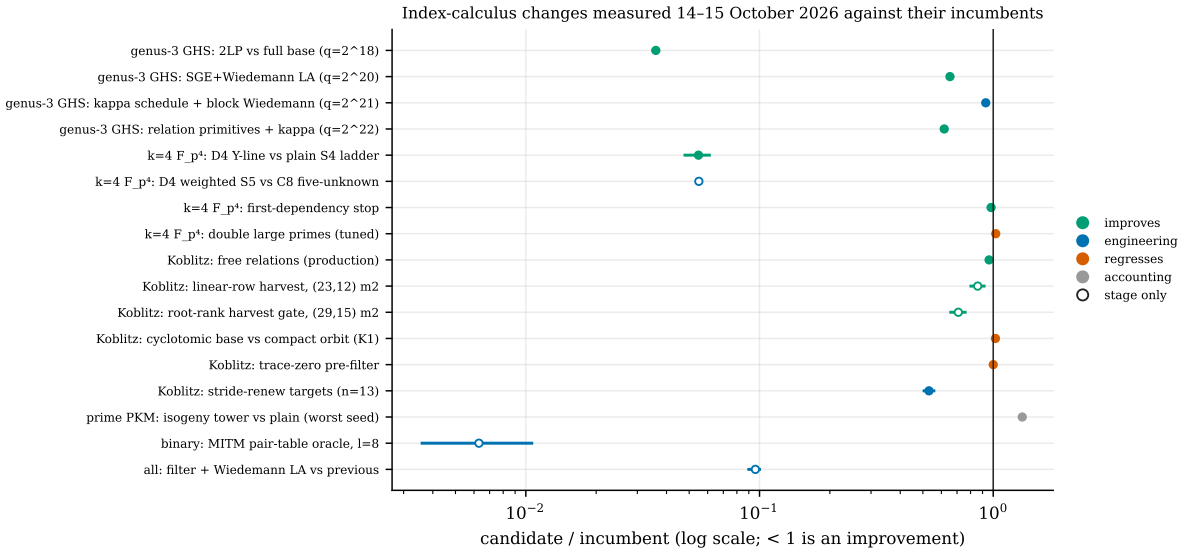


Figure 8: Candidate over incumbent for the changes of 14–15 October, with 95% intervals where the study reports one (the binary oracle's range is the spread over three runs). Filled: whole solve; open: one stage. Values: `papers/ecdlp-index-calculus-program/figures/ladder.tsv`; sources in Table 4.

run of it on a family the repository can build ($4 \mid \#E_0(\mathbb{F}_{q^3})$, split arm, $q = 2^8$–2^{22}, every logarithm checked against the planted scalar). Adding double large primes changed the exponent from $q^{1.910}$ to $q^{1.385}$ $[1.379, 1.391]$, and keeping the large primes as columns, merging them with structured Gaussian elimination and finishing with Wiedemann moved it to $q^{1.380}$ and the constant to 0.653×. This was the first index-calculus family in the programme whose fitted exponent lies below rho's (the $k = 4$ binary family of §12.4 is the second), and the only one where measured IC beats a rho arm: the item-7 rho, which walks at $1.75\sqrt{\ell}$ steps, at $q = 2^{15}$ (0.69 $[0.55, 0.83]$). Two further rounds kept going. A frozen schedule for the factor-base fraction $\kappa(q)$ plus block Wiedemann brought the whole solve to 0.605× the first double-large-prime arm at $q = 2^{21}$. Cheaper relation primitives (one-inversion Cantor addition, a Stickelberger pre-filter, reused exponentiations) then cut the relation phase to 0.47× without changing a single relation, and with κ re-tuned the whole solve to 0.617 $[0.616, 0.619]$ at $q = 2^{22}$, in a unit that also charges inversions. The distance to rho shrank from 2.58× the negation-map rho model at $q = 2^{20}$ (first double-large-prime arm) to 1.15× a measured negation-map rho there after the second round ($2.31 \cdot 10^{10}$ against $2.01 \cdot 10^{10}$ $\mathbb{F}_q$ multiplications; a reference on overlapping but not identical curves). The third round's 0.62× has no matched rho in its unit yet, so whether it crosses is not established. The local exponents sit at 1.38–1.41, still above 4/3, because the linear algebra grows faster than $q^{4/3}$ and the tuned κ falls with q. The family is narrow: the transfer needs the right 2-power torsion, and $\mathbb{F}_{q^3}$ curves are not deployed. *Next improvement*: a negation-map rho measured in the third round's unit on the same instances, then the run past $q = 2^{22}$ that would show a crossover or rule it out.

The $k = 4$ chain. The D_4 Macaulay system of §10.9 entered a complete ladder ($p = 101$–1033, 21 curves): 18.2× cheaper than plain S_4 at the same exponent, with the handover with negation rho extrapolated to 2^{126}–2^{137}. Writing the D_4-symmetrised S_5 in four weighted unknowns cut the per-solve cost another 17.8–18.5×, to $1.42 \cdot 10^9$ $\mathbb{F}_p$ multiplications, 854× below the plain form; in the 2-torsion frame that moves the derived handover from $2^{157.9}$ to $2^{118.9}$. Double large primes cut the linear algebra (0.827×) but regress the whole solve; stopping at the first dependency gains 2.1%. No $k = 4$ run below rho exists; every handover is an extrapolation of fitted exponents through a measured constant. *Next improvement*: put the four-unknown solve into the complete ladder, then keep double large primes' linear-algebra gain without its relation-phase cost.

Koblitz. Free relations—root-table self-collisions used as homogeneous rows—are now the production default (0.960× counted, 0.941× wall); the linear rows the Gröbner oracle used to discard are harvested (0.86–0.93× where the gate fires, and a root-rank gate then made it fire on more cells: 1/1.41 at (29, 15), 1/1.20 end to end); the gated fall-row and equivariant passes pay only on overdetermined systems the production pipeline does not build. The cyclotomic base and the trace-zero filter were refuted. A census

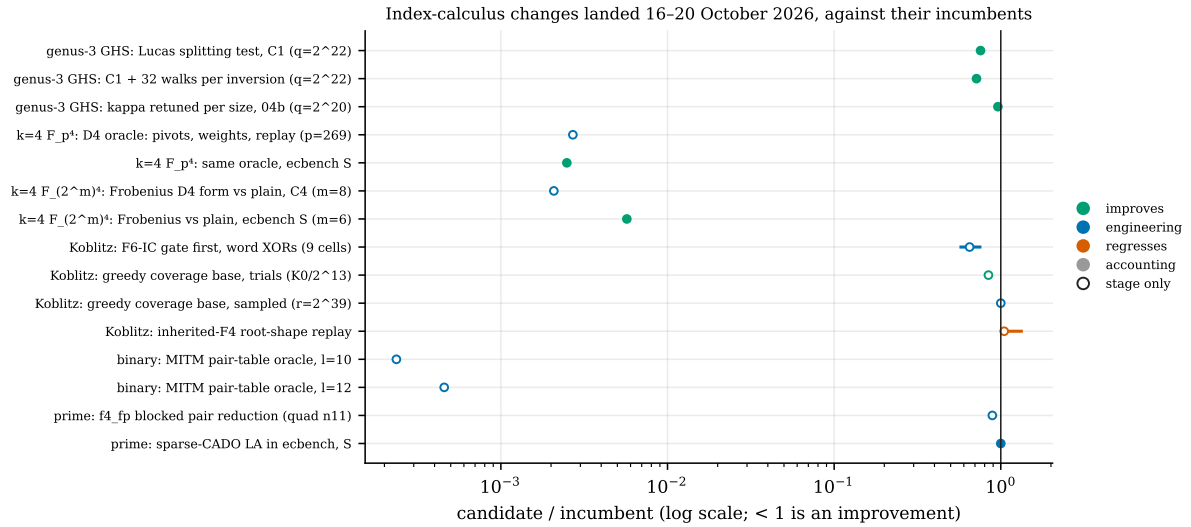


Figure 9: Candidate over incumbent for the changes that landed 16–20 October, as in Figure 8. Values: `papers/ecdlp-index-calculus-program/figures/ladder2.tsv`; sources in Table 5.

closes the free-relation lever at the rungs run: at $n = 61$, $K = 600$ the self-collisions span 49 independent relations of the 599 needed, and full rank would need $K \approx 2{,}050$. The distance to signed-Frobenius rho is still 3.8–15× cold at $w = 0$, and the fitted Koblitz ratio grows as about $r^{1/6}$ (slope 0.165 $[0.117, 0.213]$ per $\ln r$ on eight public curves, every native priced). *Next improvement*: relation yield per probe, which none of these changes touched.

The online result is retired. An equal-advice audit gave Bernstein–Lange preprocessing rho [2] a distinguished-point table no larger than the index-calculus pair table and ran both on the same 64 unseen targets at $n = 41$, 53 and 61. Rho with the table is 53–317× faster online (at $n = 41$, with exactly equal advice, 317 $[137, 739]$), and even at 1/1,024 of the bytes it stays 7.7–32× faster. The 13.9× online win of §8 and the later 7.3× and 12.3× hosted single-target points at $n = 71$ and $n = 83$ compared index calculus *with* a table against rho *without* one. They are measurements of the preprocessing model, not index-calculus improvements, and the online frontier belongs to preprocessing rho.

The reference moved too. Two rho studies shortened the distances above. On prime curves, choosing the better of $P + R$ and $P - R$ at each step takes 1.129× fewer steps under distinguished points (1.081–1.184); each step costs more, so the predicted net gain is 1.03–1.12× with 8–64 walks per inversion and a net loss (0.95) at 1,024. On ECC2K-130, signed σ steps cut GPU time per solve to 0.961–0.970 (derived; not rolled out). Neither is yet in a campaign arm. A comparison against a rho that has not adopted them would overstate index calculus's position by that much.

12.4 The ladder continued, 16–20 October

Work that reached `main` between 16 and 20 October, after the survey was first assembled (some of it measured on 14–15 October), added a genus-3 round, two $k = 4$ oracles, a long binary-oracle ladder, Gröbner-engine work and a repository-wide validation sweep (Figure 9, Table 5). The reading rules of §12.3 apply: each row is against its own incumbent, on its study's boundary, unpriced across the band.

Genus three, round four. A field-level splitting test (a Lucas ladder on the Cardano resolvent's cubic character, with the same relations) removes most of the per-trial log factor (9.75 to 2.71 units per bit) and moves the whole solve to 0.756× at $q = 2^{22}$; with 32 walks sharing one inversion, to 0.715×, with local exponent 1.356 (fitted 1.368 over 2^{16}–2^{22}), 0.023 above 4/3. Chained with round three, the whole solve is about 0.44× the second round's at $q = 2^{22}$ (derived, in the inversion-charging unit). Rounds three and four ran no rho arm; a 23 October study then measured round 4 against the strongest rho at $q = 2^{14}$–2^{20}: under the registered pricing it is below rho at 2^{16} at every band weight and unresolved at the other sizes (geometric means 0.70–1.52), and above rho under the protocol's heavier sensitivity

Table 5: Changes that landed 16–20 October 2026, against their own incumbents; columns and classes as in Table 4. Sources under `research/`: `ic_improve4_20261015/`, `ic_improve_20261014/04b-genus3-2lp-rebalance/`, `gaudry_quartic_d4_cost_20261014/`, `gaudry_quartic_d4_char2_20261015/`, `joux_vitse_n1_20261015/`, `binary_mitm_oracle_l9_l10_20261015/`, `binary_mitm_oracle_l11_l12_20261016/`, `ic_m3_algebraic_vs_mitm_20261014/`, `secg_small_x_20261014/`, `greedy_factor_base_yield_20261014/`, `f6_gate_first_x86_20261016/`, `pdp_shape_replay_20261015/`, `f4_fp_blocked_pairs_20261016/`, `ic_linalg_pipeline_20261019/`, `k_conversion_20261018/`.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer			
Lucas splitting test (round 4, C1)	Ie	0.756 [0.755, 0.756] at 2^{22}; local 1.408 → 1.364	no rho arm since round 2
C1 + 32 walks per inversion	Ic	0.715 at 2^{22}; local 1.356	as above
Factor base retuned per size (04b)	Ic	0.961 at 2^{20}	0.16× rho on Jac(C) at 2^{14}, not on E_0
$k = 4$ over $\mathbb{F}_{p^4}$ and $\mathbb{F}_{(2^m)^4}$			
D_4 oracle: pivots, weighted form, replay, $p = 269$	E	1/370 [1/371.6, 1/367.9]	—
Same oracle, ecbench S	Ic	1/402 [1/406, 1/398]	$1.39 \cdot 10^6$× rho
Frobenius-reduced D_4 form, characteristic 2, $m = 8$	E	1/481 [1/483, 1/479]	—
Same, ecbench S ($m = 6$)	Ic	0.0057	6.1× rho model, 4.3× measured, at $n \approx 2^{56}$
Koblitz and binary oracles			
F6-IC gate-first order	E	1/1.73–1/1.34 word XORs	default on
Greedy coverage base, $K_0/2^{13}$	Ic	0.843 (expected trials)	gain 1.000 at $r = 2^{39}$
Inherited-F4 root-shape replay	R	1.015–1.328	switch stays off
Binary MITM oracle, $\ell = 9..12$	E	1/4,244–1/928 per target	relation collection $\Theta(2^n)$
Algebraic $m = 3, 4$ oracles vs MITM floor	R	slope 0.53–0.81 vs 0.21–0.36	$3 \cdot 10^2$–$3 \cdot 10^4$× floor
Prime fields and shared stages			
`f4_fp` blocked pair reduction	E	0.78–0.89 (wall, quad $n10, 11$)	beats msolve there
Sparse-CADO LA inside ecbench	E	S 1.000; LA wall 0.25–0.72	149× rho
S_3 solve on SECG small-x bases	—	table 2,600–313,000× cheaper	52–280 bits over rho
Column-touch conversion K (ECC2K-130 $m = 4$)	A	+7.62/ + 10.32 → +9.78/ + 12.48 bits ($w = 1$ floor; +45.71 at $w = 2$, unconverted)	accounting

pricing; §11). Re-tuning κ on training again steepened the schedule and raised the exponent. A separate rebalance study measured IC at 0.16× (0.25× renormalised) a rho on $\mathrm{Jac}(C)$ at $q = 2^{14}$: that rho walks on the Jacobian (four walks per size, at 1.6–2.8 steps per $\sqrt{\ell}$ against an ideal 1.25), not on $E_0(\mathbb{F}_{q^3})$, so it is not the matched reference and the ratio is not a distance to rho. Memory now limits the ladder: $q = 2^{22}$ peaks at 3.9 GB of a 4.5 GB cap.

$k = 4$: two oracles and a closing gap. Over $\mathbb{F}_{p^4}$, sparsest pivots, the weighted grading $\sigma_4 = \pi^2$ and shape replay cut one D_4 decomposition from $2.55 \cdot 10^{10}$ to $6.89 \cdot 10^7$ $\mathbb{F}_p$ multiplications (370×; 648/648 solves exact) and ecbench S by 398–406×; index calculus is still $1.39 \cdot 10^6\times$ rho at the measured sizes and the model handover moves from 2^{135} to 2^{103}. In characteristic two, a Frobenius-reduced form of the same system (weights $(2, 2, 1, 1)$, Macaulay degree 11, a 384×448 matrix) is 481× below the plain form. End to end on $E(\mathbb{F}_{(2^m)^4})$ the ratio of IC to the negation-map rho model falls from 5,080 at $m = 6$ to 6.06 at $m = 14$ ($n \approx 2^{56}$; 4.3× a measured rho), with fitted slope -0.248 in $\log_2 n$—Gaudry's $n^{-1/4}$—and an extrapolated crossover near $n \approx 2^{68}$ (2^{63}–2^{69} under the model). After genus three, this is the nearest any index calculus in the programme has come to rho, and its extrapolated crossover lies where rho itself takes about 2^{34} steps, so the extrapolation can be tested directly. A Joux–Vitse $(n-1)$-decomposition at $n = 5$ on $E(\mathbb{F}_{p^5})$ completed verified DLPs at $4.5 \cdot 10^7$–$1.8 \cdot 10^8\times$ rho, with a crossover extrapolated to groups near 2^{291}.

Binary oracles and their floor. The MITM pair-table oracle on a non-Koblitz curve reached $\ell = 12$ over $\mathbb{F}_{2^{37}}$ (2,474–4,244× over pairs-and-solve at $\ell = 9, 10$; 928–2,321× at $\ell = 11, 12$, the last as recomputed by the validation sweep; 0 disagreements on 256 targets). Against that floor, no algebraic oracle (inherited F4, F6-IC, SAT, split S_4 hybrids, a wide chained S_3 solver) grows more slowly in n: the cheapest is $3 \cdot 10^2\times$ the floor at $n = 13$ and $3.8 \cdot 10^4\times$ at $n = 31$ for $m = 3$, and the $m = 4$ chain grows at slope 0.805 against the floor's 0.333. A greedy coverage base needs 1.08–1.18× fewer trials on small Koblitz curves, but its gain falls to 1.0000 at $r = 2^{39}$. On the 33 SECG curves an S_3 solve over a small-x base, with no table, is 52–280 bits over rho.

The reference moved again. The strong signed-Frobenius rho of every Koblitz comparison executed 718 instructions per walk step after three rounds, down from 1,196; a fourth round of batched inversion took main from 707.2 to 679.7 Callgrind instructions per step at $n = 53$, with no established native-time gain (outputs identical; `research/rho_batchinv_20261022/`; steps per ideal 0.92–0.99 on 400 fixtures), and ecbench's standing `rho.signed_frobenius_strong` runs at 0.802× its former instruction count with identical GAE. On hyperelliptic Jacobians the programme's rho was slow at large N until full-width branch coefficients brought S_{walk} from 2.10 to 1.48; the negation fold saves only 0.97× at $N \geq 10^5$. Counted-unit comparisons (S) are unaffected by the instruction counts; wall-time comparisons against the older rho overstate index calculus by up to the same factor.

Corrections from the validation sweep. A repository-wide validation sweep (19–20 October) contradicted five recorded results, all now struck through in their sources with dated notes. Three bear on this paper. (i) The one-point $n = 83$ online ratio (18.5×, §6) and the hosted $n = 71/n = 83$ panels (7.3×, 12.3×) compared against rho with an empty table; with a Bernstein–Lange table matched to the index calculus's set-up time or memory, precomputed rho is faster in every matched $n = 83$ pair (3.7× time-matched and 40.6× memory-matched against $K = 3300$; 82–396× against $K = 600$). They join the $n = 61$ point as retired. (ii) The root-finder session's replay-all, recorded as 120/168 identical with the 48 mismatches confined to inherited F4, replays 72/168 on current `main`, with mismatches in all four IC arms, root finding included. The sweep later isolated the cause: an x86-only BMI2 rewrite in inherited F4 charges a different word-XOR count from the portable path, and with it disabled x86 and emulated aarch64 both replay 168/168 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). F4 counts are therefore comparable across instruction sets only when the path is stated, and until the method identity records it the root-finder rows rest on the recorded receipt. (iii) The equal-bytes preprocessing control at $n = 73$ and $n = 83$ is built and frozen but not run. *Next improvement* (from the 18 October leads review): a large-memory host. The two exponent-level questions named then—the Weil-descent cell $(m, \ell) = (4, 6)$ and the PKM tower at $m = 3$, $N = 21$—have since run on rented hosts: the $(4, 6)$ cell refuted at degree 4 at $N = 54$, and the PKM cell finished degree 7 and stopped on memory inside degree 8, implying $D \geq 8$ but read as inconclusive (§§9.1,10.2). The $(3, 7)$, $\ell = 8$ and $m = 4$, $N = 20$ cells remain.

12.5 The ladder continued, 21–24 October

Work that reached `main` between 21 and 24 October falls into four groups: factor bases closed under cheap known-logarithm maps, a paired re-run of the Koblitz leaderboard recipe, a guess-and-determine attack on the Weil-descent degree question, and transfer measurements, among them a third family whose measured gap to rho is now single-digit (Table 6). The reading rules of §12.3 apply: each row is against its own incumbent and on its study's boundary. The companion ledger (`papers/ecdlp-ic-improvement-ledger/`) records the same period as a ratchet of incumbents; this section keeps to what moves the frontier or corrects it.

One mechanism, four factor bases. The week's measured gains on the index-calculus side are all the same idea: put a point and its images under maps of known logarithm—the Frobenius τ on a Koblitz curve, the CM endomorphism ϕ, small integer multiples, the cofactor $[h]$—into one column with a known coefficient, so the linear-algebra stage needs fewer relations. On Koblitz curves the $\mathbb{Z}[\tau]$ base needs $K+1$ relations instead of about one per orbit: child CPU of the complete process falls 1.9–4.4× at $n = 41$–61 and 6.2–10.5× at $n = 71$. These are measured against the repository's own tuned producer, on a host at load about 20 (child CPU is the less affected number), with three targets up to $n = 61$ and one at $n = 71$. The prime CM base ($j = 0$, $j = 1728$, $D = -7, -8$) cuts S 10–20×, and the integer-multiple base on random binary curves 3–4×; both at 36–40-bit or degree-26–28 sizes, where the best bases have 3–10 columns over 512–3,480 points and, as their sources note, tend to a baby-step giant-step table over multiples as the column count falls. The cofactor-projected base (LEAD-19) is the only one measured on leaderboard rows, with 20 repeats and the order stable across the band: 1.29–3.36× in cold S, leaving 6.1–14.4× to matched rho. Every source states that the exponent is unchanged; the gain is bounded by the size of the multiplier set and shrinks as linear algebra's share of the cost shrinks. The known-coefficient columns are those of GGMP's Frobenius-orbit bases [12] (§8); what is new here is extending them from automorphisms to small-norm and integer multipliers. Whether that extension is in the literature has not been checked. All four are ENGINEERING, and none is measured against a published index calculus; the $\mathbb{Z}[\tau]$ source notes that the summation-polynomial methods have no table-based counterpart at these parameters.

Koblitz: the recipe is near its optimum, the distance is larger. Rule v4 retuned columns, collection window and unit size on disjoint training seeds and re-ran Table A paired against rule v3 (32 pairs per size, 16 seeds, A/A and a positive control that detects a 10–15% slowdown). One size improves (0.953 $[0.904, 0.998]$ at $n = 57$) and five are not separated, so it is not a Table A improvement under its own rule. The free-relation lever costs 5–9% in this pipeline. More consequential is the reference: with rho priced in the same process at the canonical batched step, the cold distance is 9.9–43×, 1.76–1.94× the as-built figure that rule v3 used. The ledger's rule v8 reads 4.67–18.07× on the same sizes in its own unit; the two are different domains in the sense of Definition 1 and are not combined here.

Weil descent: a hybrid closes one cell, the degree question stays open. The $(m, \ell) = (3, 7)$ cell had no refutation at degree 4. Fixing one or two variables restores degree 4: all branches refute with $k^* \le 2$ on 10 of 10 unsatisfiable draws, at 1/116–1/260 of the projected degree-5 cost, in 3–6 GB (planted-solution and placebo controls passed). For the ECC2K-130 $m = 4$ row this changes less. Its often-quoted "+10.32 bits over rho if degree 5 holds" prices one column touch per Macaulay column ($w = 1$), a floor no elimination beats; at $w = 2$, the exponent the measured closure tracks, the same row is +45.71 bits, and the measured conversion $K = 4$–16 rho steps per touch raises both (§12.4). With the hybrid, the row is +8.43 bits if the number of guesses stays flat in ℓ and +58.27 bits if it grows linearly; one value of ℓ cannot separate them, and the run at $\ell = 8$ that would did not complete. In no scenario does the row go below the degree-4 floor of +7.62 bits, so the route does not reach rho.

Prime fields: further behind. A new 30-bit Kummer-tower row in ecbench (160/160 verified, loaded host) is 158.9× $[121.6, 221.4]$ rho at $w = 0$, against 141× at 28 bits in the campaign; its source calls it "a measurement, not an improvement". Two falsifiers for the Certicom 131-bit programme (random-b binary degrees of regularity; structured bases) were null. A frozen-protocol test of a Coppersmith lattice on S_3 recovered no planted decomposition in any of 96 cells with $\beta \ge 0.20$ at 24–56 bits, with positive controls passing below, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3.

Table 6: Changes that landed 21–24 October 2026, against their own incumbents; columns and classes as in Table 4 (N = null or diagnostic, no cost row). Sources under `research/`: `ic_improve_20261023/` (`README.md`, `PRIME_README.md`, `BINARY_README.md`), `lead19_projected_base/`, `ic_tool_program/rule/v4/`, `dreg_hybrid_guess_20261024/`, `jv_cover_p_scaling_20261024/`, `hcdlp_trial_cost_20261024/`, `ghs_cover_census_n4_20261024/`, `coppersmith_s3_theta2_20261023/`; and `docs/ic/CERTICOM131_IC_PROGRAM.md`.

Change	Cl.	Ratio [95%]	To rho
Factor bases closed under known-log maps			
$\mathbb{Z}[\tau]$ small-norm multipliers, Koblitz $n = 41$–61	Ic	1/1.9–1/4.4 child CPU (3 targets, loaded host)	not measured
Same, wide path, $n = 71$	Ic	1/6.2 equal points; 1/10.5 retuned (1 target)	not measured
$\mathbb{Z}[\phi]$ multipliers, prime CM curves, 36–40 bit	Ic	1/10–1/20 in S	7–12×
Integer-multiple closure, random binary, degree 26–28	Ic	1/3.1–1/3.9 vs plain	6–12×
Cofactor-projected base (LEAD-19), $\mathbb{F}_{2^{15}}, \mathbb{F}_{2^{21}}, \mathbb{F}_{2^{27}}$	Ic	1/1.29 [1/1.52, 1/1.12], 1/1.58 [1/1.82, 1/1.35], 1/3.36 [1/4.02, 1/2.80]	6.1, 11.5, 14.4×
Koblitz leaderboard recipe (Table A, $n = 41$–61)			
Rule v4 recipe vs v3, paired, six sizes	Ic	0.953 $[0.904, 0.998]$ at $n = 57$; five sizes not separated	9.9–43× at the canonical step
Weil descent, degree question			
Hybrid guess, $(m, \ell) = (3, 7)$, $k^* \le 2$ guesses	E	1/116–1/260 of projected degree-5 closure	—
Implied ECC2K-130 $m = 4$ row (derived)	N	+8.43 (flat k) to +58.27 bits (linear k)	vs +10.32 ($w = 1$), +45.71 ($w = 2$)
Prime fields			
Kummer tower, 30 bit (new row)	N	—	158.9× $[121.6, 221.4]$
Coppersmith lattice on S_3, $\ell \le 3$	N	0/10 recovered in all 96 cells with $\beta \ge 0.20$	ceiling $\beta = 1/4$ stands
Transfer and covers			
Joux–Vitse cover of $E(\mathbb{F}_{p^6})$, $p = 53$–1009	N	slope −2.05 $[-2.28, -1.83]$ in $\ln p$	1.59–1.98× at $p = 1009$ (LB)
Measured trial cost, genus-255 binary Jacobian	A	103 curve additions per trial ($2^{6.69}$)	$N = 130$ margin $+4.71 \to -1.83$ bits
GHS covers of $E/\mathbb{F}_{q^4}$, odd q, census	N	genus floor 13 on odd-order curves	model exponent gap 0.154

A third family approaches rho: the Joux–Vitse cover. The genus-3 cover of $E(\mathbb{F}_{p^6})$ over $\mathbb{F}_{p^2}$ with Nagao decompositions (full rational 2-torsion; a weak class, not prime-order curves) ran at $p = 53$–1009, three seeds per prime, every logarithm checked. The gap to rho falls from a geometric mean of 620× at $p = 53$ to 1.73× at $p = 1009$ (1.59–1.98), with fitted slope -2.05 $[-2.28, -1.83]$ in $\ln p$ against the cost model's -2; the line meets rho near $p \approx 1330$ ($[1170, 1540]$, an extrapolation). Two limits keep it a lower-bound comparison: the cover arm does not emit its unpriced F4 and Wiedemann work, so it cannot be ranked under the band; and rho cannot be run at $p \geq 503$ (subgroups of 2^{52}–2^{58}), so those ratios use the pooled rho constant $S = 1.344$ measured at $p \leq 251$. If that constant kept falling at its (insignificant) point trend, the ratio at $p = 1009$ would be about 20% higher. A separate sieve arm on the same cover read far lower ratios, but its comparisons are withheld by review (§11).

Covers: censuses and a repriced counterfactual. Measuring the cost of one relation trial (a Cantor addition plus a smoothness test) on a genus-255 binary Jacobian gives 103 additions of a point over $\mathbb{F}_{2^{131}}$ ($2^{6.69}$), which turns the $N = 130$ GHS counterfactual's modelled margin over rho from +4.71 bits to −1.83 bits; only a trial below about 29 additions would preserve it, and the smoothness test alone costs 16.5. Two censuses fix what covers exist at toy sizes: for $E/\mathbb{F}_{q^4}$ in odd characteristic the genus floor is 13 on odd-order curves ($2 - 2/g$ against rho's 2 leaves an exponent gap of 0.154, constants excluded), and for binary curves over $\mathbb{F}_{2^{3l}}$ the genus-3 class is a 2^{-l} fraction (0.78% at $l = 7$).

The validation gate. On 24 October the leaderboard's gate ran for the first time: all 13 Table B rows (eight prime, five random binary) passed—every counted field replays identically, every scalar checks in an independent verifier, and IC stays above rho at every weight of the band (4.05–54.76×)—and all nine Table C rows failed; Table A was not in scope (`research/ic_validation_gate_20261024/`). The study staged its manifest rather than installing it, and the leaderboard now marks the Table B rows validated. The ledger paper, current to 23 October, records that no row had passed; for Table B that is now out of date. Validation says the numbers are what the archive says and the reference is fair, not that index calculus approaches rho.

What is still not measured. No arm in the programme has yet been compared with a published index calculus at the same (q, n, m): FGHR-style symmetrised summation polynomials over the standard base $\{P : x(P) \in \mathbb{F}_q\}$, solved by msolve, or WDSat for binary decompositions. Every ratio in Tables 4–6 is against rho or against the programme's own previous arm. Defining and running that reference arm (open problem 10) is the precondition for any claim of improving on the state of the art.

12.6 Three frontiers

Cold, one target, set-up included. Unchanged: generic rho. Every measured index-calculus arm is above its matched rho (Figure 10); the closest complete count is root-finding negfold at 1.55× strong signed-Frobenius rho at $n = 13$; the closest lower-bound arms are 1.08× (compact orbit with a counted Frobenius table) and 6.97× (matrix F4/F5 against rho-negation at K13). Prime-field arms are 12–239× rho and diverge with size (the 30-bit Kummer-tower row of §12.5 is 159×). Off the prime and prime-degree binary fields, three families are within a single-digit factor, all in weak classes: genus three after cubic GHS (below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, above it under a heavier sensitivity pricing), $k = 4$ over $\mathbb{F}_{(2^m)^4}$ (4.3× a measured rho at $n \approx 2^{56}$, closing as $n^{-1/4}$), and the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (plain Nagao arm; 1.59–1.98× a pooled rho constant at $p = 1009$, closing as about p^{-2}; a lower-bound comparison).

Online, after retained set-up. Generic again: Bernstein–Lange preprocessing rho. The admitted compact-orbit point at Koblitz $n = 61$, 13.9× $[9.8, 19.1]$ over strong rho without a table (cold median 83.4× rho), is 53–221× slower online than preprocessing rho given equal or less advice (§12.3), and the $n = 83$ points lose every pair run with rho given a table matched on set-up time or memory (§§12.4,8.5).

Stage. Moved substantially: the decomposition oracle is two to five orders of magnitude cheaper than at the start of the programme on the workloads where root finding applies, the $\mathbb{F}_p$ F4 is at msolve parity, the Boolean F4 family is one to two orders of magnitude cheaper in counted matrix work, and the $k = 4$ extension-field solve is 48× and then a further 370× cheaper over $\mathbb{F}_{p^4}$ (481× below the plain form in characteristic two). None of this, by itself, moves the cold frontier: the stages it leaves untouched—relation yield, factor-base size, linear algebra, table set-up—now dominate.

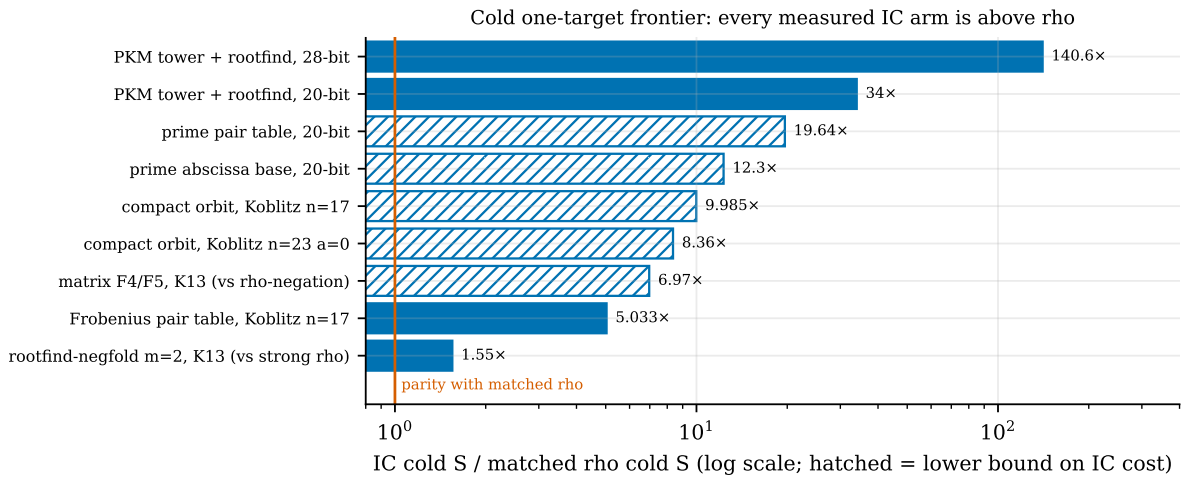


Figure 10: Cold one-target S of index-calculus candidates over their matched rho arm (each study's own estimator; curves differ, so bars are not ranked against each other). Hatched: lower bound on IC cost (unpriced native or solver work). Sources: `research/ecbench_frontier_20261013/`, `research/ecbench_rootfinding_assessment_20261012/`, `research/pkm_tower_ecbench_20261013/`, `research/ecbench_rootfind_faster_20261014/`.

12.7 What would move the cold frontier

From the frontier synthesis and the problem records: (i) register the torsion-symmetrised bases and the $n = 83$ dense-root engine in ecbench under immutable method identities and emit cold and online records; (ii) resolve the inherited-F4 cross-host counter drift before using that counter in a bound; (iii) run a four-size native challenge against the standing rho bound—two sizes cannot establish an exponent; (iv) obtain an isolated cold session at $n = 83$ with every set-up phase and retained byte charged; (v) price F6-IC's batched geometry with a pinned conversion. Since then three items have moved to the front: (vi) run $k = 4$ over $\mathbb{F}_{(2^m)^4}$ at $m = 15$–17 against a measured negation-map rho, where the fitted trend predicts parity near $n \approx 2^{68}$; (vii) measure the matched rho on $E_0(\mathbb{F}_{q^3})$ in the genus-3 rounds' unit (done on 23 October: below rho at 2^{16} under the registered pricing, unresolved elsewhere, §11); (viii) obtain a large-memory host for the $(4, 6)$ Weil-descent cell and the PKM tower. The 21–24 October work adds two more and re-weights one: (ix) measure rho on the Joux–Vitse cover family at $p \geq 503$ (or a size where it can be run) and emit the cover arm's unpriced counters, so that the p^{-2} trend can be ranked under the band and tested past $p \approx 1330$; (x) define and run a published-index-calculus reference arm (open problem 10); and (viii) is worth less than it looked for the binary question, since the $m = 4$ row at $w = 2$ is about +46 bits over rho even if degree 5 holds (§12.5). A result in the cold domain that beats rho at four sizes would be the first ADVANCE of the programme.

13 Open problems and proposed experiments

1. The solving degree of Weil-descended summation systems. Every subexponential claim for prime-degree binary curves rests on a bound for this degree (§3); none is proved and the experimental evidence is contested. *Experiment:* measure first-fall, last-fall and solving degree of S_{m+1} restricted to ℓ-dimensional subspaces over a grid of (n, ℓ, m) with $m \geq 3$, at least four values of n per (ℓ, m), against the semi-regular prediction, with multi-graded regularity (one block per summand) as a second model. *Settles:* whether the degree grows with n at fixed (ℓ, m); a growth consistent with zero would revive the Petit–Quisquater and Semaev heuristics, a positive slope would close them for this model.

2. Cold index calculus against rho with every operation priced. No measured arm beats matched rho on the cold one-target boundary once native work is priced on both sides (§12). *Experiment:* the frontier campaign of §12 extended to the largest sizes each arm can reach, with exponents fitted per arm and the crossover reported only where the fitted exponents separate across the whole pricing band. *Settles:* whether any implemented arm has an exponent below $1/2$ in r on these families.

3. Relation yield is now the dominant stage. Root finding made the decomposition oracle cheap; what remains is the number of trials per relation, governed by the factor-base density. *Experiment:* for each factor-base family, the measured trials-per-relation law against r and base size, compared with the counting prediction $m!\,r/N^{m-1}$; the residual (e.g. the unexplained factor 5 in the ECC2K-130 extrapolation, §8) is the target.

4. Root finding beyond two summands on prime curves. The two-summand oracle is prior art in mechanism (McGuire–Mueller) and costs $\Theta(N)$ per trial; three summands cost a pair table of N^2 entries. *Question:* is there a sub-N^2 structure for S_4 membership over a prime-field base—for example from the D_m-invariant coordinates of 2-torsion curves—and does it change the $r^{2/3}$ cost of two-summand prime index calculus? A negative answer with a counting argument would be as useful as a positive one.

5. The preprocessing regime. Index calculus with a retained table beats rho without one, and loses by 53–317× to Bernstein–Lange rho with an equal or smaller table (§12.3). *Experiment:* measure the advice–time curve (S, T) of the compact-orbit arm across sizes against preprocessing rho's $S\,T^2 \approx N$; an index-calculus online result counts only if it lies below that curve at equal S.

6. Symmetry at $k = 4$, end to end. The D_4 reduction and its four-unknown weighted form move a *modelled* handover over $\mathbb{F}_{p^4}$ from about 2^{150} to 2^{119}–2^{137} depending on frame (§12.3). Since then the odd-characteristic oracle is 370× cheaper again (index calculus still $1.39 \cdot 10^6\times$ rho), and in characteristic two a Frobenius-reduced form brings the measured gap to 6.1× rho at $n \approx 2^{56}$, closing as $n^{-1/4}$ (§12.4). *Experiment:* continue the characteristic-two ladder to $m = 15$–17 with a measured negation-map rho at every rung, cold and priced across the band; a measured crossover near $n \approx 2^{68}$ would confirm the fit, and its absence would locate the term the model misses.

7. Pricing and calibration infrastructure. Rankings now require every arm's native work to be counted (AGENTS.md §5a). *Work:* count the three uncounted categories of the shared-rank arm; calibrate every ladder size on one fixed host; record the inherited-F4 code path (BMI2 or portable) in its method identity, now that the cross-host counter drift is traced to it; register the $n = 83$ dense root index in the harness.

8. Genus three after cubic GHS, to the crossover. The one family whose fitted exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$ was 1.15× a measured negation-map rho at $q = 2^{20}$ before rounds three and four cut a further 0.44× (§§12.3,12.4), and is now below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, and above it under a heavier sensitivity pricing, with no measurable pooled trend (§11). What would settle it is the same comparison at $q = 2^{21}$–2^{24} on instances of one cofactor, with the subgroup multiplication's price measured rather than chosen; its local exponent is down to 1.356, still above 4/3 because the linear algebra grows faster than $q^{4/3}$, and $q = 2^{23}$ needs more than the 4.5 GB cap. *Experiment:* measure negation-map rho on the same instances in the latest unit (inversions and mod-ℓ multiplications priced), and run both past $q = 2^{22}$, in the harness and across the pricing band. A measured crossover there would be the programme's first ADVANCE against rho, on a narrow and undeployed family.

9. Plane quartics: Diem's $\tilde{O}(q)$. Genus-3 GHS covers whose Jacobian is that of a smooth plane quartic admit Diem's index calculus [4, 6] at $\tilde{O}(q)$, against $\tilde{O}(q^{4/3})$ for the hyperelliptic route and $q^{3/2}$ for rho on $E(\mathbb{F}_{q^3})$. The repository builds such quartics directly ($m = 2$ GHS, toy index calculus at $q \le 11$) and through Smith's $(2, 2, 2)$-isogeny from the hyperelliptic case (18.6% of random curves); a four-workstream programme was scoped on 18 October (`research/plane_quartic_program_20261018/PROGRAM.md`). *Experiment:* the hyperelliptic ladder's instances, carried to plane quartics where Smith's conditions pass, with Diem's relation search and the same linear algebra, against the same matched rho. The existing toy plane-quartic S values are unverified (their degree-3 reduction merges distinct classes; §4.5), so the arithmetic must be rebuilt first; the 23 October literature round ranks this branch as its one exponent candidate (LEAD-15, `research/pdp_literature_leads_20261022/`).

10. A published index calculus as the reference arm. Every ratio in this paper is against rho or the programme's own previous arm; none is against the state of the art in index calculus at the

same (q, n, m) (§12.5). *Work:* fix, per curve and size, the standard base $\{P : x(P) \in \mathbb{F}_q\}$, the FGHR-symmetrised summation polynomial (2-torsion and, on Koblitz curves, Frobenius invariants), msolve as the solver and WDSat for binary decompositions, and run it in ecbench on the existing ladders (binary $n = 13$–37, Koblitz $n = 41$–61, $k = 4$ extension fields). *Settles:* whether any engineering result of §§6–11 improves on published index calculus, as opposed to on this programme's own baseline. Stage ratios quoted against the in-house F4 say nothing about published Gröbner engines until this is done.

11. The Joux–Vitse cover, to the crossover. The cover of $E(\mathbb{F}_{p^6})$ closes on rho as about p^{-2} and is 1.6–2.0× a pooled rho constant at $p = 1009$, with the line meeting rho near $p \approx 1330$ by extrapolation (§12.5). *Experiment:* emit the cover arm's unpriced F4 and Wiedemann counters, measure rho at the largest p it can run (or with a memory-light rho at $p \geq 503$), and run both at $p \approx 1300$–2000, across the band. A measured crossover would be the programme's first ADVANCE against rho, on a weak and undeployed class; its absence at the predicted p would locate the term the cost model misses.

14 Conclusion

Sixteen weeks of work produced three kinds of result. The first is infrastructure: a single unit and harness in which rho, BSGS, the kangaroo and every index-calculus variant are charged the same way on the same one-target workloads, with sealed records, exact replays and admitted intervals; and a ledger that refuses a claim with a missing field. The second is a set of constant-factor improvements, the largest of which—the closed-form roots of S_3—began as one sentence in a manuscript we were asked to check, and ended as a proved identity, an oracle family covering prime and binary fields, two to four summands, CPU and GPU, plain, tower and dense-index bases, and the default decomposition strategy of the codebase. The third is a map of what does not work, which is longer than the list of what does: SAT that does not scale, symmetry that pays at $n = 17$ and costs at $n = 31$, transfers that provably cannot change decomposition counts, fibre-aware bases whose gap to rho grows as $\#E^{1/6}$, literature attacks that are exhaustive search.

By family the record reads as follows. On *Koblitz curves* (§8) every September result below rho reversed against a stronger rho, the $n = 83$ online wins were retired by a matched preprocessing rho, Table A sits at 4.67–18.07× (rule v8) or 9.9–43× (rule v4 recipe, canonical step), and every constructed or priced route to ECC2K-130 is at least about 5 bits above rho (a hypothetical genus-130 Jacobian would not be, but none is known). On *random and prime-degree binary curves* (§9) the solving degree of Weil-descended systems grows with ℓ and tracks the semi-regular prediction on every toy cell measured, while the first and last fall degrees stay at 3–4; no algebraic oracle beats the pair table; and the five validated Table B rows are 10.6–38.2× rho. On *prime fields* (§10) the eight validated Table B rows are 5.5–13.5× rho and rising, the Kummer tower's solving degree reaches 7 and is at least 8 at $N = 21$, 24 rounds of structured bases on the P-256 field found no relation for an unplanted target, and walks over a million curves isogenous to P-256 fired no structural screen. The single-digit gaps all belong to weak extension-field classes (§11).

The honest summary is the campaign of §12.1. On the question the survey asks—does index calculus solve one unseen elliptic-curve target, cold, for less than rho?—the answer on every curve we measured, at every weight of the pricing band, is no. The closest arm is a compact Frobenius-orbit Koblitz index calculus at 4–18× signed-Frobenius rho in the campaign (separately, the leaderboard's Koblitz pipeline is 9.9–43× at 36.6–47.2 bits once rho is priced at the canonical step), prime-field index calculus falls behind as $r^{1/6}$, and no fitted exponent of an index-calculus arm lies below rho's. The online 13.9× at $n = 61$ compared a table against no table and loses 53–317× to preprocessing rho with equal advice; it is retired.

Measured against its own baseline, which is the criterion the programme now works to, index calculus keeps improving: about twenty changes landed on 14–15 October, among them a genus-3 index calculus after cubic GHS whose exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$, an 18× cheaper $k = 4$ ladder and a further 18× cheaper $k = 4$ solve, and production free relations on Koblitz curves (0.960×). The genus-3 family is the nearest to rho: 1.15× a measured negation-map rho at $q = 2^{20}$ after the second round; after rounds three and four, a preregistered comparison put it below the strongest rho at $q = 2^{16}$ at every band weight under the registered pricing, unresolved at the other sizes to 2^{20}, and 2.7–12.4× rho under a heavier sensitivity pricing of one operation. That is the programme's only decision of index calculus below the strongest rho; it is in a weak structured class, depends on that pricing, and is not yet reviewed. The second is $k = 4$ over $\mathbb{F}_{(2^m)^4}$, whose gap closes from 5,080× to 6.1× rho between $n \approx 2^{24}$ and 2^{56} at

the $n^{-1/4}$ the theory predicts; its extrapolated crossover near 2^{68} is the first in the programme at a size where rho itself can be run, so it is the next experiment. Both families are known to be weak in theory (Gaudry, Diem, GHS); what the programme adds is a measured, verified approach to the crossover, and neither is a deployed curve shape. The validation sweep of 19–20 October also retired the remaining online wins at $n = 83$ and reopened a replay regression in the root-finder session, since traced to an x86-only code path in inherited F4.

The week to 24 October added a third family of the same kind: a Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (plain Nagao arm) whose gap closes as about p^{-2} and is 1.6–2.0× a pooled rho constant at $p = 1009$, with the crossover extrapolated near $p \approx 1330$; measuring rho and the cover's unpriced work at that size is, with the $k = 4$ run near 2^{68}, a direct test of a predicted crossover. It also added four factor bases closed under known-logarithm maps, worth 1.3–20× against the programme's own arms and nothing in the exponent, and a re-priced rho that nearly doubled the Koblitz leaderboard's distance. The stages that now dominate are relation yield, factor-base size and linear algebra. One thing the paper cannot yet say is how any of this compares with published index calculus at the same parameters: every ratio here is against rho or against the programme itself. Building that reference arm (open problem 10) comes before any claim of improving on the state of the art, and the other open problems of §13 say what would move the stages that dominate.

A Artefact index

Paths are relative to the root of the `crypto` repository. Each study directory holds its protocol, raw records, run reports with graphs and a PDF; ecbench studies also hold sealed sessions and audit receipts.

Measurement
`docs/ecbench/README.md`; `src/cryptanalysis/ecbench/`; `docs/ic/BOUNDARY_TARGETS.md`; `docs/ic/boundary_targets.json`; `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`; `docs/bounds/frontier.json`; `research/ecbench_frontier_20261013/`.

Root finding
`RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`; `papers/rootfinding-decomposition/`; `research/ecbench_rootfinding_assessment_20261012/`; `research/semaev_root_vs_groebner_20261012/`; `research/semaev_rootfind_nist_20261012/`; `research/semaev_rootfind_e2e_20261012/`; `research/pdp_root_finder_20261012/`; `research/pdp_root_finder_ecbench_20261013/`; `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`; `research/ecbench_rootfind_engine_20261012/`; `research/ecbench_rootfind_improve_20261013/`; `research/ecbench_rootfind_parity_20261013/`; `research/ecbench_rootfind_faster_20261014/`; `research/ecbench_rootfind_vs_rho_20261014/`; `research/ecbench_rootfinding_tuning_20261013/`; `research/ecbench_rootfinding_arity_20261013/`; `research/ecbench_rootfinding_batchtable_20261013/`; `research/ecbench_rootfinding_lanes_20261013/`; `research/n83_dense_root_20261013/`.

Gröbner engines
`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`; `research/f4_f5_f6_cross_family_20261011/`; `research/inherited_f4_eager_rewrite_20261011/`; `research/f4_f5_fused_readback_20261012/`; `research/f4_f5_hash_20261012/`; `research/f4_f5_dense_packing_20261012/`; `research/f4_f5_row_build_20261011/`; `research/f4_fp_msolve_parity_20261011/`; `research/gb_cas_comparison_20261011/`; `research/ic_semiregular_law_20261005/`; `research/f6_ic_ecbench_ladder_20261005/`.

Koblitz
`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`; `RESEARCH_ECC2K130_IC_FEASIBILITY.md`; `docs/ic/COMPACT_ORBIT_PRODUCER.md`; `research/ecbench_factor_bases_20261013/`; `research/koblitz_symmetrised_e2e_20260927/`.

Prime and extension fields
`research/pdp_oracle_speedup_20261011/`; `research/pdp_split_dreg_20261011/`; `research/pkm_tower_ecbench_20261013/`; `research/gaudry_quartic_d4_20261013/`; `research/gaudry_quartic_speed_20261013/`; `research/lit2026_speedup_program/`; `research/lit2026_w12_flagged_claims_20261011/`; `research/lit2026_w11_generic_engineering_20261011/`.

Transfer and isogeny
`docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md`; `docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`; `RESEARCH_ISOGENY_FIBERS_ECDLP.md`; `research/ecbench_conductor_gap_20261012/`; `research/isogeny_conductor_gap_round3_20261011/`; `research/fiber_ic_scaling_20261011/`; `research/isogeny_loop_glv_20261012/`.

21–24 October

`research/ic_improve_20261023/`; `research/lead19_projected_base/`; `research/pdp_literature_leads_20261022/`; `research/ic_tool_program/rule/v4/`; `research/dreg_hybrid_guess_20261024/`; `docs/ic/CERTICOM131_IC_PROGRAM.md`; `research/coppersmith_s3_theta2_20261023/`; `research/jv_cover_p_scaling_20261024/`; `research/hcdlp_trial_cost_20261024/`; `research/ghs_cover_census_n4_20261024/`; `research/ghs_n3_genus3_census_20261024/`; `research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`; `papers/ecdlp-ic-improvement-ledger/`.

This paper

`papers/ecdlp-index-calculus-program/figures/data.json` (every plotted value with its source); `figures/make_figures.py`; browsing copies in `benchmarks/20261014-ic-program-paper/`.

B Work in open pull requests

At the paper's date, 111 open pull requests in the repository concern elliptic-curve index calculus or its baselines. Their results are not on `main`, have not been reviewed here, and are not used anywhere in the paper; the titles below are quoted as their authors filed them and are claims, not findings. Two bear on the text: #2359 reports a plane-quartic IC/rho of 0.024, but the plane-quartic S values on `main` are unverified because their degree-3 reduction merges distinct classes (§4.5); and #2324 records that a Joux–Vitse sieve replication failed its acceptance rule and withheld its decisions. As these merge, they are folded into the sections above and the catalogue under the same rules as everything else. Source: `papers/ecdlp-index-calculus-program/data/open_prs.tsv`.

PR	Updated	Title as filed
#2514	2026-10-24	Record N83 Spot replay failure and AWS resume custody
#2513	2026-10-24	isoclass-rho: same-host repeat runs of P0 and K (data only)
#2511	2026-10-24	R6.1: direct S5 systems, chained-x ic oracle plug-in, discrepancy protocol
#2509	2026-10-24	Twist IC: polynomial zero-fiber collection with independent pair and rank audits
#2505	2026-10-24	Hyperelliptic IC rounds 13–15: the negation fold reaches its $\sqrt{2}$ on the full-width rho walk and becomes the reference
#2504	2026-10-24	isoclass-rho: 46-bit replica of the prime-field class (data, null result)
#2503	2026-10-24	Weil-descent closure A3 on the EKS high-memory runner, Spot-loss resume, and xmem runner pools
#2502	2026-10-24	R6.9 GHS phase two: isogeny walk cost and relation rate on binary Jacobians
#2497	2026-10-24	Audit a noisy local N83 five-base cold screen
#2494	2026-10-24	N83 summand sweep m=2..9 across factor bases and F4/F5/F6/SAT engines; fail-closed wide-field descent
#2493	2026-10-24	Algebraic IC ladder: linear two-summand S3 decomposition, slope 0.65 vs F4's 1.08, n up to 41
#2490	2026-10-24	Factor D18 shared root bits and audit the frozen N83 comparison
#2489	2026-10-24	Refresh index-calculus dashboard for current curve registry
#2488	2026-10-24	Run frozen N83 shared-binary panel on capacity-gated Spot runners
#2479	2026-10-24	curves: bind cover report and IC roster to a cover-catalog subset of the registry
#2470	2026-10-24	Record exploratory Certicom 131-bit rho and S3 stage benchmarks
#2465	2026-10-23	ecbench accounting: v2 method ids, automorphism-aware floor, scaled strong-rho reference
#2462	2026-10-24	rho.negation_lanes (batched-inversion rho, 1.5–2.1× faster wall) and prime IC lane prefetch
#2460	2026-10-24	ecbench verify and db: tamper-evident, fail-closed audit and index
#2441	2026-10-24	ecbench compare: add –weight (unblocks prime-base-oracle-tuning card)
#2437	2026-10-24	Certify exact ECCp-131 structured bases and auxiliary tower
#2433	2026-10-24	ecbench: Cheon DLP-with-auxiliary-inputs methods with a counted static-DH oracle
#2431	2026-10-23	Round 2 identical-output speedups: Koblitz IC set-up 0.80–0.92×, prime rho 0.90–0.95×
#2430	2026-10-24	ECDLP across a large prime conductor gap on a deep 2-volcano: theorems, Lean, and a preregistered rho measurement
#2429	2026-10-24	ECC2K-130 descent: four guessed variables bring every $\ell = 8$ draw back to degree 4
#2427	2026-10-24	IC orbit-layout and generator-orbit optimizations; n29 like-for-like crossing, validator band unresolved
#2426	2026-10-24	research(pkm): Kummer tower at P-224's Solinas 2-adic shape, priced against matched rho
#2422	2026-10-24	Measure Cheon's attack on P-224-shaped curves (7 sizes, both variants, matched rho)
#2420	2026-10-24	Stream isogeny class progress and resume SQLite snapshots from S3
#2416	2026-10-24	Resume icx runs at completed repeats from local and S3 state
#2415	2026-10-24	Stream icx progress and resume isogeny walks from local or S3 state
#2407	2026-10-24	ecbench: parse the curve registry once per process; allocate counter keys once
#2384	2026-10-24	research: native representation joins and verified IC row adapter
#2383	2026-10-23	Benchmark target-blind known-log recovery on arbitrary toy-curve targets
#2381	2026-10-24	Quadratic twists: prove relation-yield/rank transport and conditional IC cost bounds
#2380	2026-10-24	Verify twist gluing geometry and run matched IC relation controls
#2375	2026-10-24	Run the priced inversion and residue tests; one self-check policy for bitop/v1 (accounting)

PR	Updated	Title as filed
#2371	2026-10-24	IC leaderboard rigor: recompute, reference-currency and replay checks, drift bisect, Table B validation
#2370	2026-10-23	Add archive IC symmetry options and verified useful rank
#2369	2026-10-24	Factor-base designs: Amendment 8, the m = 4 cells on the EKS high-memory runner (registered, ready to dispatch)
#2368	2026-10-24	Twist IC: complete 189+21 controls and independent zero-fiber rank audits
#2366	2026-10-24	Factor-base designs: retain m=4 runs and correct memory-limit attribution
#2359	2026-10-24	N4 iteration 2: double large primes cut the plane-quartic IC cost to 0.22-0.36 of the baseline (IC/rho 0.024 at q = 1,048,507, GHS-weak family)
#2357	2026-10-24	Repair Lean proof gates and refresh cover-transfer evidence
#2356	2026-10-24	Export exact ecbench plan documents and repair runner dependencies
#2344	2026-10-24	Tuned GAE unit (proposed) and a fast, step-efficient prime rho reference
#2333	2026-10-23	research: census of the NIST binary and prime-curve seed runs for skipped candidates; register the skipped m = 277 Koblitz curve
#2332	2026-10-24	volcano-rho: negation map by default; audit of every rho walk
#2329	2026-10-24	T21: wide producer arithmetic cuts (no 1.2x on the K=600 scan) and early-stop factorise (8 s per process start)
#2324	2026-10-23	JV sieve vs strong rho, replication V2: thresholds fixed before data; acceptance rule A4 failed, decisions withheld
#2304	2026-10-24	Publish genus-2 follow-up cohorts and review packet
#2302	2026-10-23	Plain S5 C4 re-measured with #1792's solver options: 1.21e12 → 8.69e10 (13.9×); cost.rs updated, no registry verdict changes
#2297	2026-10-23	Known-log factor base and E1 dry run (generator-effect study)
#2286	2026-10-24	R10: pinned IC baseline ledger per curve family and size; round claims re-rated against it
#2282	2026-10-24	N83 K1120: padd rank stage (cryptanalysis #970 port) cuts cold time 12.5 s (10.4%); paired panel vs matched rho
#2275	2026-10-23	H2 settled: genus-2 IC on glued h=1 instances does not beat negation rho in any of 12 price cells
#2256	2026-10-23	Track weighted Weil-restriction experiments on larger curves
#2254	2026-10-24	Explore small chart grafts with a native factor-base support census
#2253	2026-10-23	Sync IC semantic mirror and generated curve indexes
#2250	2026-10-23	Support ECC2-130/ECCp-130 solvers on CPU, CUDA and Metal
#2246	2026-10-23	Audit N83 factor-base gates and S3-custody finalist evidence
#2231	2026-10-23	Optimize Boolean F4/F5 construction and recursion; compare F6 and root solvers end to end
#2229	2026-10-23	Run fresh end-to-end checks of three ECDLP leads, including two 1,260-target genus cohorts
#2227	2026-10-23	W1: canonical Pic^0 arithmetic on smooth plane quartics
#2224	2026-10-23	jv_cover: large-prime variation (L6 of #2216) does not pay at p <= 12k
#2223	2026-10-23	ecbench: prime-field PKM tower oracle; tower F4, signature F5 and an F6 analogue on prime curves (preregistered round)
#2215	2026-10-24	Koblitz IC: integral recovery and proper source equations
#2206	2026-10-24	crypto volcano: native CLI, $\Phi_\ell(X,j_0)$ builder, tall-volcano sampler; level studies (generic rho equivalent across levels within ±5%)
#2205	2026-10-23	Conductor gap round 6: early-stopping F4 and the trace coset on every Koblitz curve
#2204	2026-10-24	Record bounded N83 K16627 m4 natural query and replayed controls
#2202	2026-10-24	Verify N83 K16627 m4 selector-only phase freedom
#2192	2026-10-24	R5.3: three-prime NTT for isogeny_eval; √élu crossover re-measured
#2191	2026-10-24	IC: two orbit columns under the lazy oracle: cheaper on 1 of 5 curves, dearer on 4 (stacked on #2158)
#2187	2026-10-24	papers: isogeny-class rho historical measurements, checked algebra and review limits
#2186	2026-10-24	isoclass-rho: whole-class Pollard rho study — protocol, run records, report, benchmarks entry (null result)
#2185	2026-10-23	Conductor gap round 5: the cofactor penalty is self-inflicted (cleared base 0.28–0.71× the fiber-closed table; best cell 7.29× rho)
#2182	2026-10-23	The faster f4_fp (#1717) in the end-to-end Jacobian-cover index calculus: S 5.0× lower, same relations
#2170	2026-10-24	Weil descent and Weil restriction: Teske variants, the isogeny census, four measured angles, and round-4 follow-ups (genus-2 IC on prime-order E/F_{p^2})
#2168	2026-10-23	ECDLP pass: exponent assessment, leads and their runs (L1–L5), ecbench report/sweep, comprehensive generic sweep
#2167	2026-10-24	Verify N83 K16627 m4 selector-pinned constructed relation
#2166	2026-10-24	N83 K16627 m4 post-root search progress and capped outcome
#2165	2026-10-24	N83 K16627 m4 root fixpoint and capped post-root search
#2160	2026-10-24	research: large-memory IC runs on a 124 GB host (PKM m=3 N=21; shape G1 frozen cells)
#2159	2026-10-23	dreg closure slope (undecided), L1 closed, eliminated-S4 oracle 5-6x at toy cells, Lean for Lemmas 1-2, L2 and K-233 rows
#2158	2026-10-24	IC: lazy orbit-member index in the Frobenius-folded pair table (1.12–1.44× cheaper, engineering)
#2156	2026-10-23	IC leads assessment with tracked follow-ups (T01–T17)
#2155	2026-10-23	Refresh the IC registers: G2 record, lit2026 statuses, benchmarks index, generic-bound column
#2152	2026-10-24	GLV-invariant IC, G2: k = 4 end to end (E19), the V4 line (E20–E24, 35–240× below D4) and GLS on the V4 line (E25)

PR	Updated	Title as filed
#2148	2026-10-23	Run bounded controls for the 20-item ECDLP audit
#2147	2026-10-24	N83 K16627 m4 complete-witness phase control and capped result
#2141	2026-10-24	Blind-spot campaign: twenty frozen ECDLP index-calculus experiments
#2140	2026-10-24	N83 K=16,627 m=4 planted relation control and capped receipt
#2138	2026-10-24	Measure N83 K16627 retained-base m4 model capacity
#2124	2026-10-24	Construct and retain a 16,627-column N83 symmetric factor base
#2121	2026-10-24	N83 retained structured base: bounded m4 gate and m5 control
#2119	2026-10-24	Enable checked four-summand chained S3 for retained N83 bases
#2106	2026-10-23	Verify full selected-order C and deliver 7937-vertex volcano survey checkpoint
#2103	2026-10-23	Add backend frontiers with pinned reproduction contracts
#2101	2026-10-23	Extend Rust IC plug-ins and harden shared crypto primitives
#2092	2026-10-24	Screen N83 root11 30-bit indexed IDs through K5086
#2084	2026-10-24	Seal fresh N83 root11 fine-grid confirmation fixture
#2036	2026-10-24	N83: indexed XOR, validated arena reclamation and bounded native continuation
#2035	2026-10-24	Freeze N83 finalist holdout public points
#2022	2026-10-24	Screen N83 root22 against root11 at cold frontier
#1978	2026-10-23	ic programme R10–R12 and the rule at v6/v7: baselines v4–v8, 1.90–1.98× faster cold, distance to rho 3.43–11.51×
#1970	2026-10-23	bitop/v1: one unit of work for every ECDLP method, solvers and linear algebra charged
#1853	2026-10-24	ecbench: ten IC questions and versioned orbit/shape follow-ups
#1830	2026-10-24	Construct binary genus-2 covers and add full-subgroup rho comparisons
#1828	2026-10-23	Measure matched degree-83 IC versus optimized rho and publish unified report
#1790	2026-10-23	Measure Koblitz six-sum extraction and end-to-end bottleneck against rho
#1740	2026-10-23	N53 fixed-base S3 root batching: nine verified one-target pairs

C Catalogue of the programme's studies

This appendix lists every study directory under `research/` on `main` at the paper's date, every benchmark row under `benchmarks/`, every open-problem record under `problems/`, and the research notes and `docs/ic/` programme documents, 1444 entries in all, grouped by curve family. Campaigns of many directories are collapsed to one row, with every member path listed under it. Each row gives the study's own headline number with its class and status (the study's own class:
emphexponent vs own incumbent is a change of exponent against the index-calculus incumbent, not the paper's
clsadvance); a number marked superseded or corrected must be read through the correcting path. Rows were compiled from each study's README and results files and are summaries, not re-measurements; where a study states no number, none is given. Generated tables (leaderboard rows, the registry) are summarised through `docs/ic/LEADERBOARD.md` rather than listed file by file. The machine-readable source is `papers/ecdlp-index-calculus-program/data/catalogue.tsv`, built by `papers/ecdlp-index-calculus-program/tools/catalogue.py`, which also checks that no directory is missing.

C.1 Koblitz curves

519 entries: 42 campaigns collapsed from 474 entries, and 45 single entries.

Study	Class	Result	Status
decomposition oracle bounds and solver panels (13 entries, 2026-09-12..2026-10-06)	accounting	13 members. No oracle below 2^131 per family; corrected bottom 2^124.99 (2^64.18x rho); relation sweeps best +13.59 bits; Crossbred/RR/WDSat engineering only.	current
Members: `notes/ecc2k130/RESEARCH_ECC2K130_CROSSBRED.md`, `notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION.md`, `notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION_RUNS.md`, `notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION_TARGETS.md`, `notes/ecc2k130/RESEARCH_ECC2K130_RELATION_SWEEPS.md`, `notes/ecc2k130/RESEARCH_ECC2K130_RR_SOLVER_PANEL.md`, `notes/ecc2k130/RESEARCH_ECC2K130_WDSAT.md`, `notes/ecc2k130/RESEARCH_KOBLITZ_INDEX_CALCULUS.md`, `notes/ecc2k130/RESEARCH_KOBLITZ_SCALING_TARGET.md`, `notes/ecc2k130/RESEARCH_TRIMOSKA_BENCHMARKS.md`, `notes/ecc2k130/RESEARCH_WDSAT_IC_UNIFICATION.md`, `notes/ecc2k130/WDSAT_CHILD_STATUS_PROTOCOL_20260925.md`, `notes/ecc2k130/WDSAT_CHILD_STATUS_RESULT_20260925.md`			
early Groebner oracle engineering and accounting (6 entries, 2026-09-14..2026-09-25)	engineering	6 studies; suffix-XOR kernel 1.39x; matrix-F4 stage 1.597x word XORs (holdout 1.577x); instruction rounds 1.42-1.43x giving S 964x rho (K0_2^13); polynomial cache no end-to-end gain; degree naming corrected.	current
Members: `degree_reporting_20260922`, `f4_linear_algebra_20260914`, `groebner_base_sweep_20260915`, `groebner_instructions_20260925`, `groebner_stage_20260915`, `polynomial_reuse_20260914`			

Study	Class	Result	Status
AutoLab instruction-cost campaigns on small Koblitz (3 entries, 2026-09-15)	engineering	3 members; first campaign 26.9% fewer cold Ir, second 47.72% fewer (0.522831359); final candidate still 4.98x matched rho instructions.	current
Members: `ic_autolab_evidence_20260915`, `ic_autolab_harness_20260915`, `ic_autolab_next30_20260915`			
IC candidate tournament and its rho control (4 entries, 2026-09-15..2026-09-23)	accounting	4 members; tournament best 0.5347 of worker rho Ir (round 0010); triple table 0.409 at n43a1; rho with IC canonical form beats IC at all five cells (1.19-2.88x): sub-rho claims reversed.	corrected; see `research/ic_triple_counted_20260923`
Members: `ic_candidate_tournament_20260915`, `ic_candidate_tournament_20260921`, `ic_triple_counted_20260923`, `ic_triple_table_20260923`			
Groebner/F4 stage engineering of decomposition oracle (6 entries, 2026-09-15..2026-10-11)	engineering	6 members. 1.60x word XORs (stage), inherited F4 20.95x, chain order up to 7.25x holdout, support-local 1.41x whole log; Ir re-pricing: whole log 964x rho at n13.	current
Members: `notes/ecc2k130/RESEARCH_CHAIN_SPLIT_ORDER.md`, `notes/ecc2k130/RESEARCH_F4_CACHE_LIMIT_AUDIT.md`, `notes/ecc2k130/RESEARCH_GROEBNER_STAGE.md`, `notes/ecc2k130/RESEARCH_GROEBNER_STAGE_INSTRUCTIONS.md`, `notes/ecc2k130/RESEARCH_INHERITED_F4.md`, `notes/ecc2k130/RESEARCH_SUPPORT_LOCAL_MULTIPLIERS.md`			
boundary ledger sections 17-23 studies (4 entries, 2026-09-22..2026-09-30)	accounting	4 members; matched negation rho re-pricing raised vs-rho 2.1-6.5x (best rows 13.5x, 38.2x); single-target online 8.84-17.5x vs table-less rho but cold rho wins at all six sizes.	current
Members: `ic_constructions_20260926`, `ic_framework_engines_20260922`, `ic_rho_reference_20260923`, `ic_single_target_20260930`			
ECC2K-130 IC program plans and ladders (5 entries, 2026-09-23..2026-09-29)	accounting	5 members. Routes closed 2026-09-23 (X4'); decision ladder 2026-09-25 supersedes earlier gates; no plan yields a crossover.	current
Members: `notes/ecc2k130/NEXT_EXPERIMENT_DECISIONS_20260925.md`, `notes/ecc2k130/RESEARCH_DECISION_LADDER_20260925.md`, `notes/ecc2k130/RESEARCH_ECC2K130_ROUTES.md`, `notes/ecc2k130/RESEARCH_ECC2K130_ROUTE_TARGETS.md`, `notes/ecc2k130/RESEARCH_NEXT_EXPERIMENTS_20260924.md`			
sparse compact bases and four-sum support (5 entries, 2026-09-25)	diagnostic	5 members. Sparse n37/41 bases give 0/3 holdout relations; complete four-sum oracle agrees with extractor (n37 R3 164/512); support is the lever.	current
Members: `notes/ecc2k130/COMPACT_BASE_SWEEP_PROTOCOL_20260925.md`, `notes/ecc2k130/COMPACT_BASE_SWEEP_RESULT_20260925.md`, `notes/ecc2k130/FOUR_SUM_MEMBERSHIP_RESULT_20260925.md`, `notes/ecc2k130/four_sum_membership_20260925/PROTOCOL.md`, `notes/ecc2k130/four_sum_membership_20260925/evidence/source/PROTOCOL_before_prefix.md`			
paired n37/n41 point-defined full-rank control (3 entries, 2026-09-25)	null	3 members. Source-overlay panel: IC far slower than rho (rho/IC CPU 0.029-0.634); clean panel archived elsewhere.	superseded; see `research/notes/ecc2k130/paired_fullrank_clean_evidence_20260925`
Members: `notes/ecc2k130/PAIRED_N37_N41_FULLRANK_PROTOCOL_20260925.md`, `notes/ecc2k130/paired_fullrank_20260925/EVIDENCE.md`, `notes/ecc2k130/paired_fullrank_20260925/source_snapshots/README.md`			
n13 rotated support/row/rank toys (9 entries, 2026-09-25)	diagnostic	9 members. Five rotated spaces cover 1,591/2,003 vs 61; 17,917 row witnesses certified; full rank on holdouts.	current
Members: `notes/ecc2k130/rotated_rank_holdout_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_rank_holdout_20260925/RESULT.md`, `notes/ecc2k130/rotated_rank_holdout_20260925/evidence/README.md`, `notes/ecc2k130/rotated_row_certificate_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_row_certificate_20260925/RESULT.md`, `notes/ecc2k130/rotated_row_certificate_20260925/evidence/README.md`, `notes/ecc2k130/rotated_subspace_support_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_subspace_support_20260925/RESULT.md`, `notes/ecc2k130/rotated_subspace_support_20260925/evidence/README.md`			
n19 rotated base/phase support toys (24 entries, 2026-09-25)	diagnostic	24 members. Beta sweep best +6.11% (fails 10% gate); five-beta union 96.48%; phase selector cap-8 128,751/130,873; joint rank recovers 58/64 vs 29/64; slot placement saturated.	current
Members: `notes/ecc2k130/rotated_beta_portfolio_20260925/EVIDENCE.md`, `notes/ecc2k130/rotated_beta_portfolio_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_beta_portfolio_20260925/RESULT.md`, `notes/ecc2k130/rotated_beta_sweep_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_beta_sweep_20260925/RESULT.md`, `notes/ecc2k130/rotated_beta_sweep_20260925/evidence/README.md`, `notes/ecc2k130/rotated_four_base_joint_rank_20260925/EVIDENCE.md`, `notes/ecc2k130/rotated_four_base_joint_rank_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_four_base_joint_rank_20260925/RESULT.md`, `notes/ecc2k130/rotated_m56_export_gate_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_m56_export_gate_20260925/RESULT.md`, `notes/ecc2k130/rotated_m56_export_gate_20260925/evidence/README.md`, `notes/ecc2k130/rotated_m56_export_gate_20260925/evidence/failure_0/README.md`, `notes/ecc2k130/rotated_nearthreshold_slot_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_pdp_corpus_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_pdp_corpus_20260925/RESULT.md`, `notes/ecc2k130/rotated_pdp_corpus_20260925/evidence/README.md`, `notes/ecc2k130/rotated_phase_charged_successor_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_phase_selector_20260925/EVIDENCE.md`, `notes/ecc2k130/rotated_phase_selector_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_phase_selector_20260925/RESULT.md`, `notes/ecc2k130/rotated_slot_placement_20260925/PROTOCOL.md`, `notes/ecc2k130/rotated_slot_placement_20260925/RESULT.md`, `notes/ecc2k130/rotated_slot_placement_20260925/evidence/README.md`			

Study	Class	Result	Status
n13 O-aware SAT and DRAT certificates (15 entries, 2026-09-25..2026-09-29)	diagnostic	15 members. Three SAT engines 5 SAT/27 UNSAT on 32 targets; 5 negatives DRAT-certified; sparse/ dense pairing and S5/S6 panels held.	open
Members: notes/ecc2k130/n13_oaware_remaining_unsat_proof_20260925/PROTOCOL.md, notes/ecc2k130/n13_oaware_sat_benchmark_20260925/PROTOCOL.md, notes/ecc2k130/n13_oaware_sat_benchmark_20260925/RESULT.md, notes/ecc2k130/n13_oaware_sat_benchmark_20260925/evidence/README.md, notes/ecc2k130/n13_oaware_sat_benchmark_20260925/evidence/failed_smoke_sandbox/README.md, notes/ecc2k130/n13_oaware_sparse_dense_sat_20260925/PROTOCOL.md, notes/ecc2k130/n13_oaware_sparse_dense_sat_20260925/evidence/README.md, notes/ecc2k130/n13_oaware_sparse_dense_sat_20260925/evidence/RUN_RELEASE_20260929.md, notes/ecc2k130/n13_oaware_unsat_proof_20260925/EVIDENCE.md, notes/ecc2k130/n13_oaware_unsat_proof_20260925/PREOUTCOME_PREFLIGHT_FAILURE.md, notes/ecc2k130/n13_oaware_unsat_proof_20260925/PROTOCOL.md, notes/ecc2k130/n13_oaware_unsat_proof_20260925/RESULT.md, notes/ecc2k130/n13_s5_fullpoint_control_20260925/PROTOCOL.md, notes/ecc2k130/n13_s5_solver_panel_20260925/PROTOCOL.md, notes/ecc2k130/n19_s6_fullpoint_exporter_20260925/PROTOCOL.md			
recursive-S3 semantics, CNF export, solver admission (27 entries, 2026-09-25..2026-09-29)	tooling	27 members. Finite-fibre S3 exactness proved; O-aware CNF exact; sparse CNF 252.7x smaller at n13; n19 sparse replay PASS; zero solver arms admitted from existing interfaces.	current
Members: notes/ecc2k130/ROTATED_M5_M6_SOLVER_ADMISSION_20260925.md, notes/ecc2k130/rotated_s3_candidate_20260925/PROTOCOL.md, notes/ecc2k130/rotated_s3_candidate_20260925/RESULT.md, notes/ecc2k130/rotated_s3_candidate_20260925/evidence/README.md, notes/ecc2k130/rotated_s3_candidate_20260925/evidence/failure_0/README.md, notes/ecc2k130/rotated_s3_fibre_theorem_20260925/PROOF.md, notes/ecc2k130/rotated_s3_fibre_theorem_20260925/PROTOCOL.md, notes/ecc2k130/rotated_s3_fibre_theorem_20260925/RESULT.md, notes/ecc2k130/rotated_s3_fibre_theorem_20260925/evidence/README.md, notes/ecc2k130/rotated_s3_o_branch_20260925/PROOF.md, notes/ecc2k130/rotated_s3_o_branch_20260925/PROTOCOL.md, notes/ecc2k130/rotated_s3_o_branch_20260925/RESULT.md, notes/ecc2k130/rotated_s3_o_branch_20260925/evidence/README.md, notes/ecc2k130/rotated_s3_sparse_cnf_20260925/PROTOCOL.md, notes/ecc2k130/rotated_s3_sparse_cnf_20260925/RESULT.md, notes/ecc2k130/rotated_s3_sparse_cnf_20260925/evidence/README.md, notes/ecc2k130/rotated_s3_sparse_n19_20260925/PROTOCOL.md, notes/ecc2k130/rotated_s3_sparse_n19_20260925/RESULT.md, notes/ecc2k130/rotated_s3_sparse_n19_20260925/evidence/README.md, notes/ecc2k130/rotated_s3_sparse_n19_linear_20260929/PROTOCOL.md, notes/ecc2k130/rotated_s3_sparse_n19_linear_20260929/RESULT.md, notes/ecc2k130/rotated_solver_admission_20260925/PROTOCOL.md, notes/ecc2k130/rotated_solver_admission_20260925/RESULT.md, notes/ecc2k130/rotated_solver_admission_20260925/evidence/README.md, notes/ecc2k130/rotated_solver_admission_20260925/evidence/path_only_initial/README.md, notes/ecc2k130/rotated_solver_admission_replay_20260925/PROTOCOL.md, notes/ecc2k130/rotated_solver_admission_replay_20260925/RESULT.md			
symbolic full-point DAG and DIMACS export (20 entries, 2026-09-25..2026-09-29)	tooling	20 members. Toy full-point DAG exact; macOS launch failure then Linux v2 outcome replay PASS incl. n131 single-edge DIMACS; n131 symbolic PDP not admitted (u64 ceiling).	current
Members: notes/ecc2k130/symbolic_dag_dimacs_gate_20260925/PROTOCOL.md, notes/ecc2k130/symbolic_dag_dimacs_gate_20260925/RESULT.md, notes/ecc2k130/symbolic_dag_dimacs_gate_20260925/preoutcome_failure_0/README.md, notes/ecc2k130/symbolic_dag_dimacs_gate_20260925/preoutcome_release_1/PROTOCOL.md, notes/ecc2k130/symbolic_dag_dimacs_gate_20260925/preoutcome_release_1/README.md, notes/ecc2k130/symbolic_dag_dimacs_linux_attempt2_20260926/PREPARATION_RESULT.md, notes/ecc2k130/symbolic_dag_dimacs_linux_attempt2_20260926/PROTOCOL.md, notes/ecc2k130/symbolic_dag_dimacs_linux_v2_20260928/PROTOCOL.md, notes/ecc2k130/symbolic_dag_dimacs_linux_v2_20260928/RESULT.md, notes/ecc2k130/symbolic_dag_dimacs_linux_v2_outcome_replay_20260929/PROTOCOL.md, notes/ecc2k130/symbolic_dag_dimacs_linux_v2_outcome_replay_20260929/RESULT.md, notes/ecc2k130/symbolic_dag_fullpoint_20260925/PROTOCOL.md, notes/ecc2k130/symbolic_dag_fullpoint_20260925/RESULT.md, notes/ecc2k130/symbolic_dag_fullpoint_20260925/evidence/README.md, notes/ecc2k130/symbolic_dag_fullpoint_20260925/evidence_failure_0/README.md, notes/ecc2k130/symbolic_oaware_width_gate_20260925/DESIGN.md, notes/ecc2k130/symbolic_oaware_width_gate_20260925/EVIDENCE.md, notes/ecc2k130/symbolic_oaware_width_gate_20260925/FAILURES.md, notes/ecc2k130/symbolic_oaware_width_gate_20260925/PROTOCOL.md, notes/ecc2k130/symbolic_oaware_width_gate_20260925/RESULT.md			
n131 rotated normal-basis factor-base census (13 entries, 2026-09-25..2026-10-01)	accounting	13 members. Exact F0 census 1,048,134 columns (proxy 2x high); only balanced m10,d13 passes N>=q (F^m/q 1.5329); m10 capacity attempt pending review.	open
Members: notes/ecc2k130/challenge_point_import_20260925/PROTOCOL.md, notes/ecc2k130/challenge_point_import_20260925/RESULT.md, notes/ecc2k130/m10_export_capacity_20260925/ATTEMPT_CONTINUATION_20261001.md, notes/ecc2k130/m10_export_capacity_20260925/PROTOCOL.md, notes/ecc2k130/n131_beta3_f0_census_20260925/EVIDENCE.md, notes/ecc2k130/n131_beta3_f0_census_20260925/PROTOCOL.md, notes/ecc2k130/n131_beta3_f0_census_20260925/RESULT.md, notes/ecc2k130/rotated_arity_screen_20260925/EVIDENCE.md, notes/ecc2k130/rotated_arity_screen_20260925/PROTOCOL.md, notes/ecc2k130/rotated_arity_screen_20260925/RESULT.md, notes/ecc2k130/rotated_unequal_arity_20260925/EVIDENCE.md, notes/ecc2k130/rotated_unequal_arity_20260925/PROTOCOL.md, notes/ecc2k130/rotated_unequal_arity_20260925/RESULT.md			

Study	Class	Result	Status
compact-orbit S3 IC vs strong rho, cold panels (59 entries, 2026-09-25..2026-10-10)	null	59 members. Best: point panel n41/L1024 0.309x v2 rho (withdrawn); final orbit-disjoint v2: CPU/rho 1.314 (n41/L1024) to 22.675 (n53/L1); every eligible cell above strong rho.	current
Members: notes/ecc2k130/base_window_screen_20261001/INPUT_FREEZE.md, notes/ecc2k130/base_window_screen_20261001/MEASURE_DISPATCH.md, notes/ecc2k130/base_window_screen_20261001/PROTOCOL.md, notes/ecc2k130/base_window_screen_20261001/SOURCE_LOCK.md, notes/ecc2k130/base_window_screen_20261001/development/README.md, notes/ecc2k130/base_window_screen_20261001/outcome_run_36827211814/RESULT.md, notes/ecc2k130/base_window_screen_outcome_20261001/RESULT.md, notes/ecc2k130/compact_frozen_source_replay_20260929/PROTOCOL.md, notes/ecc2k130/compact_ir_cold_gap_20260930/PROTOCOL.md, notes/ecc2k130/compact_ir_cold_gap_20260930/RESULT.md, notes/ecc2k130/compact_ir_cold_gap_20260930/evidence/run_36694337420_fillin_b8606c7ee/README.md, notes/ecc2k130/compact_ir_ledger_20260930/DEVELOPMENT.md, notes/ecc2k130/compact_ir_ledger_20260930/PROTOCOL.md, notes/ecc2k130/compact_ir_ledger_20260930/RESULT.md, notes/ecc2k130/compact_k_boundary_20260929/PROTOCOL.md, notes/ecc2k130/compact_k_boundary_20260929/RESULT.md, notes/ecc2k130/compact_orbit_point_panel_20260929/AMENDMENT.md, notes/ecc2k130/compact_orbit_point_panel_20260929/PROTOCOL.md, notes/ecc2k130/compact_orbit_point_panel_20260929/RESULT.md, notes/ecc2k130/compact_orbit_producer_merge_repair_20261010/LIBRARY_VALIDATION.md, notes/ecc2k130/compact_orbit_producer_merge_repair_20261010/RESULT.md, notes/ecc2k130/compact_orbit_rank_evidence_20260929/PROTOCOL.md, notes/ecc2k130/compact_orbit_rank_evidence_20260929/RESULT.md, notes/ecc2k130/compact_orbit_strong_rho_20260929/PROTOCOL.md, notes/ecc2k130/compact_orbit_strong_rho_20260929/RESULT.md, notes/ecc2k130/compact_s3_batch_20260929/PORTABLE_REPLAY.md, notes/ecc2k130/compact_s3_batch_20260929/PROTOCOL.md, notes/ecc2k130/compact_s3_batch_20260929/RESULT.md, notes/ecc2k130/compact_s3_batch_20260929/development/README.md, notes/ecc2k130/compact_s3_prefilter_20260930/CI_MATERIALIZATION_AMENDMENT.md, notes/ecc2k130/compact_s3_prefilter_20260930/DEVELOPMENT.md, notes/ecc2k130/compact_s3_prefilter_20260930/PROTOCOL.md, notes/ecc2k130/compact_s3_prefilter_20260930/REPEAT_PROTOCOL.md, notes/ecc2k130/compact_s3_prefilter_20260930/RESULT.md, notes/ecc2k130/compact_shared_log_20260925/CI_COMPARISON_CORRECTION.md, notes/ecc2k130/compact_shared_log_20260925/PROTOCOL.md, notes/ecc2k130/compact_shared_log_20260925/RESULT.md, notes/ecc2k130/compact_swap_quotient_20260929/PROTOCOL.md, notes/ecc2k130/compact_swap_quotient_20260929/RESULT.md, notes/ecc2k130/disjoint_cold_outcome_20261001/PROTOCOL.md, notes/ecc2k130/disjoint_cold_outcome_20261001/RESULT.md, notes/ecc2k130/disjoint_cold_q_20260930/PROTOCOL.md, notes/ecc2k130/disjoint_cold_q_20260930/SMOKE.md, notes/ecc2k130/disjoint_cold_q_20260930/SOURCE_REPLAY_AMENDMENT.md, notes/ecc2k130/disjoint_cold_v2_20261001/PROTOCOL.md, notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md, notes/ecc2k130/pdp_difficulty_features_20261001/PROTOCOL.md, notes/ecc2k130/pdp_difficulty_features_20261001/RESULT.md, notes/ecc2k130/pinned_x_witness_20260930/VERIFICATION_CONTRACT.md, notes/ecc2k130/point_sum_cold_20260930/DISPATCH_AMENDMENT.md, notes/ecc2k130/point_sum_cold_20260930/GENERATION_AMENDMENT.md, notes/ecc2k130/point_sum_cold_20260930/LOCAL_SMOKE.md, notes/ecc2k130/point_sum_cold_20260930/PROTOCOL.md, notes/ecc2k130/point_sum_cold_20260930/VALIDATION_AMENDMENT.md, notes/ecc2k130/point_sum_cold_outcome_20260930/RESULT.md, notes/ecc2k130/point_sum_cold_outcome_20260930/STATUS.md, notes/ecc2k130/point_sum_query_20260930/DEVELOPMENT_RESULT.md, notes/ecc2k130/point_sum_query_20260930/PROTOCOL.md, notes/ecc2k130/selector_phase_admission_20261001/DECISION.md			
target-only four-point decompositions on ECC2K-130 model (2 entries, 2026-09-26..2026-09-30 (2 members))	diagnostic	Counting claims verified (2^-96.87 lifts per tuple at n=131); toy fields n=13,19 saturate so test nothing; independent SAT: n=13 solved, n=19 0 of 4 in 1,200 s.	current
Members: frobenius_quotient_followup_20260926, frobenius_quotient_sat_20260930			
Koblitz m=3 thread vs batch rho (ledger 20-23) (3 entries, 2026-09-26..2026-09-30 (3 members))	accounting	No crossing: best 4.86x batch rho at 2^39; descent fixed cost cut up to 3.11x at n=19; ledger 23 withdrawn and redeclared single-target.	current
Members: ic_descent_20260930, ic_exponent_20260926, ic_exponent_top_20260930			
m83 S4 solver-stage matrix (9 entries, 2026-09-27..2026-09-30)	diagnostic	9 members. Native-XOR SAT fastest SAT stage, FES faster at six vars; fixed bases cover <=8.565e-18 of subgroup so zero natural hits uninformative.	current
Members: notes/ecc2k130/m83_relation_capacity_20260930/PROTOCOL.md, notes/ecc2k130/m83_relation_capacity_20260930/RESULT.md, notes/ecc2k130/m83_solver_matrix_20260927/AMENDMENT.md, notes/ecc2k130/m83_solver_matrix_20260927/PROTOCOL.md, notes/ecc2k130/m83_solver_matrix_20260927/PROTOCOL_EXTENSION.md, notes/ecc2k130/m83_solver_matrix_20260927/PROTOCOL_ORBIT_RELATIONS.md, notes/ecc2k130/m83_solver_matrix_20260927/PROTOCOL_PLAIN_SUBSPACE.md, notes/ecc2k130/m83_solver_matrix_20260927/RESULTS.md, notes/ecc2k130/m83_solver_matrix_20260927/results/README.md			
compact-orbit IC vs strengthened Koblitz rho (8 entries, 2026-09-28..2026-10-01)	accounting	8 members. Best reported IC/rho 0.2403 (PR #955, withdrawn); final with strong rho R3: 0.794, 1.520, 1.923, 2.297 at n=37..61 and direct arm 44.0x strong single-target rho; line closed (IC loses).	current
Members: notes/index-calculus/RESEARCH_CACHEGRIND_N41_20260929.md, notes/index-calculus/RESEARCH_FIXED_REFERENCE_RHO_20261001.md, notes/index-calculus/RESEARCH_MATCHED_RHO_ORBIT_DLP_20260928.md, notes/index-calculus/RESEARCH_PRECOMPUTED_START_RHO_20260929.md, notes/index-calculus/RESEARCH_SINGLE_TARGET_STRONG_RHO_N53_20260930.md, notes/index-calculus/RESEARCH_STRONG_RHO_LADDER_20260929.md, notes/index-calculus/RESEARCH_STRONG_RHO_SWEEP_PROTOCOL_20260929.md, notes/index-calculus/strong_rho_ladder_20260929_run/ladder_table.md			
Linearization/XL reach on S3 core (ECC2K-130) (3 entries, 2026-09-30..2026-10-06)	null	3 dirs. Boundary results: linearization gap 70.19 - b_ max bits; best charged XL gap 71.82 bits (omega 2); trilinear 87.16 bits. No oracle closes the gap to rho 2^60.8090. Closed.	current
Members: linearization_reach_20260930, xl_bilinear_core_20261006, xl_trilinear_t_unknown_20261006			
solving-degree cost model at n=131 (1 entries, 2026-09-30..2026-10-17)	accounting	1 members. Final optimistic m=4 row deg-5 2^71.13 (+10.32 w=1; +45.71 w=2); deg-4 floor +7.62 contradicted; no row beats rho.	current
Members: notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md			

Study	Class	Result	Status
n37 ecbench cold single-target IC chain (15 entries, 2026-10-03..2026-10-04)	diagnostic	15 members. Folded table 66,822 vs 4,831,386 additions (72.3x); full-rank gate passes but IC/rho 22.600x; shared-rank ecbench comparison preregistered, no result.	open
Members: notes/ecc2k130/cold_compact_orbit_ic_20261003/PARITY_FAILURE.md, notes/ecc2k130/cold_compact_orbit_ic_20261003/PROTOCOL.md, notes/ecc2k130/cold_compact_orbit_ic_20261003/RESULT.md, notes/ecc2k130/full_rank_column_check_20261003/DECISION.md, notes/ecc2k130/full_rank_compact_orbit_ic_20261003/PROTOCOL.md, notes/ecc2k130/full_rank_compact_orbit_ic_20261003/RESULT.md, notes/ecc2k130/n37_complete_table_floor_20261003/PROTOCOL.md, notes/ecc2k130/n37_complete_table_floor_20261003/RESULT.md, notes/ecc2k130/n37_frobenius_fold_gate_20261003/PROTOCOL.md, notes/ecc2k130/n37_frobenius_fold_gate_20261003/RESULT.md, notes/ecc2k130/n37_shared_rank_ecbench_20261004/PROTOCOL.md, notes/ecc2k130/n37_shared_rank_targets_20261003/PROTOCOL.md, notes/ecc2k130/n37_shared_rank_targets_20261003/RESULT.md, notes/ecc2k130/shared_rank_folded_table_20261003/PROTOCOL.md, notes/ecc2k130/shared_rank_folded_table_20261003/RESULT.md			
is IC cost a curve or presentation property (10 entries, 2026-10-03..2026-10-12)	diagnostic	10 members. Yield spread explained by cofactor balance (corr 0.850); no curve effect at n=19 (ICC<=0.006); n=31 m=3 H1 on work per call only; overall inconclusive/no curve effect.	current
Members: notes/koblitz-isogeny/full-solves-20261004.md, notes/koblitz-isogeny/m3-presentation-vs-curve-design-20261011.md, notes/koblitz-isogeny/m3-presentation-vs-curve-n31-design-20261012.md, notes/koblitz-isogeny/m3-solver-design-20261004.md, notes/koblitz-isogeny/presentation-vs-curve-design-20261003.md, notes/koblitz-isogeny/presentation-vs-curve-pilot-20261003.md, notes/koblitz-isogeny/presentation-vs-curve-results-20261004.md, notes/koblitz-isogeny/subspace-structure-20261004.md, notes/koblitz-isogeny/yield-spread-design-20261003.md, notes/koblitz-isogeny/yield-spread-results-20261003.md			
ecbench-n37-n53 (6 entries, 2026-10-04..2026-10-05)	mixed	6 dirs. n37 IC 3.6446x (Ir) to 13.591x (counted S) rho, cold; n37 parameters fail at n41/n53.	current
Members: ecbench_n37_k8_k16_20261004, ecbench_n37_native_online_wall_20261004, ecbench_n37_online_ir_20261004, ecbench_n37_rank_columns_20261004, ecbench_n37_shared_rank_20261004, ecbench_n41_n53_shared_rank_20261005			
f6-n83 (24 entries, 2026-10-04..2026-10-05)	mixed	24 dirs. F6 micro-studies on K0 (f6_wide_n83 on K1): retained kernel gains (query up to 2.998x, RSS -52%), SAT encodings smaller but every ordinary solve timed out; four-summand coverage <= 4.662e-12; no relation.	current
Members: f6_n83_basis_sat_20261005, f6_n83_batch_size_20261004, f6_n83_compact_pairs_20261005, f6_n83_domain_20261004, f6_n83_factored_s3_sat_20261005, f6_n83_five_source_prolongation_20261005, f6_n83_fivesum_d18_20261005, f6_n83_karatsuba_sat_20261005, f6_n83_lazy_branches_20261005, f6_n83_mixed_base_20261004, f6_n83_pmull_index_build_20261005, f6_n83_pmull_inverse_20261005, f6_n83_pmull_xonly_20261005, f6_n83_private_cubic_peel_20261005, f6_n83_private_quartic_core_20261005, f6_n83_reduction_20261004, f6_n83_sign_fold_20261004, f6_n83_sixsum_prefix_20261005, f6_n83_sixsum_wide_20261005, f6_n83_valid_coordinates_20261005, f6_n83_wide_block_20261005, f6_n83_xfilter_20261005, f6_n83_xonly_query_20261005, f6_wide_n83_20261004			
n83-k0-sat-factor-base (7 entries, 2026-10-05..2026-10-19)	mixed	7 dirs. a=0 K0 81-bit gate curve: SAT/XCNF decomposition only with known witness; no natural relation, no m=83 rho S, no cold runtime; prebuilt-base problem open.	open; see problems/n83-prebuilt-base-quality-m2-m4.md
Members: ecbench_m83_ic_20261009, ecbench_m83_rho_20261005, ecdlp_surfaces_n83_20261012, ic_wide_shared_rank_plan_20261005, koblitz_n83_factor_base_sweep_20261008, n83_chain_xcnf_control_20261019, n83_prebuilt_factor_base_20261016			
ECC2K-130 leads, conductor gap and transfer notes (6 entries, 2026-10-07..2026-10-12)	diagnostic	6 members. Hypotheses, protocols and toy measurements; transfers change only constants; no D* drop for ECC2K-130 total work.	open
Members: docs/ic/EXPERIMENTS_ISOGENY_CONDUCTOR_GAP_20261007.md, docs/ic/IC_TECHNIQUE_INTEGRATION_20261012.md, docs/ic/LEAD_LADDER_METHODOLOGY_20261010.md, docs/ic/RESEARCH_IC_NOVEL_DIRECTIONS_20261007.md, docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md, docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md			
koblitz-pipeline-engineering (3 entries, 2026-10-08..2026-10-23)	mixed	3 dirs. Constant-factor kernel changes to the Koblitz pipeline: laned keys 1.092-1.310x portable, u64 root batch 3.26x median (cells corrected), aimed gather 1.037x set-up.	current
Members: canon_laned_keys_20261008, ic_aimed_gather_20261023, u64_root_batch_20261015			
n53 K220 index/rank engineering (8 entries, 2026-10-10)	engineering	8 members. Pair-swap halves index S3 calls (complete-cold 0.969); rank restarts 5.189% probe cut, ratio 1.020, gate failed.	current
Members: notes/ecc2k130/n53_fixed_base_rank_restarts_20261010/HELDOUT_RESULT.md, notes/ecc2k130/n53_fixed_base_rank_restarts_20261010/PILOT_RESULT.md, notes/ecc2k130/n53_fixed_base_rank_restarts_20261010/PROTOCOL.md, notes/ecc2k130/n53_fixed_base_rank_restarts_20261010/README.md, notes/ecc2k130/n53_pair_symmetry_20261010/DEVELOPMENT.md, notes/ecc2k130/n53_pair_symmetry_20261010/PROTOCOL.md, notes/ecc2k130/n53_pair_symmetry_20261010/RESULT.md, notes/ecc2k130/n53_pair_symmetry_20261010/VALIDATION.md			
equal-advice preprocessing rho control (1 entries, 2026-10-11)	accounting	1 member here (plus ic_next10/09-equal-advice); derived 48x / 3,400x, measured 53.11-317x BL faster online.	current
Members: ic_preprocessing_rho_equal_advice_20261011			
Koblitz m=3 S4 presentation diagnostics (3 dirs) (3 entries, 2026-10-11)	diagnostic	PinX3 0.052/0.154 ms vs geometric 9.702/123.595 ms per call; n=31 16.6 ms/call; no sub-rho claim	current
Members: 20261011-koblitz-m3-1955z, 20261011-koblitz-m3-2150z, 20261011-koblitz-m3-2357z			

Study	Class	Result	Status
split/hybrid PDP and orbit-union slots (3 entries, 2026-10-11..2026-10-18 (3 members))	diagnostic	Best projected or priced split/hybrid arm 22.8x strong rho (relation phase only) to 36x (priced, n=11); closure base 0.35-0.54x standard; round 3 registered, not run.	open
Members: pdp_split_dreg_20261011, pdp_split_round3_20261018, pdp_split_slots_20261012			
PDP decomposition oracle programme (8 dirs) (8 entries, 2026-10-11..2026-10-19)	engineering	PS2 scalar kernel 0.0668-0.1225x baseline; root-finder 3,007-115,160x PS-0; split/hybrid best 22.8-36x strong rho; shape replay null	current
Members: 20261011-pdp-ps0-baseline, 20261011-pdp-ps1-algebra-gates, 20261011-pdp-ps2-scalar-group-law, 20261011-pdp-split-hybrid, 20261012-pdp-root-finder, 20261015-pdp-root-shape-replay, 20261018-pdp-ps3-bitsliced, 20261019-pdp-split-closure-ecbench			
root-finding oracles in ecbench on Koblitz E0 (7 entries, 2026-10-12..2026-10-15 (7 members))	engineering	Best complete negfold S 38.8 -> 21.8-23.0 (1.54-1.55x strong rho); vs priced F4 0.0545; stride-renew 0.770x rho-strong at n=13 (lower-bound reference) but 4.68x at n=19; early F4 'parity' used unpriced F4.	current
Members: ecbench_rootfind_engine_20261012, ecbench_rootfind_faster_20261014, ecbench_rootfind_improve_20261013, ecbench_rootfind_parity_20261013, ecbench_rootfind_speedup_20261014, ecbench_rootfind_vs_rho_20261014, rootfind_faster_e2e_20261015			
n83-a1-rho-reference (2 entries, 2026-10-12..2026-10-16)	mixed	2 dirs. K600 re-eval (12 Oct): IC slower online (1.382) and 218.0x cold vs default strong rho; precomputed-rho control (16 Oct): rho 3.74x-395.5x faster in every matched pair.	current
Members: 20261016-n83-precomputed-rho-control, n83_evaluation_20261012			
n83-a1-root-index-capacity (25 entries, 2026-10-13..2026-10-16)	mixed	25 dirs. Root-index layouts grew the a=1 base from K512 (84,992 pts) to K3596 (596,936 pts) within 14 GiB; 18.53x and other one-point online ratios retired; probe reductions stand.	current
Members: n83_dense_root_20261013, n83_indexed5_20261013, n83_indexed5_id30_20261016, n83_indexed5_load75_20261013, n83_indexed5_load75_results_20261014, n83_indexed5_tier1_20261013, n83_indexed5_tier1_results_20261013, n83_large_local_20261016, n83_load75_20261013, n83_root11_20261014, n83_root11_results_20261014, n83_root21_20261014, n83_root21_results_20261014, n83_root4_20261014, n83_root4_fixed30_20261014, n83_root4_fixed30_results_20261014, n83_root4_fixed30_root11_20261014, n83_root4_results_20261014, n83_root_only_20261014, n83_root_only_results_20261014, n83_state26_20261014, n83_state26_results_20261014, n83_state28_20261013, n83_symmetric_root_20261013, n83_symmetric_root_results_20261014			
N83 Koblitz one-target online campaign (6 dirs) (6 entries, 2026-10-13..2026-10-17)	accounting	online 18.528x and 12.29x IC-over-rho readings retired: precomputed-table rho faster in every matched pair (IC/rho 3.74-395.53x); K screens BOUNDED_UNKNOWN	corrected; see benchmarks/20261016-n83-precomputed-rho-control
Members: 20261013-n83-dense-root, 20261014-n71-n83-hosted-single-target, 20261016-n83-finegrid, 20261016-n83-prebuilt-factor-base, 20261016-n83-precomputed-rho-control, 20261017-n83-e2e-precision			
n83-a1-fb-selection-online (56 entries, 2026-10-14..2026-10-16)	mixed	56 dirs. Within-IC online selection on 100-20,000 points; best means ~26-28.7 ms (K3425/K3450/ K3500), many BOUNDED_UNKNOWN; K3072 one-point 6.6-7.4 ms; A/A noise open.	current
Members: n83_best_e2e_20261015, n83_best_e2e_results_20261015, n83_bloom_profile_20261015, n83_bloom_profile_results_20261015, n83_compact4_crosslayout_20261016, n83_compact4_crosslayout_results_20261016, n83_edge_confirmation_20261015, n83_edge_confirmation_results_20261015, n83_factor_base_100_20261014, n83_factor_base_100_results_20261014, n83_factor_base_confirm_20261015, n83_factor_base_confirm_results_20261015, n83_factor_base_panel_20261014, n83_factor_base_panel_results_20261014, n83_factor_base_refine_20261014, n83_factor_base_refine_results_20261014, n83_final_tournament_20261015, n83_final_tournament_results_20261015, n83_k3250_confirm_20261015, n83_k3250_sparse_fresh_20261015, n83_k3250_sparse_fresh_results_20261015, n83_k3350_reverse_20261015, n83_k3350_reverse_results_20261015, n83_k3350_sparse_fresh_20261015, n83_k3350_sparse_fresh_results_20261015, n83_k_abba_20261015, n83_k_abba_results_20261015, n83_k_neighborhood_20261015, n83_k_neighborhood_results_20261015, n83_legacy_challenger_20261015, n83_legacy_challenger_results_20261015, n83_online_finegrid_20261016, n83_online_k3114_20261016, n83_online_range_20261016, n83_online_rescreen_20261016, n83_root11_bloom_20261015, n83_root11_bloom_results_20261015, n83_root11_bloom_sparse_20261015, n83_root11_bloom_sparse_results_20261015, n83_root11_edgek_20261015, n83_root11_edgek_results_20261015, n83_root11_highk_20261015, n83_root11_highk_results_20261015, n83_root4_bloom_20261015, n83_root4_bloom_results_20261015, n83_root4_highk_20261015, n83_root4_highk_confirm_20261016, n83_root4_highk_confirm_results_20261016, n83_root4_highk_results_20261015, n83_root4_memory_frontier_20261015, n83_root4_memory_frontier_results_20261015, n83_root4_vs_root11_20261015, n83_root4_vs_root11_results_20261015, n83_root4_wide3_20261016, n83_sparse_canonical_20261015, n83_sparse_canonical_results_20261015			
n83-a1-tier1-online-vs-rho (14 entries, 2026-10-14..2026-10-16)	mixed	14 dirs. Online rho/IC medians 54.10x, 210.25x, 113.011x, 104.23x, 35.10x, 413.34x, 208.72x, all vs empty-table rho; all retired as IC wins (precomputed-rho control, ledger audit para 1).	superseded; see research/20261016-n83-precomputed-rho-control;papers/ecdlp-ic-improvement-ledger/sections/audit.tex
Members: n83_final_tier1_20261015, n83_tier1_bloom_reserve_20261015, n83_tier1_bloom_reserve_results_20261015, n83_tier1_combined_20261015, n83_tier1_combined_results_20261015, n83_tier1_final_20261014, n83_tier1_final_results_20261014, n83_tier1_online_20261016, n83_tier1_refine_reserve_20261015, n83_tier1_refine_reserve_results_20261015, n83_tier1_reserve_20261014, n83_tier1_reserve_results_20261014, n83_unresolved_tier1_20261015, n83_unresolved_tier1_results_20261015			

Study	Class	Result	Status
n83-a1-cold-factor-base (11 entries, 2026-10-16..2026-10-21)	mixed	11 dirs. Cold one-target process fell from ~150 s (K1024) to 119.826 s (K1120 holdout) and 113.755 s (K1107, 21 Oct); rho online ~2.2-2.6 s; no matched cold rho ratio admitted.	current
Members: `n83_cold_base_confirm_20261021`, `n83_cold_base_confirm_visualization_20261021`, `n83_cold_factor_base_20261016`, `n83_cold_neighborhood_20261016`, `n83_cold_reserve_20261016`, `n83_e2e_factor_base_20261016`, `n83_e2e_finegrid_20261016`, `n83_e2e_holdout_20261016`, `n83_e2e_precision_20261016`, `n83_e2e_root22_neighbors_20261016`, `n83_root11_dynamic_20261016`			
IC tool programme Table A rounds, rules, explorations (17 dirs) (17 entries, 2026-10-16..2026-10-24)	engineering	baseline v3 to v8 via R10/R06/R08/R11/R12 (1.90-1.98x over v4 cold); rule v7 rho lead 3.43-11.51x; matched GFNI rho at v8 widens to 4.67-18.07x; rule v4 null, distance 9.9-43.5x	current
Members: `20261016-ic-r06-scan-key`, `20261016-ic-r08-scan-fold`, `20261016-ic-r10-main-head`, `20261016-ic-r11-build-streams`, `20261016-ic-rule-v6`, `20261016-ic-rule-v7`, `20261017-ic-a14-prefetch-size-gate`, `20261017-ic-a15-filter-slots`, `20261017-ic-a16-huge-pages`, `20261017-ic-a17-filter-huge-pages`, `20261017-ic-a18-wider-base`, `20261017-ic-a19-filter-pages-build-parts`, `20261017-ic-a20-build-line-streams`, `20261017-ic-r12-admitted-prefetch`, `20261017-ic-scan-stages-v8`, `20261019-ic-matched-rho-v8`, `20261024-table-a-rule-v4`			
`research/sat_factor_base_review_20260908` 2026-09-08 Koblitz factor-base SOTA gate campaign	engineering	Through Stage 201: all-seven and SOTA gates false; contains n=71/n=83 hosted panels, online 7.29x/12.29x vs empty-table rho, cold IC 133.7x slower than rho.	corrected; see `benchmarks/20261016-n83-precomputed-rho-control/(precomputedrhofaster)`
`research/fixed_parameters_20260915` 2026-09-15 fixed-parameter persistent IC (ECC2K-130 basis)	engineering	Engineering capability: persistent ONB pair tables and relations (ic fixed); 72 matched cases accepted; no operation-cost, runtime or exponent claim.	current
`research/notes/index-calculus/RESEARCH_FACTOR_BASE_SOLVE_COST.md` 2026-09-15 factor-base selector ranks bases backwards for solver	diagnostic	At icv1-f2m15-t275-b7f03703 trials-optimal base costs 12.12x the cheapest base's F4 word-XORs (Spearman -0.90 over 5 subspaces, 60 verified DLPs); at n=131 only two usable invariant subspaces; accounting correction 2026-09-22. (measured)	corrected; see `research/notes/index-calculus/RESEARCH_FACTOR_BASE_SOLVE_COST.md(accountingcorrection2026-09-22)`
`research/rho_parity_20260915` 2026-09-15 full-cost IC vs signed-Frobenius rho parity iterations	engineering	Rho parity not established; on fresh confirmation targets the candidate uses 1.06-1.59x rho's instructions, 72.6-94.0% below the original chart implementation (toy fields n=7-15).	current
`docs/ic/FIXED_PARAMETERS.md` 2026-09-17 ic fixed workflow on explicit K_0 incl. ECC2K-130	tooling	[usage] Usage doc for resumable fixed-parameter workflow (select, pairs, relations, logs) on ECC2K-130 profile; no result. (n.a.)	current
`docs/ic/SOLVER_SELECTION.md` 2026-09-17 learned solver/budget selection	tooling	[usage] Cost-sensitive decision tree for decomposition policy; on 288 natural queries pair-table policy always cheapest: no routing headroom, no speedup. (measured)	current
`docs/ic/ci/README.md` 2026-09-20 ic end-to-end CI benchmark gate	tooling	[narrative+generated] CI gate v6: whole-process wall rho/IC 10.28 (n=41), 27.28 (n=53), 'crosses rho e2e', on a 32-target batch, mixed units; IC S null; rho step count gated within 3x of floor; not a one-target or unit claim. (measured)	current
`research/crossbred_20260920` 2026-09-20 Crossbred decomposition oracle vs F4 and brute force	engineering	Crossbred oracle agrees with exhaustive/F4 root sets; bit ops xb/F4 0.002-0.023 and wall xb/F4 0.020-0.706 on K_0 rungs n=7-13, m=2-3; whole-pipeline panel shows enumerate fastest; relation collection stays Theta(2^n).	current
`docs/ic/COMPACT_ORBIT_PRODUCER.md` 2026-09-24 compact Koblitz orbit producer source promotion	tooling	[usage] Promotes koblitz_s5_sat_instance producer (n=7..53 rungs); build and tests only; no PDP/DLP gain or rho speedup established. (n.a.)	current
`research/chain_split_order_20260924` 2026-09-24 inherited-F4 splitting order on S3 chains	engineering	No README; tables: candidate order cuts word XORs 1.04x (frozen), 1.82x (chain), 7.25x (chain-holdout), 4.91x (chain-holdout-2) vs reference.	current
`research/notes/index-calculus/KOBLITZ_SUBFIELD_EXPERIMENT_TRACKER.md` 2026-09-24 Koblitz/subfield IC experiment intake queue	tooling	Implementation queue (P0-P2) for Koblitz orbit-coordinate decomposition, Frobenius telemetry, trace-zero bases etc.; trace-zero D* side settled by EXP-H; no rho-beating claim. (plan)	open

Study	Class	Result	Status
`research/support_local_multipliers_20260924` 2026-09-24 support-local multipliers in decomposition oracle	engineering	Oracle cost 1.02x total on frozen rungs, 2.01-3.12x on m=3/4 chain rungs; word ops 2.19x/2.08x but registered wall 0.82x until hash fix (1.41x/1.24x).	current
`research/koblitz_symmetrised_e2e_20260927` 2026-09-27 torsion-symmetrised Koblitz IC end to end	null	S_sym/S_x 0.84 (n=17), 0.76-0.77 (n=23), 4.01 (n=31); all arms far above rho (e.g. sym 987x at n=17).	current
`research/ic_rr_norm_ladder_20260930` 2026-09-30 Riemann-Roch norm form solving degree at m=3	null	Constant lever only: s_rr = 1.167 (threshold 0.35); degree ~ l+2; rr below x4 on all 36 resolved draws by mean -2.44.	current
`research/notes/index-calculus/abliterated_tricks_20261002/README.md` 2026-10-02 three model-suggested IC tricks checked	null	Three classical tricks from an abliterated-model reasoning trace checked on E_0 over GF(2^m); none beats rho; no S measurement, no speedup claimed. (derived)	current
`research/ic_solver_online_20261003` 2026-10-03 F5 vs inherited F4 online panel, n17	diagnostic	Median F5/inherited-F4 online ratio 208.49x (181.29-223.86); inherited F4 ~315x slower than rho; capture defect (wrapper exit 1) makes panel diagnostic only.	current
`research/ecbench_wide_gf2m_20261004` 2026-10-04 wide GF(2^m) curves in ecbench, m=83 gate	tooling	Infrastructure: ecbench builds the m=83 gate curve and runs strong rho and pair claw; all 26 committed sessions re-audit ok, 6,449 replays identical; no m=83 measurement reported.	current
`research/notes/ecc2k130/n83_rotated_m10_20261004/PROTOCOL.md` 2026-10-04 n83 rotated-m10 complete-point gate	diagnostic	Frozen protocol: exact public n83 rotated-m10 complete-point PDP gate (slot dims [9,9,9,8x7]).	current
`research/notes/ecc2k130/n83_rotated_m10_20261004/RESULT.md` 2026-10-04 n83 rotated-m10 complete-point gate	diagnostic	BOUNDED_NATURAL_TARGET_TIMEOUT: native n83 m10 relation exact and replayable, but public PDP hit 120 s CaDiCaL cap (270,483 decisions); no natural relation or DLP.	current
`benchmarks/20261005-ic-semiregular-law` 2026-10-05 registered m=3 refutation-degree law, Macaulay solves (copy of research/ic_semiregular_law_20261005) [adm:not-listed; sweep:-; lb:no]	diagnostic	68/68 units resolved; x4 meets its generic degree window on 32/32 draws, rr falls below on 8; degree-10 solves 239k-247k rows x199k columns, 42-51 min on 4 threads; registered verdict neither; n=131 cost conditional model	current
`research/f2m83_pmull_20261005` 2026-10-05 ARM64 PMULL degree-83 field ops in shared F2m	null	Rejected by small-query gate: exact query 0.897x at 258 base points despite 1.68-2.02x faster index build; patch preserved, baseline restored.	current
`research/geometric_v_linear_20261006` 2026-10-06 linear (A,P) oracle over geometric factor base	accounting	Geometric V resolves pairs up to l=floor((n+1)/3)=44 at n=131 (random V stops at 14); n=131 gap moves 56.19 to 42.47 bits-only, 71.82 to 55.22 charged (omega 2); still above rho, pair-oracle floor about 40 bits.	current
`research/cheap_closures_20261007` 2026-10-07 torsion invariants and additive energy closures	null	Both ideas closed: torsion/model invariants change decomposition by at most 2 bits per summand (vs 42-55-bit gap); geometric factor base additive-energy ratio R 0.94-1.01 in 6 of 7 cells (1.34 in one), no free relations.	current
`research/graph_cycle_ic_20261007` 2026-10-07 graph-cycle IC with linear pair oracle	null	End-to-end toy Koblitz ECDLP: rho solved first in 20 of 20 walk-mode runs; fresh-point mode median S 8.9-35.9 vs rho 1.0-2.0; at n=131 about 2^85.7 oracle solves vs rho 2^60.81; asymptotically 2^(2n/3).	current
`research/ecbench_sat_pdp_20261009` 2026-10-09 SAT/PDP matched pipeline on small Koblitz curve	accounting	count-qualified-main: Frobenius MITM S 17.073 (4.271x strong rho); native-XOR SAT 3.413 counted (0.854x rho) is a lower bound omitting 2,755,966 conflicts, not a speedup; IC MITM 96.88x.	current
`research/transfer_aware_ic_20261010` 2026-10-10 multiplier closure in 4-sum pair-table IC	null	Yield law holds (frob arm first-hit probes within 10% of 2r/L^2); Z[tau] multipliers are cosmetic (tau vs int arms 0.88-1.56x, decisive tau <= 0.67 int never occurred); coincidence loss real at K=1 (probes 1.49-2.15x).	current

Study	Class	Result	Status
`benchmarks/20261011-koblitz-4e-historical` 2026-10-11 closed-form Koblitz 4E membership test cost (copy of research/isogeny_fibers_ 20261010/round4/16_2_fast_ 4e) [adm:not-listed; sweep:U; lb:no]	diagnostic	n=131/233 table check 0.124/0.122 us vs tau-NAF 160/98 us (order-of-magnitude only); counted multiplications primary; sweep: not like-for-like, unvalidated	current
`research/koblitz-isogeny` 2026-10-11 m=3 presentation vs curve solver pilots	diagnostic	Reports only: e3 per-call ~16-28 ms at n=17; n=31 curve-effect H1 (2357Z) did not replicate on 2026-10-18, verdict inconclusive.	superseded; see `research/koblitz-isogeny/reports/run-20261018T1200Z`
`research/ic_techniques_20261012` 2026-10-12 IC techniques: cyclotomic, double large prime, sixsum	diagnostic	Subdirectories (a2 cyclotomic/delta, double large prime on pair-table pipeline, ecbench smoke, sixsum extraction); no top-level README and headline numbers not read (no number).	current
`benchmarks/20261014-ic-m3-algebraic-vs-mitm` 2026-10-14 algebraic m=3/4 oracles vs MITM floor (copy of research/ic_m3_algebraic_vs_ mitm_20261014) [adm:listed-pre-gate; sweep:VC; lb:no]	null	no arm grows slower than floor: m=3 floor slope 0.207, f4-chain 0.534, f6-ic 0.583; cheapest/floor 3.0e2 (n=13) to 3.8e4 (n=31); SAT and split hybrid censored	current
`research/ecbench_frobenius_orbit_vs_rho_20261014` 2026-10-14 Frobenius-orbit IC fully priced vs rho-frob	engineering	Constant-only: orbit IC never below rho-frob on eight Koblitz curves; IC/rho-frob 2.96 (n=13) to 18.2 (n=41); slope beta 0.165 [0.117, 0.213] per ln r.	current
`research/ic_m3_algebraic_vs_mitm_20261014` 2026-10-14 algebraic/SAT decomposition vs MITM floor	diagnostic	No algebraic arm beats the pair-table floor: cheapest is 303x floor at n=13 and 3.8e4x at n=31 (m=3); slopes F4 0.534, F6-IC 0.583 vs floor 0.207; m=4 wide-chain 0.805 vs 0.333; SAT undetermined.	current
`benchmarks/20261015-ecdlp-speedups` 2026-10-15 orbit fast path and j=0 relation hoists (copy of research/ecdlp_speedups_ 20261015) [adm:listed-pre-gate; sweep:U/U; lb:no]	engineering	orbit fast path whole-process 60.9 s to 37.5 s (1.63x); j=0 relation stage 13.99 to 11.90 s (1.17x); shared host, 3 reps, no A/A, unvalidated	current
`benchmarks/20261015-ic-m4-wide-chain-vs-mitm` 2026-10-15 wide u128 chained-S3 solver m=4 vs floor (copy of research/ic_m3_ algebraic_vs_mitm_20261014) [adm:listed-pre-gate; sweep:VC; lb:no]	null	slope 0.805 [0.717, 0.872] vs floor 0.333 on n=11-23; / floor 5.5e2 (n=11) to 3.1e4 (n=23), >=1.2e5 at n=29 (censored); 70/70 exact	current
`benchmarks/20261015-u64-rootbatch` 2026-10-15 u64 RootBatch shared inversion (copy of research/u64_root_batch_ 20261015) [adm:listed-pre-gate; sweep:U/X; lb:no]	engineering	whole-process median 76.4 s to 23.4 s (3.26x; 1.87x min-to-min); rank stage 2.95x and per-target 6.09x corrected to 2.47x and 5.00x (median vs median); no A/A, L0	corrected; see `research/validation_sweep_20261019/B-benchmarks/verdicts.jsonl(row44.2)`
`research/cyclotomic_falsifier_20261015` 2026-10-15 cyclotomic gamma-weight base yield falsifier	null	Yield clause survives as yield-neutral: cyclotomic base inside 1999 random Frobenius-stable bases (smallest p 0.363, max deviation 0.65%); vs compact-orbit-scan it is costlier in 2 of 4 pairs (+2.2%, +12.2%), unresolved in 2.	current; see `research/validation_sweep_20261019/B-benchmarks/recompute/SWB-RECOMPUTE-09.jsonl(positive-controlpmismatchonw=1cells)`
`research/ecdlp_speedups_20261015` 2026-10-15 engineering speedups: compact-orbit fast path, j=0 IC	engineering	Koblitz compact-orbit fast path 1.63x median process wall (targets 1.95x), identical outputs; prime j=0 IC ~1.17x median relation-stage wall; wall-clock medians of 3 reps, S-ratio to rho unchanged; sweep B: unvalidated.	current; see `research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md(rows79.1,79.2:unvalidated)`

Study	Class	Result	Status
`research/linear_harvest_followup_20261015` 2026-10-15 linear-harvest follow-ups (draft protocol)	blocked	Draft, not frozen; no data collected.	open
`research/decomposition_programme_20261016` 2026-10-16 ECC2K-130 summation decomposition plan	accounting	Plan, not a protocol: best decomposition (geometric V linear pair oracle) stays 42.47 bits (bits only) and 55.22 bits (charged, omega=2) above matched rho 2^60.8090 at n=131.	current
`research/ic_probe_latency_20261017` 2026-10-17 n83 IC probe latency vs pointer-chase null	blocked	Frozen launcher only; no run has happened.	open
`benchmarks/20261018-k-conversion` 2026-10-18 K = Macaulay column touch per rho step (copy of research/k_conversion_ 20261018) [adm:listed-pre-gate; sweep:-; lb:no]	accounting	K pooled m=4 median 4.48 [3.96, 16.77]; 83 at 17,902 columns; +7.62 and +10.32 bit rows become +9.78 and +12.48 at median K (derived, not an improvement)	current
`research/koblitz_ic_new_leads_20261018` 2026-10-18 new Koblitz IC leads literature screen	literature	Four new leads; L1 lambda-coordinate bases close on arrival (verified); distance unchanged: m=4 row +10.32 bits (2^71.13 vs 2^60.81); quotes Table A 5.23-21x.	current
`research/advance` UNREAD N83 five-divisor profile script (no README)	tooling	no number (directory holds only five_divisor.py and n83_five_divisor_profile.json; no README).	current
`research/ecc2k130_relations` n.a. ECC2K-130 three/ four-point relation sweeps	diagnostic	Derived boundary (results/README read; no top-level README): section-1 m=4 total 2^99.19 vs rho 2^60.809 (+38.38 bits); four-point orbit sweeps complete for 8 normal bases.	current
`research/ic_tool_program` n.a. ic tool programme rounds and rule comparisons	accounting	Rule v4 (24 Oct): recipe PARTIAL, only n=57 improves (0.953 [0.904, 0.998]); cold IC/rho 9.9 to 43x with rho at canonical batched step; rule v8 matched rho 4.67x-18.07x; v3 5.23-20.97x.	current

C.2 Random and prime-degree binary curves

181 entries: 23 campaigns collapsed from 167 entries, and 14 single entries.

Study	Class	Result	Status
first-fall vs solving degree programme (8 entries, 2026-05-29..2026-10-08)	null	8 members. FFD 3 throughout while refutation degree grows (6 at l=3, 7 at l=5); Delta_low predicts D* at family level; offensive levers no closer than 9 bits to 2^N; line status: defensive evidence, open.	open
Members: `notes/index-calculus/RESEARCH_DEGREE_REDUCTION.md`, `notes/index-calculus/RESEARCH_DESCENT_CROSSOVER.md`, `notes/index-calculus/RESEARCH_DREG_MEASUREMENT.md`, `notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md`, `notes/index-calculus/RESEARCH_FALL_DEGREE_BOUNDS.md`, `notes/index-calculus/RESEARCH_FFD_MEASUREMENT.md`, `notes/index-calculus/RESEARCH_FFD_PROOF_COMPLEXITY.md`, `notes/index-calculus/RESEARCH_FFD_WORKFLOW.md`			
toy binary IC, isogeny neighbours, quotients (7 entries, 2026-09-22..2026-09-24 (7 members))	accounting	Toy IC slower than plain rho (best 2.33x); no consistent isogeny-neighbour or coefficient advantage; quotient helps 3 summands (0.26-0.31) but not 2 (1.06-1.32).	current
Members: `toy_binary_elimination_20260924`, `toy_binary_formula_audit_20260924`, `toy_f5_neighbors_20260924`, `toy_ic_e2e_candidates_20260922`, `toy_quotient_controls_20260924`, `toy_subfield_relation_geometry_20260922`, `toy_symmetry_quotient_20260924`			
standalone Boolean solver and Macaulay construction studies (22 entries, 2026-09-22..2026-10-03)	engineering	22 studies; best: packed direct construction 3.48-4.90x (stage), linear-tail kernel 3.59x, quadratic state 2.40-2.51x, packed residuals 1.56-1.67x complete solve; final graded tail reuse 1.779-1.826 n24, pivot replay 0/64; no universal 2x.	current
Members: `boolean_affine_substitution_20260923`, `boolean_basis_transport_20260923`, `boolean_byte_sieve_20260923`, `boolean_construction_reduction_20260922`, `boolean_degree_replay_20260922`, `boolean_envelope_indexed_20261002`, `boolean_graded_tail_reuse_20261003`, `boolean_halfword_20260923`, `boolean_implicit_minors_20260923`, `boolean_initial_restriction_20260923`, `boolean_linear_fibers_20260923`, `boolean_linear_tail_20260923`, `boolean_packed_construction_20260922`, `boolean_pivot_screen_20261003`, `boolean_product_schedule_20260922`, `boolean_projected_fibers_20260923`, `boolean_quadratic_frontier_20260923`, `boolean_quadratic_state_20260923`, `boolean_rank_coordinates_20260922`, `boolean_schedule_construction_20260923`, `boolean_solve_trace_20260923`, `boolean_support_envelope_20260922`			

Study	Class	Result	Status
Inherited F4 engine and its accounting (8 entries, 2026-09-22..2026-10-22)	engineering	8 dirs. Same-tree 26.1x fewer word ops than matrix F4 (round 1); overhead 0.847x and remap 0.923x instructions; eager rewrite 1.40-1.62x; shift runs 1.22-1.70x; #1774 pricing makes pre-#1774 F4 S lower bounds; BMI2 path causes cross-ISA count drift. Defaults hold on Arm.	current; see `research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`
Members: `arm64_f4_defaults_20261022`, `f4_pricing_jump_20261018`, `inherited_f4_20260922`, `inherited_f4_eager_rewrite_20261011`, `inherited_f4_eager_runs_20261016`, `inherited_f4_overhead_20261005`, `inherited_f4_remap_20261007`, `inherited_f4_replay_determinism_20261018`			
plain Macaulay/XL refutation degree, m=3,4 (5 entries, 2026-09-23..2026-10-06 (5 members))	diagnostic	Plain degree tracks l not n: m=3 reads 5 (l=2), 6 (l=3,4 at S>=-2), exactly 7 at l=5; m=4 reads 5, 6, >=7 at l=1..3; within one of semi-regular; FFD 2-3 throughout.	current
Members: `dreg_degree7_ell5_20260930`, `dreg_ell_grid_20260925`, `dreg_fixed_surplus_20260923`, `dreg_m4_chain_20261006`, `dreg_surplus_control_20260925`			
Boolean F4 solver-stage micro-optimisations (5 entries, 2026-09-26..2026-09-27)	engineering	5 studies; kept: bitmap seen-set (full F4 1.199), batched pair inserts (1.063), five-column tables (1.079); refuted: unsorted products (0.857), degree-bucket queue (1.002); solver-stage only, no IC/DLP claim.	current
Members: `boolean_f4_batched_pairs_20260926`, `boolean_f4_bitmap_20260926`, `boolean_f4_degree_queue_20260927`, `boolean_f4_unsorted_products_20260926`, `boolean_f4_wide_tables_20260926`			
shared GF(2) M4RI-style eliminator kernels (16 entries, 2026-09-26..2026-09-29)	engineering	16 dirs; kept: branchless AVX2 strip (default, 1.210x one-thread), AVX2 table XOR (opt-in, 1.088x), table reuse (opt-in, 2.030x vs default in combination); 10 pivot/panel/batching variants refuted; 2 phase profiles.	current
Members: `gf2_avx2_table_20260928`, `gf2_avx2_table_build_20260929`, `gf2_bounded_pivot_20260929`, `gf2_branchless_strip_20260929`, `gf2_column_panels_20260927`, `gf2_deferred_rref_20260926`, `gf2_eight_tables_20260929`, `gf2_elim_phase_profile_20260929`, `gf2_fused_weighted_pivots_20260929`, `gf2_minweight_pivots_20260929`, `gf2_pivot_phase_profile_20260929`, `gf2_sampled_pivot_20260929`, `gf2_stratified_pivot_20260929`, `gf2_table_reuse_20260929`, `gf2_word_batch_20260927`, `gf2_zero_table_index_20260929`			
matrix-F5 GF(2) solver-stage micro-optimisations (24 entries, 2026-09-27..2026-10-17)	engineering	24 dirs; best kept stack: combined options 1.530x, direct pack prior/new 1.210x (default since 2026-10-17), all defaults 1.11-1.27x largest cases; 16 unpack/pivot/layout/packing variants rejected plus mainline row-basis port failed; 2x complete-call goal unmet by this line.	current
Members: `f5_avx2_gather_unpack_20260929`, `f5_avx2_nibble_unpack_20260929`, `f5_avx512_unpack_20260929`, `f5_bounded_weight_pivots_20260930`, `f5_bounds_free_unpack_20260930`, `f5_byte_colex_20260930`, `f5_byte_table_unpack_20260929`, `f5_combined_2x_20260928`, `f5_compact_cols_unpack_20260929`, `f5_compact_unrolled_unpack_20260929`, `f5_direct_pack_20260929`, `f5_echelon_20260927`, `f5_flat_rows_20260929`, `f5_further_2x_20260929`, `f5_gf2_defaults_20261017`, `f5_initial_density_order_20260929`, `f5_mainline_port_20261002`, `f5_nibble_subset_unpack_20260929`, `f5_one_pass_unpack_20260929`, `f5_popcnt_unpack_20260929`, `f5_projected_sparse_basis_20260930`, `f5_row_build_20260928`, `f5_stratified_weight_pivots_20260930`, `f5_x86_guarded_unpack_20260930`			
m=4 Semaev decomposition exponent and symmetry lever (3 entries, 2026-09-28..2026-09-29 (3 members))	null	m=4 cost slope 0.985-1.060 bits per unit n (closed vs 0.25 threshold) at n<=19; head engine 1.4-3.2x constant gain only; torsion symmetrisation is a constant lever (slope 1.033).	current
Members: `ic_m4_exponent_audit_20260928`, `ic_m4_head_engine_20260929`, `ic_symmetry_lever_slope_20260929`			
point-decomposition oracles for binary Semaev (4 entries, 2026-09-30..2026-10-10)	engineering	4 members. Pair-and-root-find oracle 926x over triple enumeration; SAT/CDCL ~1 conflict per triple; Hamming ideals no gain; SAT route exhausted as exponent lead.	current
Members: `notes/index-calculus/RESEARCH_HAMMING_IDEAL_PDP.md`, `notes/index-calculus/RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md`, `notes/index-calculus/RESEARCH_SAT_SEMAEV.md`, `notes/index-calculus/RESEARCH_SEMAEV_DECOMPOSITION.md`			
m=4 tree-split cut-out degree screens (3 entries, 2026-09-30..2026-10-11)	null	3 members; all rounds CLOSED: cut-out degree equals random sets (gaps 0, one degenerate -1); frob miscount (six -> five) corrected.	current
Members: `ic_tree_split_cutout_20260930`, `ic_tree_split_structured_v2_20261011`, `ic_tree_split_structured_v_20261011`			
Boolean F5 affine-batch reuse campaign (5 entries, 2026-10-02..2026-10-02)	null	5 studies; structural signature 32/32 retained; validation-inclusive ceiling 1.61 rejected, call-only ceiling 4.398-4.774 passes; actual graded reuse 1.784-1.838 fails 2x; support-aware follow-up never run (open).	open
Members: `boolean_f5_affine_signature_20261002`, `boolean_f5_amortized_ceiling_20261002`, `boolean_f5_call_only_ceiling_20261002`, `boolean_f5_graded_batch_20261002`, `boolean_f5_support_aware_20261002`			

Study	Class	Result	Status
exact support rank certificate for F5 output (5 entries, 2026-10-02..2026-10-03)	engineering	5 dirs; support-split -> fixed cut-19: reduction XORs 28.08% of selective echelon; fresh-seed Linux confirmation PASS 3.018-3.070x one-thread complete call (LB >=2.995x); earlier width-8/width-6 Linux runs failed original gates.	current
Members: `f5_support_cut19_call_20261002`, `f5_support_cut19_confirm_20261003`, `f5_support_cut19_final_recheck_20261003`, `f5_support_rank_bits_20261002`, `f5_support_split_call_20261002`			
rr/x4 Groebner refutation ladder on K0/K1 (3 entries, 2026-10-03..2026-10-05 (3 members))	diagnostic	x4 refutes at 8, 9, 10, 11 for l=3..6 (4 above sharp first-fall bound 7 at l=6), rr at 8; semi-regular window fits x4 on 32/32 draws, rr falls below on 8; registered law verdict neither.	current
Members: `ic_dense_ladder_20261004`, `ic_gb_ladder_20261003`, `ic_semiregular_law_20261005`			
F6-IC geometric-closure solver line (13 entries, 2026-10-03..2026-10-16)	engineering	13 dirs. Prepared n17 F4/F6 online 1.291-1.583x, never 2x; six follow-up micro-optimisations rejected; gate-first default (1.336-1.733x fewer XORs); ladder: charged geometry makes F6-IC a relabelling (S/rho 54.75x / 2,091x / 2.721x at m=13/19/23); n17 rho_online_ms/IC_online_ms 0.000515-0.0221.	current; see `research/f4_pricing_jump_20261018/README.md;research/validation_sweep_20261019/D-known-issues/verdicts.jsonlitem2`
Members: `f6_gate_first_x86_20261016`, `f6_ic_adaptive_pair_20261005`, `f6_ic_bitmap_columns_20261006`, `f6_ic_compact_refuted_20261006`, `f6_ic_direct_fused_pack_20261006`, `f6_ic_ecbench_ladder_20261005`, `f6_ic_generic_direct_pack_20261006`, `f6_ic_geometric_closure_20261003`, `f6_ic_inherited_basis_profile_20261006`, `f6_ic_phase_profile_20261006`, `f6_ic_rho_online_gap_20261006`, `f6_ic_support_local_profile_20261006`, `f6_ic_support_local_stream_20261006`			
XL reach on bilinear/ trilinear S3 at n=131 (0 entries, 2026-10-06 (2 members))	null	Best charged gap at n=131 stays 71.82 bits (omega=2) / 78.15 (omega=2.81) from linearization; trilinear worse (87.16 / 96.51).	current
Members:			
PDP SAT/SMT/WDSat programme and kernel speedups (6 entries, 2026-10-07..2026-10-11 (6 members))	diagnostic	SAT/SMT/WDSat are enumerators (~1 conflict per tuple); SMT within 0.44-2.63x native; pairs-and-solve kernel 50-62x PS-0 baseline; isogenous curves algebraically identical.	current
Members: `pdp_followups_20261010`, `pdp_isogeny_probe_20261008`, `pdp_oracle_speedup_20261011`, `pdp_preregistrations_20261007`, `pdp_sat_smt_survey_20261010`, `pdp_smt_oracle_20261008`			
Boolean F4/F5 instruction/allocation engineering (5 entries, 2026-10-11..2026-10-12)	engineering	5 dirs. Instruction counts on an eleven-system suite: fused readback -5.28% F4 / -11.02% F5; hashing -3.715% F5, -45.1% criterion; dense packing -3.947% F5; row build 42-50% fewer bytes; history is tooling. No wall-time claims.	current
Members: `f4_f5_dense_packing_20261012`, `f4_f5_fused_readback_20261012`, `f4_f5_hash_20261012`, `f4_f5_history_20261012`, `f4_f5_row_build_20261011`			
binary MITM pair-table oracle ladder l=8-12 (3 dirs) (3 entries, 2026-10-15..2026-10-16)	engineering	94-277x (l=8) to 2,473.9/4,244.2x (l=9,10) and 928.0/2,187.2x (l=11,12) over pairs-and-solve; sweep finds count/median mismatches, unvalidated	current
Members: `20261015-binary-mitm-ladder-l9-l10`, `20261015-binary-mitm-oracle-l8`, `20261016-binary-mitm-ladder-l11-l12`			
MITM pair table vs pairs-and-solve, l=8..12 (3 entries, 2026-10-15..2026-10-16 (3 members))	engineering	Pair-table oracle 94-277x (l=8), 2,474x (l=9), 4,244x (l=10), 928x (l=11), 2,187x (l=12) over Gaudry pairs-and-solve; 0 disagreements; recompute mismatches on l=8 and l=12 figures.	current
Members: `binary_mitm_oracle_l11_l12_20261016`, `binary_mitm_oracle_l8_20261015`, `binary_mitm_oracle_l9_l10_20261015`			
factor-base shape search (2 entries, 2026-10-15..2026-10-18)	null	2 members: G1, G3, G5 killed on scaled cells (gap 0.000); frozen G1 cells not testable (probe killed at 14,405 s).	current
Members: `shape_search_g1_frozen_cells_20261018`, `shape_search_g1_g6_20261015`			
MutantXL closure degree and ECC2K-130 m=4 rows (6 entries, 2026-10-16..2026-10-24 (6 members))	diagnostic	Closure refutes at 3-4 where XL needs 5-7; rises past 4 at m=3, l=7; (4,6) holds at 4 (l=7 survives); 1-2 guesses restore 4 at 1/116-1/260 cost; m=4 row +8.43 to +58.27 bits, never below rho.	open
Members: `dreg_closure_driver_20261016`, `dreg_closure_m4l6_20261018`, `dreg_closure_mutantxl_20261016`, `dreg_closure_scale_20261016`, `dreg_closure_specialise_20261017`, `dreg_hybrid_guess_20261024`			
GF(2^83) d18 S3 norm SAT/root-finding cycles (11 entries, 2026-10-17..2026-10-19)	diagnostic	11 members; best: cutoff-8 circuit cuts AND 60.485% / clauses 32.914%; 1237/1800 cells certified empty; final: every hard search UNKNOWN (0 witnesses); no end-to-end or 8x gain; line open.	open
Members: `autolab_cycle_20261017_v10`, `autolab_cycle_20261017_v2`, `autolab_cycle_20261017_v3`, `autolab_cycle_20261017_v4`, `autolab_cycle_20261017_v5`, `autolab_cycle_20261017_v6`, `autolab_cycle_20261017_v7`, `autolab_cycle_20261017_v8`, `autolab_cycle_20261017_v9`, `autolab_cycle_20261019_v1`, `autolab_cycle_20261019_v3`			

Study	Class	Result	Status
`research/weil_factor_composition_20260914` 2026-09-14 Weil-friendly factor-base composition	diagnostic	Component-coordinate S3 solver exact on all checks; charts_linear, enumerate, pair_table 96/120 DLPs verified vs ambient_f4 33/120; total operations, S and ratios null.	current
`research/notes/index-calculus/RESEARCH_QUASI_SUBFIELD.md` 2026-09-19 quasi-subfield polynomials census over F_2^n	literature	Exhaustive census over F_2^7 finds non-subfield quasi-subfield polynomials below the stripped HKPYY Lemma 4.1 bound (answers tightness question); argues barrier lies elsewhere; two corrections to earlier notes. (derived + census)	current
`research/wdsat_rr_20260920` 2026-09-20 WDSat on Riemann-Roch incidence systems	null	No README: WDSat solves n=5 targets (47 vs 51,425 conflicts with/without XG) but crashes rc=-11 at n=9 (static-allocation ceiling ~21-24M entries); CryptoMiniSat solves n=9 targets in 0.4-2.6 s.	current
`research/hamming_ideal_pdp_20260930` 2026-09-30 Hamming-ideal ISD-like Groebner PDP oracle	accounting	H1 refuted at every size: MultiSolve exact but tame only after 7.4-18.2 of n coordinates assigned; GroebnerSafe calls grow with exponents 1.9-2.5 vs candidate count; subspace baseline tame at root.	current
`research/nonlinear_factor_labels_20261002` 2026-10-02 nonlinear factor-base label gauges, GF(2^7)	relabelling	No nonlinear label class refutes through degree 6 (identity refutes at 6); winner cuts degree-6 columns 53% but raises input degree 3->5 and does not refute: relabelling, negative.	current
`research/notes/index-calculus/RESEARCH_SEED_GRIND_TRAPDOOR_20261002.md` 2026-10-02 seed-grind hidden GHS trapdoor protocol	tooling	Preregistered protocol: can seed freedom grind a composite-N binary curve to a hidden GHS weakness while passing a realistic rigidity check; all measurements PENDING. (plan)	open
`docs/ic/ECBENCH_LARGE_PRIME_PROTOCOL.md` 2026-10-04 ECBench n-1 / large-prime correctness protocol	tooling	[protocol] Frozen correctness protocol for m=n-1 decompositions with 0/1/2 large-prime elimination on ECBench corpus; no performance claim. (plan)	current
`research/sect113r1-weak-curve-20261006` 2026-10-06 sect113r1 weak-curve audit	diagnostic	Pre-admission diagnostic: CLASS_WEAK by size (56.325748 bits generic work); two degree-5 isogeny routes TRANSFER_ONLY_SPEEDUP_NOT_ESTABLISHED; GHS and Semaev routes no conclusion.	current
`research/fall_degree_bounds_20261008` 2026-10-08 HKY last fall degree of descended Semaev systems	diagnostic	d_F is 3 or 4 at every reachable size (m=2 to n=32; m=3 l<=6; m=4 l<=3; m=5,6 to N=54) while plain D* rises 5 to >=7; random controls rise; Semaev d_F one below semi-regular at n<=18.	current
`research/g2_equivariant_f4_20261008` 2026-10-08 Equivariant G2 block decomposition of Macaulay rank	diagnostic	Frobenius-sigma equivariant blocks reproduce the full F2 rank (identity HOLDS) at n=7-23, degree 3; block rank 31.2x-39.1x faster at n=11/13 and 104.6-128.1x at n=17/19/23 versus full rank (timing tables only, no README).	current
`research/f4_f5_f6_cross_family_20261011` 2026-10-11 F4/F5/inherited F4/ F6-IC across binary families	diagnostic	W(matrix F4)/W(inherited F4) 198.6 [174.5, 212.8] (K13) and 222.1 [173.4, 544.0] (R13), different splitting trees; F5/F4 0.6465/0.6414 same tree; F6-IC log2 saving 0.29-0.37 but lower-bound S 65-1,350x inherited F4; every IC arm above rho.	corrected; see `research/f4_pricing_jump_20261018/README.md;research/validation_sweep_20261019/D-known-issues/verdicts.jsonlitem2;research/validation_sweep_20261019/A-leaderboard/verdicts.jsonl(unvalidated: replay_okfalse,pricing_okfalse)`
`research/pdp_shape_replay_20261015` 2026-10-15 F4Remake-style root shape replay	engineering	Replay costs more on all 9 registered cells: 0.753-0.985x (chain 0.970, chain-holdout 0.954, holdout 0.983); verdicts identical; e3 enumeration 553-34,901x the table oracle.	current
`research/lead19_projected_base` 2026-10-23 LEAD-19 cofactor-projected base, random binary	engineering	Cold S(A')/S(B) 1.29 (f2m15), 1.58 (f2m21), 3.36 (f2m27) on published rungs; still 6.1x-14.4x above matched rho; not yet a leaderboard row.	current
`research/nagao_relations` UNREAD Nagao/ Riemann-Roch relation encoding prototype	engineering	Implementation and toy correctness work on function-coefficient search with support constraint; not an ECDLP speedup (no headline number read).	current

C.3 Prime-field curves

120 entries: 16 campaigns collapsed from 94 entries, and 26 single entries.

Study	Class	Result	Status
PKM tower F4 solving-degree line (10 entries, 2026-09-24..2026-10-18)	diagnostic	10 dirs. Kummer m=2: D=5 to N=18, 6 at N=20,22; m=3: 6,6,7,7 over N=9-18; m=4: 7 at N=12,16. Final (N=21, m=3): degree-7 closure exhausted without refuting on two engines, implying D>=8; read inconclusive; next step needs >=256 GB.	open
Members: pkm_dickson_torus_20261015, pkm_lead1_m3n21_20261016, pkm_m3_n21_msolve_20261018, pkm_tower_pilot_20260924, pkm_tower_round2_20260925, pkm_tower_round3_20260925, pkm_tower_round4_20260926, pkm_tower_round5_20260926, pkm_tower_round6_20260930, pkm_tower_round7_20261006			
F_p F4 (f4_fp, generic field) engine line (6 entries, 2026-09-26..2026-10-16)	engineering	6 dirs. Deferred narrow mode 2 kept (1.64x); AVX2/AVX-512 lazy kernels refuted (1.04-1.06x); parity with msolve 0.96-1.22 (1 thread) after 29.8-54.1x; blocked pairs 0.78-0.89x on quad_n10/n11; generic-field engine 2.18-2.57x. Active.	current
Members: f4_fp_blocked_pairs_20261016, f4_fp_msolve_parity_20261011, f4_generic_field_20261015, fp_f4_lazy_avx2_20260928, fp_f4_lazy_avx512_20260928, fp_f4_narrow_20260926			
P-256 Dickson factor-base rounds 1-24 (25 entries, 2026-10-04..2026-10-06)	diagnostic	24 dirs on icv1-fp256 (= P-256). Toy degree reductions (4 to 3 to local 2) via branching and split image trees; no round produced a natural relation for a non-planted target; final boundary 392-404x rho (round 24), best end-to-end projection 16.2-19.4 bits above rho (round 20).	current
Members: icv1-fp256-t89188191154553853111372247798585809583-f188c491_affine_bitbox_degree_round1_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_atomized_s5_round10_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_balanced_s5_round9_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_branched_s3_round4_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_deep_branch_round5_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_fb1_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_quadratic_s3_round3_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_residual_scaling_round6_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_s3_image_filter_round7_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_dickson_terminal_degree_round2_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_factor_base_symmetry_round22_20261006, icv1-fp256-t89188191154553853111372247798585809583-f188c491_fb_log_transport_round21_20261006, icv1-fp256-t89188191154553853111372247798585809583-f188c491_image_atomized_s5_round11_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_negation_cross_colour_round24_20261006, icv1-fp256-t89188191154553853111372247798585809583-f188c491_root_orbit_fb_round23_20261006, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_batch_collision_round20_20261005, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_batch_inversion_round17_20261005, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_image_growth_round13_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_image_width_round14_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_multilevel_selector_round19_20261005, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_outer_scan_round18_20261005, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_packed_atom_round16_20261005, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s17_target_join_round15_20261005, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s3_image_transfer_round8_20261004, icv1-fp256-t89188191154553853111372247798585809583-f188c491_s9_image_growth_round12_20261004			
P-256 isogeny-class enumeration screens (6 entries, 2026-10-04..2026-10-07)	null	6 dirs. Up to 1,064,000 distinct P-256-isogenous curves enumerated and replayed; no structural screen fired; no factor-base test selected; speedup unset.	current
Members: p256_isogeny_million_20261006, p256_isogeny_multigraph_64k_20261006, p256_isogeny_s3_cairn_20261004, p256_isogeny_shared_task_20261005, p256_isogeny_walk_20261004, p256_j_windows_20261007			
native isogeny walker on P-256/P-224/P-192 (5 entries, 2026-10-04..2026-10-08)	tooling	5 members: walker enumerated 20,000-curve classes on P-256/P-224 and P-192 (depth-0 volcanoes if C prime); Phi_l construction l^5 caps l at 61; no ECDLP cost.	current
Members: isogeny_walk_20261004, isogeny_walk_p192_20261004, isogeny_walk_traits_20261004, isogeny_walker_engineering_20261008, large_prime_isogeny_degree_20261008			
CryptoPro-B CM endomorphism GLV (3 entries, 2026-10-05..2026-10-06)	engineering	3 dirs. Constructive speedup only: GLV-2 1.252x to 1.417x faster scalar multiplication; CM map verified. Not an attack.	current
Members: cryptopro_b_chain_sweep_20261006, cryptopro_b_cm_map_checks_20261005, cryptopro_b_glv_chain_20261005			
large prime-degree isogenies from P-192/P-224 (4 entries, 2026-10-08..2026-10-12)	tooling	4 dirs. Maps up to degree 10453 (P-192) and 1471 (P-224); 100-map campaign complete; 2000-map campaign at 153 sealed maps, open. No ECDLP claim.	open
Members: p192_p224_2000_more_large_prime_isogenies_20261010, p192_p224_hundred_large_prime_isogenies_20261010, p192_p224_large_degree_isogenies_20261008, p192_p224_large_degree_isogenies_20261009			
NIST/SECG trapdoor and outlier audits (6 entries, 2026-10-08..2026-10-17)	null	6 dirs. No anomaly in P-256 coefficients, no seed hits, P-256 n-1 outlier claim not established after controls; sextic-twist weak siblings exist for j=0 SECG curves (twist fact). No ECDLP weakness claimed.	current
Members: nist_cyclotomic_controls_20261011, nist_cyclotomic_point_heights_20261011, nist_trapdoor_slot_program_20261011, p224_hypothesis_ranking_20261008, p256_trapdoor_round2_20261012, safecurves_trapdoor_slots_20261017			

Study	Class	Result	Status
pair-table root finding on 20-bit prime ecbench (6 entries, 2026-10-12..2026-10-14 (6 members))	engineering	Pair table 16-20x matched rho counted GAE (lower bounds); B64 0.83x B32; m=3 1.43x m=2; batch table and lane/buffer changes exact, counted ratio 1.000; 7.46x rho with lanes.	current
Members: `ecbench_rootfinding_alloc_20261014`, `ecbench_rootfinding_arity_20261013`, `ecbench_rootfinding_assessment_20261012`, `ecbench_rootfinding_batchtable_20261013`, `ecbench_rootfinding_lanes_20261013`, `ecbench_rootfinding_tuning_20261013`			
PKM bases priced end to end vs rho (4 entries, 2026-10-13..2026-10-15)	engineering	4 dirs. Kummer base IC 29-141x rho-negation (20-28 bits); isogeny base 1.00-1.33x plain base, 18-136x rho; supersingular aux removes search cost, same D; tower structure buys nothing for a table oracle.	current
Members: `pkm_isogeny_tower_ecbench_20261015`, `pkm_isogeny_tower_p192_20261015`, `pkm_supersingular_aux_20261015`, `pkm_tower_ecbench_20261013`			
exact two-point small-x yield on standard curves (2 entries, 2026-10-14)	null	2 dirs. Exact yield 2B^2/(n-1); loses to rho on P-192 and all 33 SECG curves by at least 2^40.9 in trials.	current
Members: `prime_small_x_p192_20261014`, `secg_small_x_20261014`			
p61-i0025 class census (2 entries, 2026-10-14..2026-10-23)	tooling	2 dirs. Certified class size 358,094,304 fully accounted by census plus terminal closure; adjacency and uniqueness open.	open
Members: `p61_isogeny_class_20261014`, `p61_terminal_closure_20261023`			
PKM isogeny-tower auxiliary search (2 dirs) (2 entries, 2026-10-15)	accounting	search 10.06 us/draw (0.326x); charging search removes advantage (isogeny/plain up to 1.331); P-192 search ~2^74 GAE vs rho 2^95.8	current
Members: `20261015-isogeny-p192-search`, `20261015-pkm-isogeny-tower`			
small-x relations on p=2^61-1 isogeny class (5 entries, 2026-10-16..2026-10-18)	null	5 dirs. Final: 1,228,800,000 target trials over 1,024 measured curves, zero relations, zero solves; class enumeration incomplete.	open
Members: `prime61_disk_continue_20261018`, `prime61_disk_extend_20261018`, `prime61_disk_walk_20261017`, `prime61_factor_base_20261016`, `prime61_isogeny_20261016`			
NIST seed and base-point provenance (4 entries, 2026-10-16..2026-10-21)	diagnostic	4 dirs. Seeds reproduced as 1997-98 drand48/lrand48 output; skip rule and later base points unexplained; one 444-bit cofactor open. No ECDLP effect.	open
Members: `nist_basepoint_provenance_20261017`, `nist_seed_provenance_20261016`, `nist_seed_skip_filter_20261017`, `nist_skipped_curve_dissection_20261021`			
SNFS analogues and special primes (4 entries, 2026-10-16..2026-10-23)	literature	4 dirs. Only small-embedding-degree pairing families open an SNFS-type route; Solinas and other special shapes change neither #E distribution nor k.	current
Members: `elliptic_snfs_survey_20261016`, `snfs_prime_families_20261023`, `solinas_distribution_20261023`, `special_prime_curves_20261023`			
`research/p224_attack` 2026-05-18 invalid-curve attack demo on P-224 ECDH oracle	tooling	Rust Biehl-Meyer-Mueller invalid-curve attack against a vulnerable (unvalidated) ECDH oracle; implementation attack, not P-224 ECDLP; no README, no number. Date is a reorg commit (created on or before 2026-05-18).	current
`research/notes/index-calculus/RESEARCH_INDEX_CALCULUS_FACTOR_BASE.md` 2026-06-07 structured factor bases over GF(p): four fronts	literature	Conditional no-go via Hilbert-function invariance; Weil-descent analogue over F_p hopeless; HKYY last-fall machinery does not port; initial-minor search Theta(n) (0.98-1.07 n over 245x range). (derived + measured)	current
`research/notes/index-calculus/RESEARCH_HIGHER_SEMAEV.md` 2026-09-19 S3-S6 Semaev polynomials over prime fields	tooling	Module computes S3..S6 univariate in last variable; degrees 2,4,8,16 match 2^{n-2}. (tooling)	current
`research/notes/index-calculus/RESEARCH_SYMMETRIZED_SEMAEV.md` 2026-09-19 FGHR symmetrised summation polynomials module	tooling	Elementary-symmetric decomposition of prime-field S3; claimed monomial reduction ~1.7x (S3) to ~100x (S8); module only. (tooling)	current
`research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md` 2026-09-24 PKM tower algebraic decomposition oracle, prime field	null	Pilot: F4 solving degree 4-5 (m=2, N<=18); round 2: not bounded, rises to 6 at N=20,22 (m=2), 7 at N=15 (m=3); linear growth undecided; null family behaves the same. (measured toy)	current
`research/prime_field_smt_20261005` 2026-10-05 cvc5 SMT PDP on 20-bit prime curve	null	All three frozen 20-bit SMT formulations (QF_BV, QF_FF coordinates, QF_FF Semaev S3) returned unknown at 120,000 ms; native exact reference found the decomposition after 146 probes.	current

Study	Class	Result	Status
`research/p192-weak-curve-20261006` 2026-10-06 P-192 singular-cubic and CM relation lanes	diagnostic	Review verdict BREAKS applies to the singular nodal-cubic lane: IMPLEMENTATION_WEAK, not weak P-192; CM lane 103 scalar relations reproduced, complete-set negative INDETERMINATE; no P-192 weakness or speedup found.	current
`research/prime_fourier_flatness_exp7_20261007` 2026-10-08 Fourier flatness of x-predicates	null	Six abscissa predicates on four toy prime-order curves vs 512 random sets: no adjusted candidate lead; run date corrected to 2026-10-08.	corrected; see `research/prime_fourier_flatness_exp7_20261007/ERRATA.md(dateonly)`
`benchmarks/20261011-fiber-ic-scaling` 2026-10-11 fiber-aware IC vs matched rho scaling (copy of research/fiber_ic_scaling_ 20261011) [adm:not-listed; sweep:V; lb:no]	null	Envelope IC/rho grows from 8.4-8.8 to 35.1-37.0 (h=4), and 13.4-20.1 to 60.4-63.1 (h=72); 816 audited records, negative whole-method result	current
`research/sec2_curve_inventory_20261012` 2026-10-12 SEC 2 curve inventory and traits	tooling	53 name-by-edition parameter blocks (33 v1.0, 20 v2.0) agree with the catalogue; all 33 transfer-screening rows are verdict boundary, not measured gains.	current
`research/ic_exponent_falsifiers_20261013` 2026-10-13 C4 exponent-sum structure on PKM tower	null	Exponent sum of decompositions on PKM Kummer tower (p=946177, t=8) indistinguishable from random at k=1..6 (differences <1%); positive control reads 1.000; C4 simplest form falsified.	current
`benchmarks/20261014-secg-small-x-s3-cost` 2026-10-14 Semaev S3 small-x decision on SECG curves (copy of research/secg_small_x_ 20261014) [adm:listed-pre-gate; sweep:V; lb:no]	null	S3 per target at B=4096: 5.4 ms sect113r1, 42 ms secp160r1; table 2-5 us; B+1 relations by S3 are 52-280 bits slower than rho over the 33 curves; not a speedup	current
`benchmarks/20261016-j0-parallel-relations` 2026-10-16 j=0 relation collection rayon-parallel (copy of research/j0_parallel_ relations_20261016) [adm:listed-pre-gate; sweep:U; lb:no]	engineering	relations stage ~1025 ms to ~234 ms (4.4x) at 16 bits, ~39.6 s to ~13.2 s (3.0x) at 20 bits on 14 threads; counted work unchanged; no raw records committed	current
`research/curve_shortlist_20261016` 2026-10-16 46-59-bit target curves and deep volcanoes	tooling	Exact PARI construction of binary (58.00-bit), Koblitz (56.26-bit) and prime (56.00-bit, p=2^56-5) targets plus deep-volcano curves; no ECDLP measurement.	current
`research/j0_parallel_relations_20261016` 2026-10-16 parallel relation collection j=0 driver	engineering	Rayon relation search in j=0 staged IC: relation stage ~4.4x (16-bit) and ~3.0x (20-bit) faster wall at 14 threads, counted work unchanged. Sweep: no raw in repo for all claims.	current
`research/large_memory_runs_20261018` 2026-10-18 large-memory PKM tower runs on 124 GB EC2	blocked	PKM m=3 N=21 main run stopped by –max-dense cap with no verdict (B' would need 7.93e10 entries ~317 GB vs cap 2.03e10); identity run passed.	open
`benchmarks/20261019-ic-linalg-pipeline-sparse-cado` 2026-10-19 sparse-cado LA in ecbench ic.pipeline (copy of research/ic_linalg_pipeline_ 20261019) [adm:listed-pre-gate; sweep:-; lb:no]	engineering	S identical (ops ratio 1.0000); LA wall 0.72x (N=1,024) to 0.25x (8,192); oracle pair-table setup 84-89% of run at N>=8,192; gap to rho 149x at best	current
`research/a11_pipeline_pricing_20261019` 2026-10-19 A11 PKM-tower pipeline pricing feasibility	blocked	Swap NOT feasible: ecbench has no method that builds or solves a PKM-tower summation system, so no incumbent presentation and no S for either arm; nothing measured.	current
`benchmarks/20261020-ic-oracle-setup-blocked` 2026-10-20 cache-blocked pair-table oracle set-up (copy of research/ic_oracle_setup_ blocked_20261020) [adm:archived-incomplete; sweep:-; lb:yes]	engineering	set-up wall 2-4x vs batched builder and 4-8x vs scalar; S unchanged (456/456 identical); archived (incomplete), not admitted	current

Study	Class	Result	Status
`benchmarks/20261020-ic-planner-validation` 2026-10-20 ic plan planner validation (copy of research/ic_planner_validation_20261020) [adm:listed-pre-gate; sweep:-; lb:yes]	tooling	at w=0 plan beats every alternative on six curves (half base 1.36-1.56x, double 1.49-1.60x, rootfind 150-637x); at w=1 doubled base 2-9% cheaper; predicted gap to rho within 18% (12.6-25.8x)	current
`research/eccp131_bsgs_ic_20261020` 2026-10-20 ECCp-131 exact-parameter CLI integration	tooling	Exact ECCp-131 parameters wired into BSGS and IC command paths; no performance claim; 2026-10-22 three-point probe: zero natural setup rows, zero target rows, rank zero at B128; full relation-to-scalar path open.	open
`research/ic_oracle_setup_blocked_20261020` 2026-10-20 cache-blocked pair-table oracle set-up	engineering	Blocked builder: same table and S (456/456 identical); set-up wall 0.24-0.47x batched and 0.12-0.25x scalar (|F|=1,024-16,384); whole run 0.13x scalar at |F|>=8,192.	current
`research/ic_planner_validation_20261020` 2026-10-20 ic planner validated on unseen prime curves	engineering	Every planner choice beat its one-change alternatives on all six curves at w=0 (e.g. full-rank stop rule 3.64-4.56x worse, rootfind-subtract 171-439x worse); base 2N not separated at w=1.	current
`research/coppersmith_s3_theta2_20261023` 2026-10-23 Coppersmith lattice ceiling on S3	null	0/10 recovered in every cell with beta>=0.20 (96 cells, 960 trials) at l<=3, all positive controls pass; beta=1/4 ceiling stands at measured level; Howgrave-Graham tracks derived beta_det.	current
`research/notes/ecdlp-general/RESEARCH_LATTICE_ECDLP_ROUTES.md` 2026-10-23 lattice routes to prime-field ECDLP evaluated	literature	Every lattice intrinsic to (E,P,[k]P) has rank 2 (4 over Z[tau]); high-dimensional lattices pay for dimension; derived ceiling p^{5/6} on relation lattices; no campaign opened. (literature + derived)	current
`research/ecdlp_autolab` UNREAD AutoLab benchmark curve resistance map (v1-v6)	diagnostic	60+ phases, 10+ attack families ruled out on LMFDB benchmark curves; Pollard rho slope 0.4972 +/- 0.005 (theory 0.5); best C-extension 86x vs Sage Python.	current

C.4 Extension fields, covers and transfers

222 entries: 30 campaigns collapsed from 213 entries, and 9 single entries.

Study	Class	Result	Status
extension fields and isogeny-class walks (3 entries, 2026-09-12..2026-09-21)	accounting	3 members. Extension fields: >=2^131 (2^70.19x rho); isogeny class 2^65.06 vertices, 263 reachable; FFD sweep scoped negative.	current
Members: `notes/ecc2k130/RESEARCH_ECC2K130_EXTENSION.md`, `notes/ecc2k130/RESEARCH_ISOGENY_CLASS_SEARCH.md`, `notes/ecc2k130/RESEARCH_ISOGENY_DEGREE_SEARCH.md`			
rho-parity routes on E(F_p^k) (7 entries, 2026-09-18..2026-10-09)	accounting	7 members. Best: JV weak-class cover 8.1x rho at p=1009 (sieve claim 0.012x at 2^61.4, strong-rho comparison withheld); generic routes never close below 2^200 (k=4 JV 6,945x flat; k=5 2^233).	open; see `research/jv_sieve_strong_rho_20261019/review-20261020/README.md`
Members: `notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md`, `notes/index-calculus/RESEARCH_EXTENSION_FIELD_BOUNDARIES.md`, `notes/index-calculus/RESEARCH_GLV_INDEX_CALCULUS.md`, `notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`, `notes/index-calculus/RESEARCH_K5_TORSION_JOUX_VITSE.md`, `notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md`, `notes/index-calculus/RESEARCH_RHO_PARITY_PROGRAMME.md`			
CM/isogeny structural audits of deployed prime curves (6 entries, 2026-09-19..2026-10-16)	null	11 members. All structural nulls (secp256k1, P-256, NIST Solinas, EDS, volcano-floor rho); structural-completeness theorem draft corrected 2026-10-16; P-256 n-1 outlier withdrawn.	current
Members: `notes/cm-isogeny/PAPER_STRUCTURAL_COMPLETENESS.md`, `notes/cm-isogeny/RESEARCH_MESTRE_HOWE.md`, `notes/cm-isogeny/RESEARCH_P256_ISOGENY_COVER.md`, `notes/cm-isogeny/RESEARCH_PKM_CRITERION.md`, `notes/cm-isogeny/RESEARCH_SECP256K1_CM.md`, `notes/cm-isogeny/RESEARCH_VOLCANO_FLOOR_RHO.md`			
genus-2/3/4 IC vs rho and round-nine attempts (4 entries, 2026-09-19..2026-10-18)	accounting	4 members. Genus 2 0.28-0.67, genus 3 0.4-0.8, genus 4 slope -0.29 vs DP rho; all ratios lower-bound and unranked, reference later shown slow by up to 1.5x; round nine attempts 1-3 failed/superseded.	corrected; see `research/notes/index-calculus/RESEARCH_HYPERELLIPTIC_IC_RHO.mdroundsnine-twelve`
Members: `notes/index-calculus/RESEARCH_HYPERELLIPTIC_IC_RHO.md`, `notes/index-calculus/round9/attempt2/ATTEMPT.md`, `notes/index-calculus/round9/attempt3/ATTEMPT.md`, `notes/index-calculus/round9/run/ATTEMPT.md`			

Study	Class	Result	Status
ecc2k130-isogeny-transport (14 entries, 2026-09-24..2026-09-28)	mixed	14 dirs. ECC2K-130 degree-263/degree-7 isogeny transport built and certified; toy transported/native bases 1.9-4.3x costlier; degree-73 descent criterion failed.	current
Members: ecc2k130_direction_review_20260924, ecc2k130_dual_transport_20260925, ecc2k130_endo_ring_263_20260925, ecc2k130_factor_base_pilot_20260924, ecc2k130_factor_base_replication_20260925, ecc2k130_leaf7_bridge_20260925, ecc2k130_leaf7_metered_v1_20260928, ecc2k130_leaf_native_m3_gate_20260926, ecc2k130_oriented_transport_20260924, ecc2k130_pdp_chain_holdout_20260925, ecc2k130_pdp_features_20260924, koblitz_isogeny_descent_37_20260925, koblitz_isogeny_descent_37_results_20260925, koblitz_structural_fields_20260924			
ECDLP cost across volcano levels (9 entries, 2026-09-24..2026-10-23)	null	10 members: no level effect for rho/BSGS (census 33 of 35 equivalent); l=2 IC yield 0.942 explained by cofactor classes; crater IC 22-25% claim withdrawn (IC 84.9x rho-cubic); synthetic simulator is invented data.	current; see research/volcano_ecdlp_census_20261017/REPORT.md(erratumA3)
Members: generator_effect_ic_20261019, generator_four_followups_20261020, isogeny-volcano-cost-v2_20261017, isogeny-volcano-cost_20261017, isogeny-volcano-rho, isogeny_volcano_ic_20260924, volcano_descent_dataset_20261011, volcano_ecdlp_census_20261017, volcano_ecdlp_census_followup_20261023			
degree-263 isogeny transport and leaf census (18 entries, 2026-09-25..2026-09-30)	accounting	18 members. Dual transport verified on all 264 lines; leaves lose cheap Frobenius (rho A 262->2, 11.45x); source m10 slot counts do not transfer; m3 rejected at m10 budget.	current
Members: notes/ecc2k130/LEAF_HORIZONTAL_SEVEN_BRIDGE_20260925.md, notes/ecc2k130/dual_all_lines_20260930/DECISION.md, notes/ecc2k130/dual_all_lines_20260930/PROTOCOL.md, notes/ecc2k130/dual_all_lines_20260930/README.md, notes/ecc2k130/leaf_m10_column_budget_20260930/DECISION.md, notes/ecc2k130/leaf_m10_support_20260930/PROTOCOL.md, notes/ecc2k130/leaf_m10_support_20260930/RESULT.md, notes/ecc2k130/leaf_m10_support_20260930/evidence/run_36707051160/README.md, notes/ecc2k130/leaf_m10_support_outcome_20260930/DECISION.md, notes/ecc2k130/native_fullpoint_edge_20260929/DECISION.md, notes/ecc2k130/native_fullpoint_edge_20260929/PROTOCOL.md, notes/ecc2k130/native_m3_chain_20260929/DECISION.md, notes/ecc2k130/native_m3_chain_20260929/PROTOCOL.md, notes/ecc2k130/native_m3_chain_20260929/evidence/attempt2/README.md, notes/ecc2k130/native_m3_chain_20260929/evidence/initial/README.md, notes/ecc2k130/native_m3_counting_admission_20260929/DECISION.md, notes/ecc2k130/native_m3_counting_admission_20260929/PROTOCOL.md, notes/ecc2k130/native_m3_counting_admission_20260929/README.md			
degree-7 GF(2^21) m3 policy/selector toys (7 entries, 2026-09-30)	null	7 members. Native selection beat transported in 1/4 cells; charged selector 2/8 source, 0/8 native wins; reordering not worthwhile.	current
Members: notes/ecc2k130/m3_base_selector_20260930/PROTOCOL.md, notes/ecc2k130/m3_base_selector_20260930/RESULT.md, notes/ecc2k130/m3_base_selector_20260930/SOURCE_LOCK.md, notes/ecc2k130/m3_four_policy_20260930/PROTOCOL.md, notes/ecc2k130/m3_four_policy_20260930/REPLAY_REPAIR_PROTOCOL.md, notes/ecc2k130/m3_four_policy_20260930/RESULT.md, notes/ecc2k130/m3_probe_order_bound_20260930/RESULT.md			
degree-73 descendant-native n37 factor bases (20 entries, 2026-10-01..2026-10-05)	diagnostic	20 members. Native K42 six-sum: 917/1024 direct, 1024/1024 with shifts; batch IC/rho CPU ~10.3-10.8 (exploratory); K16 seed selection no lead (p~0.75); one-target performance undetermined.	open
Members: notes/ecc2k130/n37_four_policy_pdp_20261003/MUTATION_AUDIT.md, notes/ecc2k130/n37_four_policy_pdp_20261003/PROTOCOL.md, notes/ecc2k130/n37_four_policy_pdp_20261003/RESULT.md, notes/ecc2k130/n37_four_policy_sparse16_20261004/PROTOCOL.md, notes/ecc2k130/n37_four_policy_sparse16_20261004/RESULT.md, notes/ecc2k130/n37_four_policy_support_20261003/PROTOCOL.md, notes/ecc2k130/n37_four_policy_support_20261003/RESULT.md, notes/ecc2k130/n37_native_basis_bridge_20261002/RESULT.md, notes/ecc2k130/n37_native_column_capacity_20261001/DECISION.md, notes/ecc2k130/n37_native_m6_mitm_20261002/PROTOCOL.md, notes/ecc2k130/n37_native_m6_mitm_20261002/RESULT.md, notes/ecc2k130/n37_native_one_target_20261003/PROTOCOL.md, notes/ecc2k130/n37_native_one_target_20261003/RESULT.md, notes/ecc2k130/n37_native_residual_20261002/PROTOCOL.md, notes/ecc2k130/n37_native_residual_20261002/RESULT.md, notes/ecc2k130/n37_native_residual_rho_20261005/PROTOCOL.md, notes/ecc2k130/n37_native_residual_rho_20261005/RESULT.md, notes/ecc2k130/n37_native_residual_rho_20261005/evidence/native/README.md, notes/ecc2k130/n37_native_residual_rho_20261005/evidence/python-pilot/README.md, notes/ecc2k130/n37_native_velu_transport_20261002/RESULT.md			
plane-quartic (Diem) IC (4 entries, 2026-10-05..2026-10-18)	diagnostic	4 members: bielliptic line relations verified at F_53; GHS quartic toy 90/90 recoveries but S (650,896-37,920,188) unverified due to class-merging degree-3 reduction; Diem W2a interrupted; no IC/rho ratio exists.	open; see papers/ecdlp-ic-improvement-ledger/sections/audit.tex
Members: bielliptic_quartic_20261005, diem_quartic_relations_20261018, ghs_cubic_quartic_ic_20261012, plane_quartic_program_20261018			
Jacobian-transfer screen and assessments (8 entries, 2026-10-06..2026-10-15)	accounting	8 members: registry screen 314/321 curves boundary; advantages only on deprecated Oakley/c2pnb; cover_descent rows repriced at genus floor (848 to 601 advantage curves); counted Cantor 2.0-3.3 bits over 10g^2; 15,791 cover maps assessed, no transfer advantage measured.	current
Members: cover_descent_genus_floor_20261015, cover_transfer_assessment_20261013, hyperelliptic_cover_infrastructure_20261006, jacobian_op_count_20261014, jacobian_transfer_lemma6_20261013, jacobian_transfer_scan_20261010, transfer-methodology-validation-20261013, twist_walk_blocked_towers_20261014			
covers/curves carrying ECC2K-130 subgroup (4 entries, 2026-10-06..2026-10-17)	accounting	4 members. Genus-130 model 2^37.17 unreachable; best explicit curve genus 2^28.92; N=130 GHS counterfactual 2^55.60 vs 2^60.31 (-4.71 bits) after MMT correction.	corrected; see docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md
Members: notes/ecc2k130/RESEARCH_ECC2K130_CURVE_CONSTRUCTION.md, notes/ecc2k130/RESEARCH_ECC2K130_HYPERELLIPTIC.md, notes/ecc2k130/curve_construction_20261006/PROTOCOL.md, notes/ecc2k130/curve_construction_20261006/RESULT.md			

Study	Class	Result	Status
JV weak class structure (3 entries, 2026-10-07..2026-10-08)	diagnostic	3 members: depth-1 classes never hold weak curves (0 of 3,690); 86-95% of depth >= 2 do; norm-one conductor theorem; JV class complete among 2-torsion hyperelliptic covers but not all genus-3 covers.	current
Members: `iso1_weak_classes_20261007`, `jv_weak_class_law_20261007`, `jv_weak_generalization_20261008`			
fiber-aware IC across conductor gaps (6 entries, 2026-10-07..2026-10-11)	null	6 members: fiber tables cut relation cost up to 200x but best IC cell 12.8x rho; fiber-IC/rho grows as #E^(1/6) (8.4x to 37x, 18 to 30 bits); two-torsion quotient saving withdrawn at matched columns; combined descent 234-529x worse.	current
Members: `fiber_ic_scaling_20261011`, `isogeny_conductor_gap_20261007`, `isogeny_conductor_gap_ic_20261010`, `isogeny_conductor_gap_round2_20261011`, `isogeny_conductor_gap_round3_20261011`, `isogeny_fibers_20261010`			
ECDLP difficulty across isogeny classes (4 entries, 2026-10-08..2026-10-17)	literature	4 members: transport theorem and JMV literature verdict (only large-prime vertical step open); ML predictor and difficulty protocols unrun; 61-bit trio selected; no measurement.	open
Members: `ec_hardness_predictor_20261010`, `isogeny_class_difficulty_20261008`, `isogeny_class_trio_20261017`, `jmv_self_reducibility_20261008`			
GHS cover censuses and binary GHS routes (11 entries, 2026-10-08..2026-10-24)	diagnostic	11 members: deployed prime and prime-degree binary curves have no usable GHS cover; genus floors (n=3: 13/5/3; n=4: 13/25/5; n=5: 33/33/25/5); N=130 counterfactual margin +4.71 then -1.83 bits; Oakley genus-16 relations 5.2 bits slower than rho.	current
Members: `cover_decomposition_gap_search_20261018`, `cover_route_prime_binary_20261008`, `ghs-c2pnb`, `ghs_counterfactual_n130_20261013`, `ghs_cover_census_n4_20261024`, `ghs_cover_census_n5_20261013`, `ghs_cubic_cover_census_20261012`, `ghs_curve_comparison_20261009`, `ghs_n3_genus3_census_20261024`, `ghs_oakley_extended_20261023`, `hcdlp_trial_cost_20261024`			
Joux-Vitse cover and decomposition (6 entries, 2026-10-09..2026-10-24)	diagnostic	6 members: (n-1)-decomposition 4.5e7-1.8e8x rho; plain cover S/rho 1.59-1.98 at p=1009, slope -2.05, crossover about 1330; sieve below strong rho (0.024x at p=1009) and e2e 0.058x vs pooled rho are historical, comparison withheld; prospective re-run planned.	open; see `research/jv_sieve_strong_rho_20261019/review-20261020/README.md`
Members: `joux_vitse_n1_20261015`, `jv_cover_e2e_30bit_20261009`, `jv_cover_p_scaling_20261024`, `jv_cover_sparse_fglm_20261018`, `jv_sieve_prospective_20261023`, `jv_sieve_strong_rho_20261019`			
weak-curve catalogue, database and atlas (7 entries, 2026-10-10..2026-10-12)	tooling	7 members: 814 exact models, 339,740 class observations, 36 published cases, 922-model atlas; selected-subgroup advantage unmeasured throughout.	current
Members: `weak_curve_case_index_20261012`, `weak_curve_catalog_20261010`, `weak_curve_conjectures_20261012`, `weak_curve_database_20261012`, `weak_curve_leaderboard_20261012`, `weak_curve_properties_20261010`, `weak_curve_properties_lean_20261011`			
large prime conductor gap crater vs floor (7 entries, 2026-10-10..2026-10-18)	null	7 members: no method sees a large prime conductor gap; final large-sample rho floor/crater 0.998 [0.986, 1.010]; only crater automorphisms (j=0 sqrt(3), 1.69-1.92x best-method) differ; l=8209 rho lead 1.22 closed; follow-ups partial.	current
Members: `conductor_gap_cm_orbits_20261015`, `conductor_gap_cm_pairs_20261010`, `conductor_gap_followups_20261016`, `conductor_gap_rho_large_sample_20261018`, `drastic_prime_conductor_gap_20261012`, `drastic_prime_conductor_gap_round2_20261012`, `drastic_prime_conductor_gap_round3_20261012`			
isogeny conductor-gap floor vs crater (3 dirs) (3 entries, 2026-10-11..2026-10-15)	null	floor/crater counted S 0.975 and 1.127 (CIs span 1); native CPU 0.980; combined Weil descent 234-529x blind F4, negative	current
Members: `20261011-conductor-gap-round3`, `20261012-conductor-gap-counts`, `20261015-conductor-gap-cm-orbits`			
Bhati-Singh weak curves are subfield curves (1 entries, 2026-10-12..2026-10-13)	null	2 members: free-relation factor base confined to a (1+sqrt q)^4-smooth subfield part; second weak line explained by affine symmetry.	current
Members: `weak_line_7_4_followup_20261013`			
cubic GHS to genus-3 Jacobian (4 entries, 2026-10-12..2026-10-23)	diagnostic	4 members: explicit genus-3 plane-quartic and hyperelliptic covers built (#Jac = #E0); odd-char descent family Theta(q^2); round-4 IC vs strong rho on E0(F_q^3) geomean IC/R_ideal 0.70 (2^16) to 1.52, 1.27 at 2^20 (ghs3_vs_rho, no README).	open
Members: `ghs3_vs_rho_20261023`, `ghs_cubic_genus3_build_20261012`, `ghs_cubic_hyperelliptic_genus3_20261013`, `odd_char_descent_genus3_20261014`			
Gaudry k=4 D4 decomposition over F_q^4 (7 entries, 2026-10-13..2026-10-15)	engineering	7 members: C4 from 1.227e12 (S4) to 2.54e10 (D4, 48x), 1.42e9 (weighted), 6.9e7 (chains); char 2 Frobenius form 481x; end to end 6.7e8x rho at E(F_29^4), finally 6.1x rho theory / 4.3x measured at n about 2^56, crossover extrapolated 2^68.1.	current
Members: `d4_symmetrised_s5_20261015`, `gaudry_quartic_d4_20261013`, `gaudry_quartic_d4_char2_20261015`, `gaudry_quartic_d4_cost_20261014`, `gaudry_quartic_d4_ecbench_20261013`, `gaudry_quartic_replay_20261013`, `gaudry_quartic_speed_20261013`			

Study	Class	Result	Status
Gaudry/FGHR k=4 decomposition cost (6 dirs) (6 entries, 2026-10-13..2026-10-15)	engineering	C4 1.21e12 to 2.54e10 (D4) to 6.89e7 (weighted+pivots) p=269; S5 weighted x17.8-18.5; char-2 IC/rho 6.06 at m=14, crossover extrapolated 2^68.1; IC 1.39e6x rho at E(F_29^4)	current
Members: `20261013-gaudry-k4-ecbench`, `20261013-gaudry-quartic-d4`, `20261013-gaudry-quartic-solve-speed`, `20261014-gaudry-k4-d4-cost`, `20261015-d4-weighted-s5`, `20261015-gaudry-k4-d4-char2`			
Smith (2,2,2) isogeny to plane quartics (8 entries, 2026-10-13..2026-10-18)	diagnostic	8 members: Smith 18.57% reproduced (0.18638); class ceiling 0.4897/0.4943, class share 0.4304/0.4750; clustering within L-classes unexplained by 2-adic invariants; sign law holds, bias bounded; walk cost unbuilt.	open; see `problems/smith-kernel-isogeny-class-walk.md`
Members: `smith_class_invariant_20261015`, `smith_elliptic_walk_20261018`, `smith_iso_granularity_20261015`, `smith_prop24_native_20261014`, `smith_quotient_composition_20261016`, `smith_serre_sign_20261016`, `smith_sign_scaling_20261016`, `weak_isogeny_classes_20261013`			
p61 selected isogeny-class census pilots (2 dirs) (2 entries, 2026-10-14..2026-10-15)	tooling	294,912 vertices generated 858.472 s; adjacency 4,096 vertices; full-class projection 3.632 TB logical	current
Members: `20261014-p61-class-pilot`, `20261015-p61-adjacency-pilot`			
IC improvement rounds 1-4, genus-3, D4, Koblitz oracle (14 dirs) (14 entries, 2026-10-14..2026-10-16)	engineering	best genus-3 IC-2LP whole q^1.38 local ~1.36 (not 4/3); D4 2LP regressed then fixed 0.979x; Koblitz free relations 0.960x, harvest/gates constants; no IC below rho	current
Members: `20261014-ic-improve-01-d4-2lp`, `20261014-ic-improve-02-equivariant-oracle`, `20261014-ic-improve-03-tracezero-freerel`, `20261014-ic-improve-04-genus3-2lp`, `20261015-genus3-2lp-rebalance`, `20261015-ic-improve2-01-genus3-linalg`, `20261015-ic-improve2-02-freerel-production`, `20261015-ic-improve2-03-fallrow-gate`, `20261015-ic-improve2-04-d4-2lp-fix`, `20261015-ic-improve2-05-linear-harvest`, `20261015-ic-improve2-06-genus3-kappa-bw`, `20261015-ic-improve3-01-genus3-relations`, `20261015-ic-improve3-02-harvest-gate`, `20261015-ic-improve4-01-genus3-relations`			
Joux-Vitse k=5 and Jacobian-cover studies (5 dirs) (5 entries, 2026-10-15..2026-10-20)	accounting	complete DLPs IC/rho 4.5e7-1.8e8 (plain), 1.24e6-5.77e6 (Edwards+replay); parity extrapolated ~2^227.4; sieve comparison withheld	current
Members: `20261015-joux-vitse-k5-n1`, `20261018-jv-cover-cost-breakdown-leads`, `20261018-jv-cover-sparse-fglm`, `20261020-joux-vitse-k5-edwards-replay`, `20261020-jv-sieve-strong-rho`			
complete volcano graphs and graph-wide IC (5 entries, 2026-10-16..2026-10-17)	blocked	5 members: three complete graphs (24,823 vertices, 51 strata); graph-wide IC survey checkpoints (99 and 70 vertices) halted; photographed-curve campaign restarted, IC open; no IC/rho ratio.	open; see `problems/full-volcano-class-storage-and-execution.md`
Members: `full_volcano_classes_20261016`, `full_volcano_classes_20261017`, `full_volcano_graphwide_ic_20261017`, `photo_volcano_recovery_20261017`, `photo_volcano_restart_20261017`			
isogeny-volcano ECDLP/IC level-effect studies (10 dirs) (10 entries, 2026-10-16..2026-10-22)	null	no volcano-level effect on rho/BSGS; IC yield floor/crater 0.942 at l=2 (v2); IC/matched rho 17-170x and 42.4-61.5x; graph-wide IC surveys descriptive, no runtime claim	current
Members: `20261016-isogeny-volcano-rho`, `20261017-isogeny-volcano-cost`, `20261017-isogeny-volcano-cost-v2`, `20261017-photo-volcano-restart`, `20261017-volcano-C-finite-field`, `20261017-volcano-ecdlp-census`, `20261017-volcano-ic-campaign`, `20261017-volcano-ic-release-stopped`, `20261017-volcano-repaired-2935`, `20261017-volcano-repaired-qa`			
`research/gaudry_allocation_20260914` 2026-09-14 Gaudry cubic solver allocation optimization	engineering	Identical outputs and field counters (ratio 1.0); about 5.4% less elapsed time on residual corpus and 7.0% less on cold full-DLP; existing partial S ratios to matched rho 816.30, 1201.32, 827.41 unchanged.	current
`research/notes/index-calculus/RESEARCH_DIEM_DESCENT.md` 2026-09-19 Diem-style IC toy on E/F_25	tooling	Toy E/F_25: subfield-x factor base of 4 points; 18/25 random targets 2-decompose; demonstration only. (measured toy)	current
`research/p192_extension_cover_20261012` 2026-10-12 exact extension-cover screen on P-192	null	Full 141-pivot census: 680,474 split lines give 1,360,948 distinct residual columns, zero selected-base overlap, empty leaf core: zero constraints on selected-class logs.	current
`research/weak_subfield_hyperelliptic_decomposition_20261012` 2026-10-12 Bhati-Singh weak genus-2 curves	accounting	Weak curves are subfield curves: factor base generates H inside J_C0(F_q) or J_C0(F_q^2), every prime factor of |H| <= (1+sqrt q)^4; Kani pullback image meets H only in 0.	current
`research/experiment_proposals_20261014` 2026-10-14 five proposals after MMT Lemma 6 correction	accounting	Index of five proposals (twist walk, cost-model calibration, Smith conditions, genus-3 descents); pilot: 13 towers on 5 catalog curves blocked on the twist, none blocked as recorded; earlier pilot a-in-{0,1} assumption corrected.	current

Study	Class	Result	Status
`research/pair46_factor_bases_20261015` 2026-10-15 pair-46 CM(-19) factor-base campaign, 14 families	null	No factor-base family gave an IC advantage on pair 46 or toys; 8 exact obstructions (scoped), item 14 refuted to empirical_negative by red team; item 01 best arm 1.54 [1.30, 1.82] x BSGS.	current
`research/cm_volcano_crt_20261016` 2026-10-16 CM modular polynomials via explicit CRT	tooling	Bounded reference implementation of Broker-Lauter-Sutherland CRT modular polynomials for odd prime degrees 3-31 with exact-equality acceptance; no speedup or asymptotic claim.	current
`research/f4_early_stop_20261018` 2026-10-18 F4 Hilbert early-stop for JV cover	tooling	Recovered opt-in Hilbert-based F4 stopping (lead L4 of #2216) with full-library validation 4,400 passed / 0 failed / 118 ignored; performance comparison never aggregated (aggregate.sh is a placeholder), draft PR; no number.	open
`research/fp4_d4_ladder_20261023` 2026-10-23 Gaudry k=4 D4 pipeline on F_p^4 ladder	blocked	Protocol and cards only (baseline S 1.950e8 to 3.835e8 lower bounds from earlier ladder); no result in this directory; no number.	open

C.5 Rho, the harness and validation

95 entries: 17 campaigns collapsed from 70 entries, and 25 single entries.

Study	Class	Result	Status
ecc2k130-rho-campaign (5 entries, 2026-09-25..2026-10-12)	mixed	5 dirs. ECC2K-130 rho campaign audit and GPU iteration screens; no candidate beat shipping DP32 walk.	current
Members: `dp32_sigma_table_20261012`, `ecc2k130_dp_ingest_rust_20261007`, `ecc2k130_iteration_redesign_20261012`, `ecc2k130_rho_audit_20260925`, `ecc2k_cycle_escape_20260926`			
tau-adic known-scalar arithmetic (CPU/GPU) (6 entries, 2026-09-29)	engineering	6 members. CPU m=131 holdout ratio 0.413; GPU compiled only; no walk/rho benefit established.	open
Members: `notes/ecc2k130/tau_adic_arithmetic_20260929/PROTOCOL.md`, `notes/ecc2k130/tau_adic_arithmetic_20260929/README.md`, `notes/ecc2k130/tau_adic_arithmetic_20260929/gpu/PROTOCOL.md`, `notes/ecc2k130/tau_adic_arithmetic_20260929/gpu/READINESS_PROTOCOL.md`, `notes/ecc2k130/tau_adic_arithmetic_20260929/gpu/README.md`, `notes/ecc2k130/tau_adic_arithmetic_20260929/gpu/UNROLL_DEVICE_PROTOCOL.md`			
autoresearcher harness run receipts (2 entries, 2026-09-29..2026-10-01)	tooling	2 members; all EXP-SEMAEV-001 records completed_valid; toy only.	current
Members: `ecdlp_harness_run_20251001`, `ecdlp_harness_run_20260929`			
ecbench harness calibration, frontier and audits (15 entries, 2026-10-02..2026-10-23)	diagnostic	15 members; harness calibrated (2384/2384 verified); best IC 4.30x strong rho (all-candidates), compact orbit 9.985x; online n61 13.9x corrected (empty-table rho); frontier v2: strong rho ops leader 1.089 [0.927, 1.261].	current
Members: `ecbench_all_candidates_20261003`, `ecbench_allocator_confound_20261023`, `ecbench_bounds_challenges_20261006`, `ecbench_calibration_20261002`, `ecbench_callgrind_solve_20261004`, `ecbench_conductor_gap_20261012`, `ecbench_factor_bases_20261013`, `ecbench_frontier_20261013`, `ecbench_frontier_v2_20261023`, `ecbench_hardware_20261009`, `ecbench_independence_20261009`, `ecbench_numa_vm_20261002`, `ecbench_pair_claw_20261003`, `ecbench_yield_sweep_20261004`, `eks_crypto_validation_20261019`			
strengthening the strong signed-Frobenius rho baseline (6 entries, 2026-10-08..2026-10-22)	engineering	6 members; Koblitz strong rho 1,196 -> 938 -> 781 -> 718 instructions/step (n53), round 4 707.2 -> 679.7 Ir/step; ecbench standing rho 0.802155x legacy Ir; outputs identical; native gains not established.	current
Members: `ecbench_rho_optimized_20261014`, `rho_assurance_20261016`, `rho_batchinv_20261022`, `rho_canon2_20261016`, `rho_step3_20261016`, `strong_rho_canon_20261008`			
GPU rho ECC2K-130 and NIST (5 dirs) (5 entries, 2026-10-13..2026-10-17)	engineering	ECC2K-130 sigma walk 14.64 audited / 15.88 B it/s RTX PRO 6000; sigma-table 1.024x; P-256 1.632 B it/s (W=32); seeding 17.4x	current
Members: `20261013-nist-gpu-rho-ladder`, `20261014-ecc2k130-signed-sigma`, `20261014-paper-figures`, `20261015-ecc2k130-modal-rollout`, `20261017-gpu-nist-jac-seed`			
ECDLP private-key-recovery audit programme (2 entries, 2026-10-14..2026-10-14)	tooling	2 members. Charter plus first checkpoint; no scientific gate promoted; cover catalog 2,100 rows with source evidence only.	open
Members: `notes/ecdlp-general/RESEARCH_COMPREHENSIVE_ECDLP_AUDIT_PLAN.md`, `notes/ecdlp-general/RESEARCH_ECDLP_AUDIT_EXECUTION_CHECKPOINT_20261014.md`			

Study	Class	Result	Status
strong signed-Frobenius rho reference rounds (5 dirs) (5 entries, 2026-10-14..2026-10-22)	engineering	instructions per step 1,196 to 938 to 781 to 718 to 679.7 (n53); steps/ideal 0.922-0.994; native gains not established	current
Members: `20261014-ecbench-rho-optimized`, `20261016-strong-rho-assurance`, `20261016-strong-rho-canon2`, `20261016-strong-rho-step3`, `20261022-strong-rho-batchinv`			
one-word prime-field backends (3 entries, 2026-10-15..2026-10-17)	engineering	3 dirs. fp_backend layer (39/39 claims pass); branch-free Fp64 rho kernel 0.436-0.466x time; neutral for two_sylow_search (max 1.083).	current
Members: `fp64_branchfree_20261016`, `fp_backend_20261015`, `two_sylow_fp64_20261017`			
hwcost time-and-dollar model (2 entries, 2026-10-16)	tooling	2 members; calibrated model 1.8-5.5% median error on rho; round 2 mostly unvalidated by sweep B (L0 wall time).	current
Members: `hwcost_model_20261016`, `hwcost_round2_20261016`			
hardware cost model and Fp64 kernels (4 dirs) (4 entries, 2026-10-16..2026-10-17)	engineering	model error 1.8-5.5% calibrated; branch-free Fp64 rho kernel 0.40-0.47x time; IC > rho wall on every ranked curve; two-sylow neutral	current
Members: `20261016-fp64-branchfree`, `20261016-hwcost-rho-model`, `20261016-hwcost-round2`, `20261017-two-sylow-fp64`			
generic DLP correctness and memory (2 dirs) (2 entries, 2026-10-17)	engineering	Pohlig-Hellman steps 26.62x fewer; BSGS 1.30-1.98x; rho start-up removal 1.14-5.84x; vs PARI elllog 0.020-0.167	current
Members: `20261017-generic-dlp-correctness`, `20261017-generic-dlp-memory-parallelism`			
generic DLP audit fixes and engineering (2 entries, 2026-10-17)	engineering	2 members; collab 2-cycle escape and Pohlig-Hellman fixes validated; BSGS prefetch / Jacobian start-up wall gains unvalidated by sweep B.	current
Members: `generic_dlp_correctness_20261017`, `generic_dlp_memory_parallelism_20261017`			
hyperelliptic Jacobian rho walk rounds 9-12 (3 dirs) (3 entries, 2026-10-18..2026-10-19)	engineering	negation fold 0.968 at N>=1e5 (falsified); hash mixing falsified; full-width branches 2.100 to 1.481 at largest row (8-bit reference was slow)	current
Members: `20261018-hec-rho-hash-mixing`, `20261018-hec-rho-negation`, `20261019-hec-rho-branch-width`			
isogeny-fibers round 6 (2 dirs) (2 entries, 2026-10-18..2026-10-19)	diagnostic	pooled rho descriptive; trace labels all E/4E cosets over 44,740,508 points; CT test 2.5-5.0x table	current
Members: `20261018-r6-pooled-rho`, `20261019-r6-3-coset-label`			
validation sweep, gate and perfindex (3 entries, 2026-10-19..2026-10-24)	accounting	3 members; sweep: 5 contradictions (A 1, B 4); gate: Table B 13/13 pass, Table C 0/9; perfindex null 1.005x [0.985, 1.019].	current
Members: `ic_validation_gate_20261024`, `perfindex_validation_20261024`, `validation_sweep_20261019`			
Certicom 131-bit rho/IC baselines (3 dirs) (3 entries, 2026-10-21..2026-10-23)	diagnostic	ECCp-131 small-x probes zero relations; step rates ECCp-131 15.06 M/s, ECC2-131 11.46 M/s; best IC 3.4-75x matched rho to 2^39; none admitted	current
Members: `20261021-eccp131-exact-probe`, `20261022-certicom131-rho-ic-baseline`, `20261023-certicom131-rho-parity`			
`research/notes/ecdlp-general/RESEARCH_BENCH_LOG.md` 2026-05-12 cryptanalysis bench exponent log	diagnostic	Toy bench (7-20 bits, 1 sample/scale): rho alpha measured 0.807 vs 0.500; IC 2-decomp alpha 0.691-0.825 vs theoretical 1.5 (toy scale, asymptotic not dominant). (measured)	current
`research/tiered_rho_compiled_walk_20260921` 2026-09-21 compiled reservoir-guided tiered rho walk	diagnostic	Pinned 321-byte schedule, exhaustive over 65,537 target logs: capped mean 19027631/65537 vs 19114086/65537 additions, a 0.4523% online reduction (theoretical, toy group).	current
`research/ic_rho_koblitz_20260923` 2026-09-23 Koblitz batch rho and pricing references	accounting	Batch rho k=32 costs 0.0300/0.0252/0.0219 per target (k32/k1 0.210/0.239/0.190); closest IC figure 1.17x reads 6.51x vs batch rho.	current
`research/notes/ecdlp-general/RESEARCH_P256_RHO_ARITHMETIC.md` 2026-09-25 P-256 rho arithmetic engineering matrix	engineering	Pre-registered engineering plan for P-256 rho iteration cost (fused Solinas reduction, batching, DP, CPU/GPU/FPGA); no performance result claimed. (plan)	open
`research/nist_gpu_arithmetic_20260926` 2026-09-26 NIST GPU rho arithmetic dispatch	blocked	Infrastructure non-result: Modal credentials absent in CI, GPU steps skipped; no throughput measurement; no number.	open
`research/p256_rho_round1_20260926` 2026-09-26 P-256 field fold and toy rho pilot	diagnostic	Python Solinas fold 5.74x slower than Python big-int (stage negative); toy rho on p=8191: r-adding table 8 to 64 cuts mean walk steps ~989 to ~263, xlow vs xor2 indistinguishable.	current

Study	Class	Result	Status
`research/notes/ecc2k130/isogeny_rho_speed_20260930/PROTOCOL.md` 2026-09-30 rho on isogenous ECC2K-130 curves	accounting	Preregistered question whether rho is cheaper on an isogenous vertex; null predicts A drop on leaves.	current
`research/notes/ecc2k130/isogeny_rho_speed_20260930/RESULT.md` 2026-09-30 rho on isogenous ECC2K-130 curves	accounting	Isogenous rho not faster: descending 263-leaves keep only A=2, S floor 0.8862 vs 0.07743 (11.45x penalty, sqrt(131)); n17 screen consistent; GPU leaf run not executed (no credentials).	current
`research/notes/ecc2k130/isogeny_rho_speed_20260930/gpu/ACCESS.md` 2026-09-30 rho on isogenous ECC2K-130 curves	blocked	GPU access notes: no Modal/RunPod credentials; RunPod inventory HTTP 403; CUDA campaign has no leaf kernel.	current
`docs/ic/measurement/README.md` 2026-10-01 ICMS v1 IC measurement standard	tooling	[narrative methodology] Comparability standard: found several quantities all called S (GAE, wall/ batched add, callgrind) and one thread reporting 0.286, 5.03 and 110.79 for different windows; enforces op counts, one-target primary. (n.a.)	current
`research/prime_fast_bench_20261008` 2026-10-08 native prime-field rho and IC bench	engineering	Native Fp64 vs BigUint: 28-bit rho 466x faster wall; native S3 2-decomposition IC 704x faster at 20 bits, but still 0.499 s vs rho 0.0193 s at 20 bits and 1744 s vs 0.51 s at 32 bits.	current
`benchmarks/20261014-rho-walk-leads` 2026-10-14 four prime-field rho walk proposals (copy of research/rho_walk_leads_ 20261014) [adm:listed-pre-gate; sweep:VC; lb:no]	null	best of P+-R steps speedup 1.129 (1.081-1.184), but step cost 1.215x on P-256 (net ~0.95); negation rule per epoch 0.828 (worse than none); most leads no net gain	current
`research/rho_walk_leads_20261014` 2026-10-14 four prime-field rho walk proposals	diagnostic	Better-of-two step 1.13x [1.08, 1.18] fewer steps but net loss at large batches; lazy-x table index 0.733x; shared index 0.073x; per-epoch class rule 0.83x, worse than no negation map.	current
`research/ic_frontier_campaign_20261015` 2026-10-15 IC frontier campaign, every family, 5a band	accounting	4,848 executions (3,178 verified, 54 exhausted): no IC arm beats rho at any band weight; compact orbit 3.8-15x signed-Frobenius rho at w=0; prime pair table 9.8->95.2x (w=0); no IC exponent below rho's.	current
`research/prime_rho_automorphism_20261015` 2026-10-15 rho on full automorphism group	diagnostic	rho.cubic/rho.negation 0.611 [0.505, 0.739] (j=0, predicted 0.577); rho.quartic/rho.negation 0.791 [0.673, 0.923] (j=1728, predicted 0.707); class-map unpriced, no wall result.	current
`benchmarks/20261016-sea-point-count` 2026-10-16 BSGS and SEA point counting speed (copy of research/sea_point_count_ 20261016) [adm:listed-pre-gate; sweep:VC/V; lb:no]	engineering	BSGS24 61.747 to 31.495 us/call (1.98652x), BSGS40 1936.899 to 591.519 us (3.24310x); SEA40 0.81228x, SEA192 1.02393x within A/A	current
`benchmarks/20261017-prime-order-search` 2026-10-17 prime-order curve search O(sqrt p) (copy of research/prime_order_search_ 20261017) [adm:listed-pre-gate; sweep:U; lb:no]	engineering	ecbench plan time 217.6 to 0.137 s at 26 bits (1,592x); same curves and 288 workload IDs re-planned; setup only, L0	current
`research/prime_order_search_20261017` 2026-10-17 O(sqrt p) prime-order curve search	engineering	Point-stepping order test cuts ecbench plan time 0.212 to 0.040 s (16 bits) up to 217.607 to 0.137 s (26 bits, 1,592x); same curves re-planned for 288 workloads. Sweep: match.	current
`research/equal_advice_wide_20261018` 2026-10-18 equal-advice preprocessing rho vs IC, n=73/83	accounting	no number: instrument built and checked (n=61 narrow/wide identical 24/24); n=73/n=83 session frozen and not run.	open
`research/k_conversion_20261018` 2026-10-18 K conversion column touches to rho steps	accounting	One rho step 6665 Ir at n=83, 725 Ir at n=61; K = 4.0-15.6 rho steps per column touch (pooled m=4 median 4.48), raising priced rows.	current

Study	Class	Result	Status
`docs/ic/CLI_PROGRESS_RESUME.md` 2026-10-21 icx live output and checkpoint/resume	tooling	[usage] icx run verbose JSON streaming, checksummed checkpoints, S3 mirroring and resume semantics; no result. (n.a.)	current
`benchmarks/20261023-ecbench-allocator-confound` 2026-10-23 ecbench allocator confound PR 2455 (copy of research/ecbench_allocator_confound_20261023) [adm:not-listed; sweep:-; lb:no]	diagnostic	counts identical (0 differences); smoke-prime solve wall 18.4-38.8x shorter on PR binary for eight rho/BSGS/kangaroo arms; IC online not distinguished (1.067); not admitted	current
`research/ext_endo_rho_20261023` 2026-10-23 endomorphism-orbit rho on F_p^2 and F_p^3 curves	tooling	No README; band tables show S(w=0) rho-frob 0.48-0.69 vs rho-neg 0.78-1.40 (F_p^3) and rho-gls 0.47-0.92 vs rho-neg 0.81-1.17 (F_p^2), but at w=1 folded arms exceed rho-neg; no pooled verdict written.	open
`benchmarks/20261024-perfindex-null-validation` 2026-10-24 perfindex null A/A validation (copy of research/perfindex_validation_20261024) [adm:not-listed; sweep:-; lb:no]	tooling	136 kernels: 134 neutral, 2 slower, 0 faster (1.5% false verdicts); performance index 1.005 (0.985-1.019), qualified false	current
`research/expspec_smoke_20261020` UNREAD experiment-spec smoke test (Go)	tooling	no number (single smoke.go file).	current

C.6 Engines, factor-base tooling and literature

228 entries: 27 campaigns collapsed from 187 entries, and 41 single entries.

Study	Class	Result	Status
factor-base selectors, designs, yield studies (8 entries, 2026-09-15..2026-10-18)	diagnostic	8 members; best: greedy orbits 1.08-1.18x fewer expected trials, trace-zero GP 15-17/128 vs 4/128 holdout decompositions; coset 1.020x [0.959, 1.083] no advantage; Lean model: smallest base optimal under measured slopes.	open
Members: `factor_base_designs_20261015`, `factor_base_dor_campaign_20261013`, `factor_base_evaluation_20261015`, `factor_base_selector_20260915`, `factor_base_shape_leads_20261018`, `factor_base_traces_20261016`, `fb_hyperparam_sweep_20261014`, `greedy_factor_base_yield_20261014`			
endomorphisms, automorphisms, auxiliary inputs (5 entries, 2026-09-18..2026-10-21)	engineering	6 members: GLV-chain scalar multiplication 1.492-1.525x counted (endo_native_bench); Cheon p-1 beats rho only for DLP with auxiliary inputs (deployed SRS lose 5.74-14.32 bits); torsion not a lever; DGM ideas unrun; nothing moves plain ECDLP or IC.	current
Members: `automorphism_ideas_20261021`, `cp6_782_endomorphism_chain_20261005`, `endo_native_bench_20261016`, `endomorphism_search_20261005`, `torsion_auxiliary_inputs_20260918`			
volcano metadata correctness (4 entries, 2026-09-22..2026-09-24)	accounting	4 members: f_pi vs f_E corrected, point-count certificate bug fixed, Koblitz End=O_-7 maximal, transported factor base leaves decomposition counts unchanged (240/240).	current
Members: `volcano_metadata_20260922`, `volcano_metadata_20260923`, `volcano_n53_n83_20260924`, `volcano_regularity_control_20260924`			
IC ledger, leaderboard and accounting docs (7 entries, 2026-09..2026-10-20)	accounting	7 members. Leaderboard: no IC row below its reference (Table B 5.47-38.2x, A 5.23-21x, C 3.62-14.9x); n61 13.9x and n83 18.5x online wins superseded 2026-10-20; BOUNDARY_TARGETS.md still states the n61 win.	current; see `research/validation_sweep_20261019/A-leaderboard/REPORT.md`
Members: `docs/ic/BOUNDARY_TARGETS.md`, `docs/ic/BOUNDARY_TARGETS_CHARGED_SUPPLEMENT_20261008.md`, `docs/ic/DASHBOARD.md`, `docs/ic/LEADERBOARD.md`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `docs/ic/STRICT_SINGLE_TARGET_HOST.md`, `docs/ic/dashboard-review-20261002/README.md`			
Alman-Vassilevska Williams thin products (5 entries, 2026-10-06..2026-10-06)	null	6 members: no IC stage or EC kernel passes the shape/ring/size gates; refuted 3SUM/APSP hardness carries no load in the repo; one plan unrun.	current
Members: `lopsided_general_regime_20261006`, `lopsided_implications_20261006`, `lopsided_other_speedups_20261006`, `lopsided_thin_product_20261006`, `wanted_thin_product_20261006`			

Study	Class	Result	Status
native isogeny catalogue tooling (3 entries, 2026-10-09..2026-10-09)	engineering	3 members: instruction-count stage gains (screen-p224 31.39x then 3.12x vs new baseline), CI closeout; 349-model catalogue; no ECDLP number.	current
Members: `isogeny_catalogue_setup_20261009`, `isogeny_ci_closeout_20261009`, `isogeny_coverage_optimization_20261009`			
isogeny-loop GLV/GLS scalar multiplication (3 entries, 2026-10-10..2026-10-12)	engineering	3 members: round-1 model sweep (char-2 erratum), round-2 4-D GLV-GLS model 1.220x, round-3 measured 1.147x/1.091x vs GLS-2 at D=-619; curve arithmetic, not DLP.	current
Members: `isogeny_loop_glv_20261010`, `isogeny_loop_glv_20261011`, `isogeny_loop_glv_20261012`			
IC leads ladders, scoping and reviews (8 entries, 2026-10-10..2026-10-18)	diagnostic	8 members; next-10 campaign ran all ten items (genus-3 DLP exponent -0.525 corrected); equal advice retires online wins; free-relation rank 49 vs 599; no lead below rho; exponent questions memory-blocked.	open
Members: `ecdlp_leads_20261018`, `ic_cover_leads_20261018`, `ic_free_relation_census_20261011`, `ic_leads_20261010`, `ic_leads_rented_20261018`, `ic_leads_review_20261018`, `ic_next10_20261014`, `ic_open_leads_scoping_20261011`			
2025-26 literature workstreams (8 entries, 2026-10-11)	engineering	8 dirs. Constant-factor engineering (gate-first 1.57-1.78x, E/<T> 1.35-1.64x, BSGS 1.15-1.67x); negatives for Coppersmith S5, zero-minor (260-6715x rho), lane batching; no cost below SGGM bound.	current
Members: `lit2026_speedup_program`, `lit2026_w10_s5_coppersmith_20261011`, `lit2026_w11_generic_engineering_20261011`, `lit2026_w12_flagged_claims_20261011`, `lit2026_w3_ggm_audit_20261011`, `lit2026_w4_early_abort_20261011`, `lit2026_w5_lane_batched_f4_20261011`, `lit2026_w8_summation_models_20261011`			
lit2026 literature-lead weeks W4/W5/W8/W11/ W12 (7 dirs) (7 entries, 2026-10-11..2026-10-12)	engineering	gate-first 1.57-2.41x fewer word XORs; quotient-S3 1.34-1.64x; early-abort null; summation models negative; zero-minor 260-6715x rho	current
Members: `20261011-lit2026-w11-generic-engineering`, `20261011-lit2026-w4-early-abort`, `20261011-lit2026-w5-gate-first-lane-roots`, `20261011-lit2026-w8-quotient-s3`, `20261011-lit2026-w8-summation-models`, `20261012-lit2026-w12-zero-minor`, `20261012-lit2026-w5-gate-first-rref`			
Comparisons against external CAS engines (2 entries, 2026-10-11..2026-10-13)	diagnostic	2 dirs. msolve 12-24x faster on larger square F_p systems (later closed to parity in fp-f4); Boolean n=17 decision 7-24 ms vs full-basis 40-52 s (different task); Magma comparison prepared but never run.	open
Members: `gb_cas_comparison_20261011`, `magma_f4_comparison_20261013`			
F_p Groebner engine vs msolve/Singular (5 dirs) (5 entries, 2026-10-11..2026-10-16)	engineering	f4_fp from 12-24x behind msolve to level single-threaded (0.96-1.42); blocked pairs 0.75-0.78x msolve at quad_n10/n11; fp_backend lazy rows 9.5x	current
Members: `20261011-groebner-vs-msolve-singular-polybori`, `20261012-f4-fp-vs-msolve`, `20261015-f4-generic-field`, `20261015-fp-backend`, `20261016-f4fp-blocked-pairs`			
inherited-F4 / F6-IC rewrite and order (3 dirs) (3 entries, 2026-10-11..2026-10-16)	engineering	lazy/eager 1.40-1.62; gate-first F6-IC 1.336-1.733x fewer XORs (default); shift runs 1.22-1.70x vs scatter	current
Members: `20261011-inherited-f4-eager-rewrite`, `20261016-f6-gate-first-x86`, `20261016-inherited-f4-eager-runs`			
Boolean F4/F5 engine instruction/allocation work (7 dirs) (7 entries, 2026-10-11..2026-10-22)	engineering	F5 Ir -11.02% (readback), -3.947% (packing), -3.715% (hashing); defaults 1.11-1.27x largest F5; Arm64 keeps defaults	current
Members: `20261011-f4-f5-row-allocation`, `20261012-f4-f5-dense-packing`, `20261012-f4-f5-fused-readback`, `20261012-f4-f5-native-hashing`, `20261012-f4-f5-performance-history`, `20261017-f5-gf2-defaults`, `20261022-arm64-f4-defaults`			
Semaev root finding vs F4/Buchberger/msolve (5 entries, 2026-10-12 (5 members))	engineering	383-1535x vs matrix-F4 per decomposition (table excluded); 3021-5802x NIST is vs Buchberger; complete binary solves F4 faster at n<=13, root finding 122x at n=23; prime 50-105x msolve.	corrected
Members: `gpu_rootfind_decomposition_20261012`, `semaev_root_vs_groebner_20261012`, `semaev_rootfind_e2e_20261012`, `semaev_rootfind_nist_20261012`, `semaev_rootfind_scaling_20261012`			
dedicated PDP root-finder requirements, build, ecbench (3 entries, 2026-10-12..2026-10-13 (3 members))	engineering	Root-finders sit on enumeration/pair-table floors (slopes 1.000, 1.995); 350x wall vs F4 at n=13 in ecbench; IC/rho grows 2.9 to 25,116 over n=13..37.	current
Members: `pdp_root_finder_20261012`, `pdp_root_finder_ecbench_20261013`, `pdp_root_finder_requirements_20261012`			
root-finding/pair-table IC in ecbench (8 dirs) (8 entries, 2026-10-12..2026-10-14)	engineering	pair-table beats subtraction (0.018861 prime); B64/ B32 0.830; m3/m2 1.43; IC 16-20x rho (prime, natives unpriced), 2.9-25,116x strong rho binary	current
Members: `20261012-rootfinding-decomposition`, `20261013-ic-rootfind-ecbench`, `20261013-prime-rootfinding-arity`, `20261013-prime-rootfinding-batchtable`, `20261013-rootfinding-pair-size`, `20261014-rootfind-stride-targets`, `20261014-rootfinding-buffers`, `20261014-rootfinding-lanes-coverage`			

Study	Class	Result	Status
named research obstructions (37 records) (30 entries, 2026-10-12..2026-10-23)	blocked	27 open, 8 investigating, 2 resolved; IC-relevant: ECCp-131 zero natural rank, k=5 JV parity ~2^55, 64-unknown Boolean cap, cover Jacobian/norm gaps, rootfinding cost calibration	open
Members: problems/binary-boolean-system-64-variable-cap.md, problems/binary-cover-arbitrary-jacobian-norm-missing.md, problems/cairn-certificate-checker-python-only.md, problems/catalog-cover-subgroup-certificates-incomplete.md, problems/certicom-binary-gpu-rho-integration.md, problems/conductor-gap-upper-floor-and-cost.md, problems/curve-traits-debug-subfield-trace.md, problems/ecbench-claim-n13-replay-accounting-version-drift.md, problems/ecbench-gpu-matched-workload-contract.md, problems/eccp131-full-parameter-index-calculus.md, problems/ecdlp-portfolio-reading-coverage.md, problems/full-volcano-class-storage-and-execution.md, problems/generalized-rootfinding-all-roots-8x.md, problems/ghs-to-smith-pipeline-characteristic-mismatch.md, problems/ic-improve-04-genus3-21p-study-missing.md, problems/k5-joux-vitse-trial-cost-and-fpk-ecbench.md, problems/n83-online-aa-noise.md, problems/n83-prebuilt-base-quality-m2-m4.md, problems/n83-recorded-query-sampler-contract-mismatch.md, problems/p61-full-isogeny-class-census.md, problems/p61-registered-root-terminal-closure.md, problems/photo-volcano-checkpoints-unavailable.md, problems/plane-quartic-jacobian-class-arithmetic-missing.md, problems/prime-cover-balanced-jacobian-and-induced-maps-missing.md, problems/prime61-horizontal-class-memory.md, problems/release-suite-jv-nagao-wall-budget.md, problems/rho-worker-tier1-host-access.md, problems/rootfinding-complete-cost-calibration.md, problems/screenshot-factor-base-driver-unlocated.md, problems/smith-kernel-isogeny-class-walk.md			
IC next-10 leads round (5 dirs) (5 entries, 2026-10-14)	mixed	D4 18.2x constant k=4; genus-3 cubic GHS q^1.300; PKM m=4 degree plateau breaks; equal-advice audit: BL rho 53-317x faster online	current
Members: 20261014-ic-next10-01-fp4-ladder, 20261014-ic-next10-03-pkm-degree, 20261014-ic-next10-06-equivariant-f4-m4, 20261014-ic-next10-07-cubic-ghs-genus3, 20261014-ic-next10-09-equal-advice			
factor-base design and selection studies (9 dirs) (9 entries, 2026-10-14..2026-10-19)	null	no factor-base advantage at size: greedy gain 1.18x vanishes by r=2^39; cyclotomic yield-neutral; orbit 2.96-18.2x rho-frob; IC 38-159x rho Ir	current
Members: 20261014-fb-hyperparam-sweep, 20261014-frobenius-orbit-ic-vs-rho, 20261014-greedy-factor-base-yield, 20261015-cyclotomic-falsifier, 20261015-factor-base-evaluation, 20261015-pair46-factor-bases, 20261017-perf-history-factor-bases, 20261018-factor-base-shape-leads, 20261019-factor-base-designs-ek1			
Relation-matrix linear-algebra engines (3 entries, 2026-10-14..2026-10-19)	engineering	3 dirs. CADO filter port 10.40x and 1.93x faster than CADO-NFS at D=262,144; LA 0.11-1.9% of runs; sparse full-rank solvers raise end-to-end S (up to 5.72x) via completion rule; in ecbench sparse-cado leaves S at 1.0000. LA not the bottleneck.	current
Members: ic_linalg_pipeline_20261019, la_engines_20261014, sparse_la_timing_20261018			
ic-improve-rounds (5 entries, 2026-10-14..2026-10-23)	mixed	5 dirs. Paired component rounds: genus-3 2LP exponent 1.910 to ~1.37; Koblitz free relations 0.960x; Z[tau] base 2-10x CPU (loaded host).	current
Members: ic_improve3_20261015, ic_improve4_20261015, ic_improve_20261014, ic_improve_20261015, ic_improve_20261023			
native norm/generalized rootfinding line (5 dirs) (5 entries, 2026-10-16..2026-10-19)	diagnostic	GF(2^83) search UNKNOWN, 0 verified witnesses; variables 58,176 to 45,427; historical separator 1.3830x; no qualified speedup	current
Members: 20261016-generalized-rootfinding, 20261017-rootfinding-native-cycles, 20261017-rootfinding-native-followup, 20261017-rootfinding-paper-synthesis, 20261019-adaptive-pair-cells			
generalized rootfinding 8x program (quintics, GF(2^83) norms) (28 entries, 2026-10-16..2026-10-19 (28 members))	tooling	Exact components with passing small controls; quintic squarefree predicate rejected as uniform 8x; every GF(2^83) hard norm search inconclusive/ UNKNOWN; transfer formula erratum; F4/F5/F6 gate unqualified.	open
Members: actual_norm_repeated_pair_20261017_v1, actual_norm_tangency_20261017_v1, actual_norm_tangency_20261017_v2, actual_norm_u_rank_20261017_v1, adaptive_pair_cells_20261019_v1, crt_support_circuit_20261017_v1, direct_support_karatsuba_20261017_v1, fixed_pair_cell_profile_20261019_v1, full_binary_norm_circuit_20261017_v1, full_binary_norm_sat_20261017_v1, generalized_rootfinding_20261016, hybrid_norm_search_20261017_v2, hybrid_norm_xor_20261017_v1, native_long_xor_interface_20261017_v1, norm_source_symmetry_20261017_v1, norm_xor_cones_20261017_v1, odd_interval_roots_20261017_v1, odd_interval_roots_20261017_v2, pair_cell_obstruction_20261019_v1, quintic_cached_frobenius_20261017_v1, quintic_domain_roots_20261017_v1, quintic_fixed_u_cells_20261017_v1, quintic_odd_oracle_20261016_v1, quintic_rational_support_20261017_v1, quintic_rational_support_20261017_v2, quintic_resultant_20261016_v1, rootfinding_publication_20261017_v1, rootfinding_publication_20261017_v2			
factor-base-sweep CLI docs (5 entries, 2026-10-17..2026-10-18)	tooling	5 members. Software only; no speed or regularity verdict.	current
Members: docs/ic/factor-base-sweep/CI_REPAIR_20261018.md, docs/ic/factor-base-sweep/IMPLEMENTATION.md, docs/ic/factor-base-sweep/MAINTENANCE_20261018.md, docs/ic/factor-base-sweep/README.md, docs/ic/factor-base-sweep/VALIDATION.md			
Certicom 131-bit rho baselines and IC program (4 entries, 2026-10-20..2026-10-23 (4 members))	accounting	Rho 2^60.81/2^64.83/2^64.92 steps; ECCp-131 step rate 15.06 M/s, ECC2-131 parity; IC gates closed at m<=3; falsifiers null; CUDA not run.	current
Members: certicom131_cuda_port_20261023, certicom131_ic_program_20261020, certicom131_rho_ic_bench_20261022, certicom131_rho_parity_20261023			

Study	Class	Result	Status
curve registry trait engine (2 entries, 2026-10-22) Members: `curve_traits_levels_20261022`, `curve_traits_tower_cost_20261022`	tooling	2 dirs. Open volcano levels 668 to 34 to 21.	current
`research/notes/index-calculus/RESEARCH_AUTOLAB_LOG.md` 2026-05-21 daily autolab log (isogeny/lattice/IC threads)	diagnostic	90 autolab sessions 2026-05-21 to 2026-09-12: LLL Gram-Schmidt NaN fix, Howe-cover/CM threads, GLV-HNP BKZ, IC boundary beats; several in-log corrections (e.g. 2026-06-10 E1[2] claim). (mixed)	current
`research/notes/ecdlp-general/RESEARCH_ECDLP_STATE_OF_THE_ART.md` 2026-06 ECDLP state of the art survey 2025-2026	literature	Survey: generic sqrt(n) still best for good curves; no subexponential prime-field ECDLP; IC gains only over F_{q^n}, FFD assumption doubted; records ~112-bit prime / ~117-bit binary. (literature)	current
`research/notes/index-calculus/RESEARCH_EXOTIC_COORDINATES.md` 2026-09-08 algorithmic search for exotic point coordinates	engineering	2-torsion frame u=1/(x+1) plus involution invariants halves S4 degrees ([4,4,4,4] to [2,2,2,2,1]) in both characteristics; constant moves 2-3 orders at toy n<=23, IC still worse than rho at deployed sizes; 'targets per solve' withdrawn s11. (measured toy)	corrected; see `research/notes/index-calculus/RESEARCH_EXOTIC_COORDINATES.mdsection11`
`docs/ic/README.md` 2026-09-09 ic research framework hub	tooling	[usage hub] Hub/manual for the ic executable: commands, factor-base encyclopedia and technique links, boundary scoreboard schema, autolab runner. (n.a.)	current
`research/index_calculus_baseline_20260914` 2026-09-14 EC IC accounting baseline and WDSat pilot	accounting	Accounting contract, budget calculator and certified six-instance WDSat pilot (60-input regression suite); full-DLP S and rho ratios null; no speedup claimed.	current
`research/summation_polynomials_20260914` 2026-09-14 summation-polynomial specialisation engineering	engineering	Grouped Horner cuts specialization time 55.4% (fixed slots 49.4%); setup 1.0-1.2%; full-DLP confirmation shows neither improvement nor regression; plus separate accounting correction.	current
`research/ic_solver_selection_20260915` 2026-09-15 ML per-query solver selection (pairs vs SAT)	null	Fitted model always chooses pairs; hindsight oracle equals best constant policy, so zero routing headroom; default pairs retained.	current
`research/notes/README.md` 2026-09-19 index of research notes by theme	tooling	Index of every research note grouped by theme (ecdlp-general, index-calculus, cm-isogeny, ecc2k130, lattice-hnp, ...); no result.	current
`research/notes/index-calculus/RESEARCH_GROEBNER_F4.md` 2026-09-19 minimal Buchberger/F4 solver module	tooling	Minimal Buchberger with Gebauer-Moller pruning: ~6 vars degree<=4 in seconds, 8+ vars hits 5,000-step brake; 6 tests pass. (tooling)	current
`research/notes/ecdlp-general/RESEARCH_REPRESENTATION_STRUCTURE.md` 2026-09-20 transfer pattern and R1-R5 admissibility test	literature	Position note: every subexponential ECDLP attack transfers to a category with a non-generic algorithm licensed by a published object; R1-R5 test; murmurations fail it; rev.2 corrected pattern (falsifier F4 fired). (derived)	corrected; see `research/notes/ecdlp-general/RESEARCH_REPRESENTATION_STRUCTURE.mdsections2.4,9(rev.2)`
`research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.md` 2026-09-21 IC boundary ledger: three regimes, one unit	accounting	No variant below rho on any instance with r>=2^20 (closest prime MITM m=3 at 14.5x at 2^23.4); every fitted total exponent >=0.54; prime base costs r^{2/3}; s3.5 accounting correction of counting ceiling. Source of LEADERBOARD Table B. (measured)	corrected; see `research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.mdsection3.5`
`research/radical_pair_image_20260921` 2026-09-21 radical unordered-pair image presentation (msolve)	engineering	Radical image lowers max F4 degree on every target (e.g. 15->11 at prime b=9); cold batch C/R 0.097-0.586 for b>=5 and binary k=2,3; prime b=3 regresses (2.3834).	current
`docs/ic/FRAMEWORK.md` 2026-09-22 IC benchmarking framework design	tooling	[narrative design] Five-stage pluggable IC harness reporting every stage in one unit against a planted log; solver trait and class vocabulary (engineering/relabelling/...). (n.a.)	current
`docs/ic/ENGINE_AUDIT_AND_DESIGN.md` 2026-09-24 curve-generic icx engine audit and design	tooling	[narrative design] Audit of every IC code path and design of icx; largest solved end-to-end: 48-bit subgroup of icv1-f2m61-t158598901-ab42b6c5; GOST tc26 cofactor corrected 1 -> 4. (n.a.)	current

Study	Class	Result	Status
`docs/ic/perf/OPTIMIZATION_PLAN.md` 2026-09-24 IC performance optimisation plan and log	engineering	[narrative plan+log] Measure-change-remeasure log: f2m53 whole pipeline 1.65 s; wide_groebner n=31 m=6 (154 unknowns) complete 2/2 planted; 2026-09-24 correction: n=31 m=3 82 s/target did not reproduce. (measured)	corrected; see `docs/ic/perf/OPTIMIZATION_PLAN.md(logrow2026-09-24correction)`
`research/env_reads_20260924` 2026-09-24 getenv reads out of F4 inner loops	engineering	Engineering on uncounted work: getenv was 1.4% of instructions; ladders 1.03-1.07x faster wall, whole logarithms flat (0.975x [0.954, 0.996]); no counter moves.	current
`research/isogeny_invariant_baseline_20260926` 2026-09-26 isogeny-derived torus fraction factor bases small field	accounting	Historical small-field stage diagnostic: torus fraction base V1 over F81 is nonlinear, 27 field values, 5-51 affine points per curve; factor-base size must come from evaluated values not tuples; no rho or full ECDLP reference.	current
`docs/ic/NIST_PARAMETER_COVERAGE.md` 2026-09-29 NIST 11-curve parameter validation	tooling	[usage] icx validate on P-256, K-163..571, B-163..571 with Sage cross-check and S3 fixtures; parameter diagnostic only. (n.a.)	current
`research/notes/index-calculus/IC_TOOL_PROGRAM.md` 2026-10-01 ic tool fidelity/speed programme charter	tooling	Programme charter (declared 2026-10-01) for making the ic binary the highest-fidelity IC emulator: goals, fidelity levels, frozen suite, round ledger; R02b withdrawn on v3 (2026-10-06). (plan)	open
`research/notes/ecdlp-general/RESEARCH_NZMS_AMS_AUSTMS_2024_IDEAS.md` 2026-10-03 2024 NZMS-AMS-AustMS ideas screened under R1-R5	literature	All 43 sessions screened; ~40 talks turned into ECDLP ideas; no new licensing datum, six closures derived, one live lever preregistered; no measurement. (literature/plan)	current
`docs/ic/ECDLP_RESEARCH_TECHNIQUES.md` 2026-10-04 IC technique inventory matrix	tooling	[narrative inventory] Inventory mapping IC techniques (factor base, relations, filtering, LA, descent) to native modules labelled integrated/ experimental/component/backlog; not a performance claim. (n.a.)	current
`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` 2026-10-10 audit of F4/F5/ F6-IC/Macaulay engines	engineering	[narrative audit] Correctness-first audit of every Groebner/Macaulay engine: one soundness bug fixed (truncated run reported as solutions), engines agree with brute force on pinned tests; speed fixes keep verdicts. (measured)	current
`research/literature_search_20261011` 2026-10-11 2025-26 literature search ECDLP, IC, Groebner	literature	No paper in 2025-01 to 2026-10 (61 classified) claims or plausibly delivers an exponent-level ECDLP improvement; summation-polynomial/GHS searches found nothing new.	current
`research/notes/ecc2k130/conversation_leads_assessment_20261011/REPORT.md` 2026-10-11 conversation-export IC leads review	literature	Source review of a private conversation export; tabulates current record (e.g. disjoint cold IC/rho 2.810/13.378/22.675 at n37/41/53 one-target); no new experiment or number.	current
`benchmarks/20261013-ecbench-frontier` 2026-10-13 ECBench evidence synthesis and replay audit (copy of research/ecbench_frontier_ 20261013) [adm:listed-pre-gate; sweep:X; lb:yes]	accounting	Cold frontier remains signed-Frobenius rho (1.089x floor [0.927, 1.261]); compact orbit 9.985x matched rho at n17; online 13.9x (n61) and 18.528x (N83) both SUPERSEDED (vs rho without table)	corrected; see `research/validation_sweep_20261019/B-benchmarks/verdicts.jsonl;benchmarks/20261014-ic-next10-09-equal-advice;benchmarks/20261016-n83-precomputed-rho-control`
`research/ecc_arithmetic_20261013` 2026-10-13 Karatsuba carryless GF(2^m) multiply (PMULL/ PCLMUL)	engineering	Two-limb multiply uses 3 carryless products instead of 4, three-limb 6 instead of 9; all outputs match reference; no qualified wall-time result (host load too variable).	current
`research/ic_speed_and_rho_scope_20261013` 2026-10-13 scope: faster IC end to end, where it beats rho	accounting	Planning note: rootfind-mitm ladder S/strong rho 2.9, 15, 61, about 3,000 at n=13,19,23,31; relation phase is the cost; linear-scan IC total ~2^(2n/3) vs rho 2^(n/2).	current

Study	Class	Result	Status
`benchmarks/20261014-ic-program-paper` 2026-10-14 IC programme paper figures (copy of papers/ ecdlp-index-calculus-program) [adm:archived-documentary; sweep:VC; lb:no]	literature	synthesis, no new measurement: admitted n=61 online 13.94x vs rho without table retired; genus-3 ~q^1.4 vs rho q^1.5; k=4 char-2 6.1x rho near 2^56, crossover extrapolated 2^68; every cold IC loses	current
`benchmarks/20261015-ic-frontier-campaign` 2026-10-15 one-target cold counts all IC families (copy of research/ic_frontier_ campaign_20261015) [adm:listed-pre-gate; sweep:V; lb:no]	null	4,848 executions; no IC arm beats rho on any curve at any weight; compact orbit 3.8-15x signed-Frobenius rho (w=0); prime pair table 10.5 to 101.8x rho-neg; no IC exponent below rho's	current
`benchmarks/20261015-ic-linear-algebra-cado` 2026-10-15 IC linear algebra vs CADO-NFS (copy of research/ la_engines_20261014) [adm:listed-pre-gate; sweep:U; lb:yes]	engineering	ours 1.86x faster than CADO bwc at D=131,072; after filter port D=262,144: 86.57 s to 8.35 s (10.40x), 1.93x faster than CADO-NFS; LA 0.11-1.9% of real runs; L0 unvalidated	current
`docs/ic/factor-base-encyclopedia/README.md` 2026-10-15 encyclopedia of every factor base tried	literature	[narrative encyclopedia] Compiles every factor-base family (PKM Kummer tower, isogeny tower, prime, F_p^k, binary, Koblitz, covers); pre-2026-10-15 x-rho ratios are lower bounds; pair-table family law S=0.75 r^{1/6}; corrected 2026-10-16 claim audit. (quoted)	corrected; see `docs/ic/factor-base-encyclopedia/README.md(2026-10-16claimaudit:PKMD>=8atm=4N=20;P-192t=48cardsuperseded)`
`research/notes/ecc2k130/RESEARCH_ECC2K130_IC_LITERATURE.md` 2026-10-17 ECC2K-130 IC literature survey	literature	No published approach outside exclusion set gives numerically explicit route beating 2^60.8 at n=131; first-fall-degree question a stalemate; 12 claims confirmed, 13 refuted.	current
`docs/ic/FACTOR_BASE_FACTORY.md` 2026-10-19 factor-base generator and object store	tooling	[narrative design] Native generator/evaluator requirements per curve family (Koblitz, binary, prime, anomalous); yield and degree are measured objectives; no result. (n.a.)	current
`docs/ic/CERTICOM131_IC_PROGRAM.md` 2026-10-20 ECC2-131 / ECCp-131 IC cost model and falsifiers	accounting	[narrative programme] Programme doc: cost model, gates, two toy falsifiers; gap to rho does not close at any measured size; structured prime bases within 3% of random-oracle expectation (verdict null); Falsifier A unfinished for n>=13. (measured toy + model)	current
`research/pdp_literature_leads_20261022` 2026-10-22 PDP reformulations beyond Semaev: 19 leads	literature	Desk review: most leads close at the desk (same ideals/oracles or constant factors); follow-up LEAD-19 projected base measured 1.29, 1.58, 3.36x at the published rung (engineering).	current
`problems/README.md` 2026-10-23 index of named research obstructions	tooling	index of 37 problem records with status, what each blocks and what would unblock it; resolved records retained	n.a.
`research/ec-index-calculus-review` UNREAD IC review scripts (audit labels, witness verify)	tooling	no number (two scripts, no README).	current
`research/gbrl` UNREAD RL pair selection for Buchberger on Semaev systems	diagnostic	PAPER.md: PPO agent cuts S-polynomials 5.5% and field ops 9.1% vs sugar on cyclic-6; on Semaev decomposition systems wins S-poly count on 99/100 held-out instances (toy Buchberger).	current
`research/notes` UNREAD notes directory (catalogued separately)	tooling	Directory of research notes; each note has its own row in this catalogue.	current
`problems/TEMPLATE.md` n.a. problem record template	tooling	template for obstruction records: status open/ investigating/resolved/superseded, kind, affected work, evidence, unblock conditions	n.a.
`research/grouped-recursion-algebra-eval` n.a. grouped recursion shared-algebra optimization plan	blocked	Experiment contract only (PLAN.md): grouped finite-field polynomial products, hash-consing and exact convolution for F4/F5 and summation-polynomial workloads; no implementation, no number.	open

C.7 Out of scope

These entries are in the repository but are not elliptic-curve index calculus or its baselines: `benchmarks/20261010-isogeny-campaign-and-profiler` (P192/P224 large-prime isogeny campaign profiler (copy of research/p192_p224_hundred_large_prime_isogenies_20261010) [adm:not-listed; sweep:V; lb:no]); `benchmarks/20261011-eigen-seta-1813z-bench` (Eigen-Seta KEM decapsulation benchmark (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1827z-sidechannel` (Eigen-Seta Toy16 instruction-count timing probe (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1944z-l1-fix` (Eigen-Seta L1 scalar-recovery follow-up (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1946z-l3-l5-fix` (Eigen-Seta L3/L5 scalar-recovery follow-up (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1955z-final-l1` (Eigen-Seta L1 after input validation (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1955z-verify` (Eigen-Seta Toy16/L1 verification costs (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1956z-pke-l1` (Eigen-Seta L1 authenticated PKE timing (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-eigen-seta-1958z-pke-l3-l5` (Eigen-Seta L3/L5 PKE timing samples (copy of research/eigen_seta_rust_20261011) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261011-glv-gls-4d-benchmark` (GLV-GLS-4 scalar multiplication, moderate-D loops (copy of research/isogeny_loop_glv_20261012) [adm:not-listed; sweep:U; lb:no]); `benchmarks/20261011-glv-gls-lattice-cost-model` (GLV-GLS lattice operation-count model (copy of research/isogeny_loop_glv_20261011) [adm:not-listed; sweep:VC; lb:no]); `benchmarks/20261011-isogeny-vertical-margin-historical` (Velu vs sqrt-Velu vertical kernel cost (copy of research/isogeny_fibers_20261010/round4/16_3_vertical_margin) [adm:not-listed; sweep:VC; lb:no]); `benchmarks/20261012-glv-gls-4d-confirmation` (GLV-GLS D=-619 second-host confirmation (copy of research/isogeny_loop_glv_20261012) [adm:not-listed; sweep:U; lb:no]); `benchmarks/20261013-ecc-arithmetic` (F2mElement::mul paired benchmark (copy of research/ecc_arithmetic_20261013) [adm:listed-pre-gate; sweep:-; lb:no]); `benchmarks/20261013-ecc-arithmetic-modal` (GF(2^m) multiply paired panels (copy of research/ecc_arithmetic_20261013) [adm:listed-pre-gate; sweep:-; lb:no]); `benchmarks/20261014-isogeny-performance` (BMSS isogeny constructor optimization (copy of research/isogeny_performance_20261014) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261015-isogeny-family-census` (isogeny_algos family coverage census (copy of research/isogeny_performance_20261014) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261015-isogeny-kernel-baselines` (seven kernel constructor baselines (copy of research/isogeny_performance_20261014) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261015-isogeny-native-cpu` (BMSS constructors first native CPU timing (copy of research/isogeny_performance_20261014) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261015-isogeny-rung4-control` (fourth BMSS candidate exact control (copy of research/isogeny_performance_20261014) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261016-endo-native-bench` (GLV-2 isogeny-chain endomorphism scalar mult (copy of research/endo_native_bench_20261016) [adm:listed-pre-gate; sweep:V/VC; lb:no]); `benchmarks/20261016-fixed-degree-isogenies` (fixed-degree isogenies (BKMPSW) end to end (copy of research/fixed_degree_isogenies_20261016) [adm:listed-pre-gate; sweep:U; lb:no]); `benchmarks/20261016-isogeny-large-ell` (sqrt-Velu at large prime degree (copy of isogeny_algos/docs/large-ell-20261016) [adm:listed-pre-gate; sweep:U; lb:no]); `benchmarks/20261016-isogeny-round5` (isogeny_algos optimisation round 5 (NTT) (copy of isogeny_algos/docs/round5-20261016) [adm:listed-pre-gate(x2); sweep:U/U/U/U; lb:no]); `benchmarks/20261016-isogeny-vs-flint-pari` (isogeny_algos vs FLINT and PARI (copy of isogeny_algos/docs/external-20261016) [adm:listed-pre-gate; sweep:VC; lb:no]); `benchmarks/20261017-eigen-seta-pk-degree-test` (Eigen-Seta public-key degree test oracle (copy of research/eigen_seta_assessment_20261017) [adm:listed-pre-gate; sweep:V/U; lb:no]); `benchmarks/20261017-eigen-seta-whitepaper` (Eigen-Seta white paper timings (copy of research/eigen_seta_whitepaper_20261017) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261017-isogeny-flint-losses` (isogeny_algos fixes vs FLINT/PARI losses (copy of isogeny_algos/docs/flint-losses-20261017) [adm:listed-pre-gate; sweep:-; lb:no]); `benchmarks/20261017-safecurves-trapdoor-slots` (sextic-twist weak sibling class census (copy of research/safecurves_trapdoor_slots_20261017) [adm:listed-no-VALIDATION; sweep:-; lb:no]); `benchmarks/20261018-eigen-seta-ifma` (Eigen-Seta AVX-512 IFMA multiplication (copy of research/eigen_seta_ifma_20261018) [adm:not-listed; sweep:-; lb:no]); `benchmarks/20261018-eigen-seta-speedups` (Eigen-Seta PR5 speed-ups (copy of research/eigen_seta_speedups_20261017) [adm:not-listed; sweep:-; lb:no]); `problems/ecc-arithmetic-platform-measurement.md` (ECC arithmetic cross-platform

performance unqualified); problems/ecc-arithmetic-pr-merge-gates.md (ECC arithmetic PR 1798 merge gates); problems/legacy-spot-process-recovery.md (recovery of legacy Spot processes); problems/nist-extension-twist-composite-cofactors.md (NIST extension-twist composite cofactor unsplit); problems/nist-factorization-raw-archive-publication.md (NIST factorization raw archive publication); problems/sea-rust-group-refinement-proof.md (SEA Rust-to-group refinement proof); problems/sqisign-statistical-root-gate.md (SQISign statistical assertion blocks root gate); research/aes_neural_distinguisher (AES neural/RL trail distinguisher code); research/cd_attack (Castryck-Decru SIDH attack reconstruction); research/csidh_toy (CSIDH toy implementation); research/eigen_seta_assessment_20261017 (Eigen-Seta v2 isogeny KEM security assessment); research/eigen_seta_ifma_20261018 (Eigen-Seta AVX-512 IFMA field multiplication); research/eigen_seta_prototype_20261011 (Eigen-Seta toy Sage prototype); research/eigen_seta_rust_20261011 (Eigen-Seta v2 Rust KEM and PKE); research/eigen_seta_speedups_20261017 (Eigen-Seta output-identical speed-ups); research/eigen_seta_toy_20261011 (Eigen-Seta toy Rust implementation status); research/eigen_seta_whitepaper_20261017 (Eigen-Seta performance and security white paper); research/fixed_degree_isogenies_20261016 (fixed-degree supersingular isogenies and sqrt-Velu); research/isogeny_degree_challenge_20261015 (largest-prime-degree isogeny challenge tool); research/isogeny_performance_20261014 (native isogeny constructor performance campaign); research/isogeny_world_frontier_20261016 (prime-degree isogeny world-frontier campaign); research/lll_degeneracy_rerun_20261016 (LLL degeneracy probes re-run (HNP)); research/lopsided_ec_arithmetic_20261006 (lopsided thin products for EC arithmetic (round 5)); research/notes/cm-isogeny/RESEARCH_EDS_RESIDUE.md (elliptic divisibility sequence residue handle); research/notes/cm-isogeny/RESEARCH_NIST_CYCLOTOMIC_POINT_HEIGHTS.md (cyclotomic factorisation of NIST p+-1, point heights); research/notes/cm-isogeny/RESEARCH_NIST_SOLINAS_EXPERIMENTS.md (NIST Solinas smoothness follow-up experiments); research/notes/cm-isogeny/RESEARCH_NIST_SOLINAS_STRUCTURE.md (cyclotomic smoothness of NIST Solinas p+-1); research/notes/cm-isogeny/RESEARCH_P256.md (P-256 CM discriminant and structural profiling); research/notes/code-based/RESEARCH_TII_MCELIECE.md (TII McEliece key-recovery challenges imported); research/notes/ecdlp-general/RESEARCH_SRS_CHEON_CENSUS.md (Cheon attack on deployed powers-of-tau setups); research/notes/ecdlp-general/RESEARCH_TORSION_AUXILIARY_INPUTS.md (auxiliary inputs and torsion vs rho/IC); research/notes/lattice-hnp/RESEARCH_CGA_HNC.md (class-group-amortised HNP cryptanalysis phase 1); research/notes/lattice-hnp/RESEARCH_GLV_HNP.md (GLV-aware HNP on secp256k1 proposal); research/notes/lattice-hnp/RESEARCH_GLV_HNP_PHASE2.md (GLV k1-only-leak lattice construction); research/notes/lattice-hnp/RESEARCH_HNP_LANDSCAPE.md (side-channel/HNP attack landscape for ECC); research/notes/lattice-hnp/RESEARCH_LLL_GS_ANALYSIS.md (Gram-Schmidt analysis of secp256k1 LLL degeneracy); research/pqc_ml_kem_profile_20261005 (ML-KEM gap profile vs pq-crystals); research/pqc_ml_kem_reference_20261005 (ML-KEM same-host pq-crystals reference); research/pqc_ml_kem_system_level_20261005 (ML-KEM system-level optimisation round); research/scripts (misc scripts (P-224, Solinas, twist demo)); research/sea_point_count_20261015 (SEA point counting review); research/sea_point_count_20261016 (SEA point counting measurements, formal checks); research/sidh_toy (SIDH toy, Castryck-Decru notes); research/srs_auxiliary_census_20260927 (Cheon exposure of deployed powers-of-tau setups); research/synthetic_performance (synthetic positive-control performance suite); research/synthetic_volcano_ecdlp_20261018 (synthetic volcano/ECDLP simulator); research/tii_mceliece (vendored TII McEliece recovered keys); research/token_efficiency_20261020 (token usage metrics for agent workflows).

References

[1] Alessandro Amadori, Federico Pintore, and Massimiliano Sala. On the discrete logarithm problem for prime-field elliptic curves. *Finite Fields and Their Applications*, 51:168–182, 2018.

[2] Daniel J. Bernstein and Tanja Lange. Computing small discrete logarithms faster. In *Progress in Cryptology – INDOCRYPT 2012*, volume 7668 of *LNCS*, pages 317–338, 2012.

[3] Henry Corrigan-Gibbs and Dmitry Kogan. The discrete-logarithm problem with preprocessing. In *Advances in Cryptology – EUROCRYPT 2018*, LNCS. Springer, 2018. Cryptology ePrint Archive, Paper 2017/1113.

[4] Claus Diem. An index calculus algorithm for plane curves of small degree. In *Algorithmic Number Theory – ANTS VII*, volume 4076 of *LNCS*, pages 543–557, 2006.

[5] Claus Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica*, 147(1): 75–104, 2011.

[6] Claus Diem and Emmanuel Thomé. Index calculus in class groups of non-hyperelliptic curves of genus three. *Journal of Cryptology*, 21(4):593–611, 2008.

[7] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases (F4). *Journal of Pure and Applied Algebra*, 139(1–3):61–88, 1999.

[8] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases without reduction to zero (F5). In *Proceedings of ISSAC 2002*, pages 75–83. ACM, 2002.

[9] Jean-Charles Faugère, Pierrick Gaudry, Louise Huot, and Guénaël Renault. Using symmetries in the index calculus for elliptic curves discrete logarithm. *Journal of Cryptology*, 27(4):595–635, 2014.

[10] Steven D. Galbraith and Pierrick Gaudry. Recent progress on the elliptic curve discrete logarithm problem. *Designs, Codes and Cryptography*, 78(1):51–72, 2016.

[11] Steven D. Galbraith and Shishay W. Gebregiyorgis. Summation polynomial algorithms for elliptic curves in characteristic two. In *Progress in Cryptology – INDOCRYPT 2014*, LNCS. Springer, 2014. Cryptology ePrint Archive, Paper 2014/806.

[12] Steven D. Galbraith, Robert Granger, Simon-Philipp Merz, and Christophe Petit. On index calculus algorithms for subfield curves. In *Selected Areas in Cryptography – SAC 2020*, volume 12804 of *LNCS*. Springer, 2021. Cryptology ePrint Archive, Paper 2020/1315.

[13] Robert P. Gallant, Robert J. Lambert, and Scott A. Vanstone. Improving the parallelized Pollard lambda search on anomalous binary curves. *Mathematics of Computation*, 69(232):1699–1705, 2000.

[14] Pierrick Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation*, 44(12):1690–1702, 2009.

[15] Pierrick Gaudry, Florian Hess, and Nigel P. Smart. Constructive and destructive facets of Weil descent on elliptic curves. *Journal of Cryptology*, 15(1):19–46, 2002.

[16] Pierrick Gaudry, Emmanuel Thomé, Nicolas Thériault, and Claus Diem. A double large prime variation for small genus hyperelliptic index calculus. *Mathematics of Computation*, 76(257):475–492, 2007.

[17] Ming-Deh A. Huang, Michiel Kosters, and Sze Ling Yeo. Last fall degree, HFE, and Weil descent attacks on ECDLP. In *Advances in Cryptology – CRYPTO 2015*, LNCS. Springer, 2015. Cryptology ePrint Archive, Paper 2015/573.

[18] David Jao, Stephen D. Miller, and Ramarathnam Venkatesan. Do all elliptic curves of the same order have the same difficulty of discrete log? In *Advances in Cryptology – ASIACRYPT 2005*, volume 3788 of *LNCS*, pages 21–40. Springer, 2005.

[19] Gary McGuire and Daniela Mueller. A new index calculus algorithm for the elliptic curve discrete logarithm problem and summation polynomial evaluation. Cryptology ePrint Archive, Paper 2017/1262, 2017.

[20] Christophe Petit and Jean-Jacques Quisquater. On polynomial systems arising from a Weil descent. In *Advances in Cryptology – ASIACRYPT 2012*, volume 7658 of *LNCS*, pages 451–466. Springer, 2012.

[21] Christophe Petit, Michiel Kosters, and Ange Messeng. Algebraic approaches for the elliptic curve discrete logarithm problem over prime fields. In *Public-Key Cryptography – PKC 2016*, volume 9615 of *LNCS*, pages 3–18. Springer, 2016.

[22] John M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

[23] Igor Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. Cryptology ePrint Archive, Paper 2004/031, 2004.

[24] Igor Semaev. New algorithm for the discrete logarithm problem on elliptic curves. Cryptology ePrint Archive, Paper 2015/310, 2015.

[25] Michael Shantz and Edlyn Teske. Solving the elliptic curve discrete logarithm problem using Semaev polynomials, Weil descent and Gröbner basis methods – an experimental study. In *Number Theory and Cryptography*, volume 8260 of *LNCS*, pages 94–107. Springer, 2013.

[26] Monika Trimoska, Sorina Ionica, and Gilles Dequen. Parity (xor) reasoning for the index calculus attack. In *Principles and Practice of Constraint Programming – CP 2020*, volume 12333 of *LNCS*, pages 774–790. Springer, 2020.

[27] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[28] Michael J. Wiener and Robert J. Zuccherato. Faster attacks on elliptic curve cryptosystems. In *Selected Areas in Cryptography – SAC 1998*, volume 1556 of *LNCS*, pages 190–200. Springer, 1999.